1. **Module Title: MATHEMATICS FOR ENGINEERS III (MA80366)**
2. **Level: 2 Semester: 1 Credits: 10**
3. **First year of presentation: 2014 Administering School of Sciences.**
4. **Pre-requisite or co-requisite modules, excluded combinations: Mathematics for Engineers I and II**
5. **Brief description of aims and content**

The Module aims to introduce students to the various properties and definitions of Fourier series, Partial Differential Equations and basic Probability and Statistics. The module will deal with these topics at a basic level, leaving more advanced techniques to the specialist courses in Engineering.

LEARNING OUTCOMES

1. Knowledge and Understanding

   Having successfully completed the module, students should be able to demonstrate knowledge and understanding of:

   1.1 Simple Fourier series and partial differential equations and the basics of probability and statistics.

   1.2 The implications of the basic mathematical theories.

2. Cognitive/Intellectual skills/Application of Knowledge

   2.1 Present simple arguments and conclusions using the mathematical theories.

   2.2 Analyse and evaluate problems in mathematics and engineering.

3. Communication/ICT/Numeracy/Analytic Techniques/Practical Skills

   Having successfully completed the module, students should be able to:

   3.1 Apply the mathematical knowledge to solve problems in range of engineering situations.

   3.2. Carry out mathematical and numerical manipulation with confidence and accuracy.

4. General transferable skills

   Having successfully completed the module, students should be able to:

   4.1 Assimilate Abstract Ideas.

4.2 Communicate information having a mathematical content.

6. INDICATIVE CONTENT

Unit I Fourier series and Introduction to Fourier Transforms

1.1. Fourier series expansion

1.2. Fourier transform

Unit II Partial Differential Equations including Boundary Value Problems

2.1. Formulation and Solution of standard types of first order equations

2.2. Lagrange's equation

2.3. Linear Homogeneous partial differential equations of second order with constant coefficients.

2.4. Classification of second order linear partial differential equations

2.5. Solutions of one–dimensional wave equation, one-dimensional heat equation.

Unit III Introduction to Probability and Statistics

3.1. Descriptive Statistics: Measures of Central tendency, *Measures of Dispersion and Measures of Forms.*

3.2. Probability: Basic concepts and definition of probability, Conditional probability.

3.3. Probability distributions including Discrete distributions e.g. binomial and Poisson distributions and Continuous distribution e.g. Normal Distribution.

3.4. Simple linear regression analysis.

7. ASSESSMENT PATTERN: CAT: 30%
Assignments+ Quizes: 20%
Final exam : 50%

8. INDICATIVE RESOURCES

1. Glyn James. (1999). Advanced modern engineering mathematics, 2nd edition Addison-Wesley
2. K.A. Stroud. (1996). Further Engineering Mathematics, 3th edition Macmillan Press ltd
3. Thomas H., Ronald J. (1977). Introductory Statistics Third edition, New York.
4. Robert V. Hogg. (2006). Probability and Statistical Inference. 7e edition. New Jersey
5. Murray R., Larry J.(1998). Schaum's outlines Statistics Third Edition. New York.

Unit I FOURIER SERIES AND INTRODUCTION TO FOURIER TRANSFORMS

1.0.Introduction

Mathematicians of the eighteenth century, including Daniel Bernoulli and Leonard Euler, expressed the problem of the vibratory motion of a stretched string through partial differential equations that had no solutions in terms of 'elementary functions'. Their resolution of this difficulty was to introduce infinite series of sine and cosine functions that satisfied the equations. In the early nineteenth century, Joseph Fourier, while studying the problem of heat flow, developed a cohesive theory of such series. Consequently, they were named after him. One important advantage of Fourier series is that it can represent a function containing discontinuities, whereas Maclaurin's and Taylor's series require the function to be continuous throughout. Fourier series and Fourier transform are investigated in this chapter.

1.1. Fourier series expansion

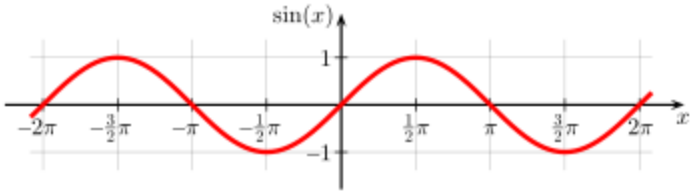
In this section we develop the Fourier series expansion of periodic functions, Dirichlet's conditions, half range sine and Cosine series.

1.1.1. Periodic functions

A function $f(x)$is said to have a ***period T*** or to be ***periodic with period T*** if for all x, $f(x) = f(x + T)$, where T is a positive constant. The least value of T > 0 is called the ***least period*** or simply ***the period*** of $f(x)$. Specifically, a function f is periodic with period. Specifically, a function *f* is periodic with period T if the graph of $f(x)$ is invariant under translation in the *x*-direction by a distance of T. A function that is not periodic is called ***aperiodic***.

Examples 1:

1. The function $sin\ x$ has periods 2π; 4π; 6π; ...; since $sin\ (x + 2\pi)$; $sin\ (x + 4\pi)$; $sin\ (x + 6\pi)$; ... all equal $sin\ x$. However, 2π is the least period or the period of $sin\ x$.

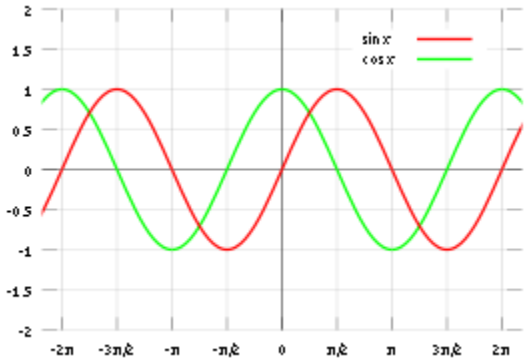


A graph of the sine function, showing two complete periods.

For example, the sine function is periodic with period 2π, since $\sin(x + 2\pi) = \sin x$; for all values of *x*. This function repeats on intervals of length 2π

2. The period of $sin\ nx$ or $cos\ nx$, where n is a positive integer, is $2\pi/n$.
3. The period of $tan\ x\ is\ \pi$.
4. A constant has any positive number as period.
5. Everyday examples are seen when the variable is *time*; for instance the hands of a clock or the phases of the moon show periodic behaviour. **Periodic motion** is motion in which the position(s) of the system are expressible as periodic functions, all with the *same* period.

If we have the following periodic function $y = A\ \sin nx$, then A is an ***amplitude;*** ***period***$= \frac{360^0}{n} = \frac{2\pi}{n}$, *n cycles in* 360^0*.The graphs of* $y = A\ \cos nx$ *have the same characteristics.*



Examples 2: Determine the amplitude and the period of the following periodic function

a) $y = 3\ \sin 5x$ b) $y = \ \sin\frac{x}{2}$ c) $y = 6\ \sin\frac{2x}{3}$

Answers:

Nº	***Amplitude***	***Period***
a)	***3***	$2\pi/_5$
b)	1	4π
c)	6	3π

1.1.2. Harmonics

A function $f(x)$ is sometimes expressed as a series of a number of different sine components. The component with the largest period is the ***first harmonic***, or ***fundamenta***l of $f(x)$.

$y = A_1\ \sin x$ is the first harmonic or fundamental

$y = A_2\ \sin 2x$ is the second harmonic

$y = A_3\ \sin 3x$ is the third harmonic

And in general

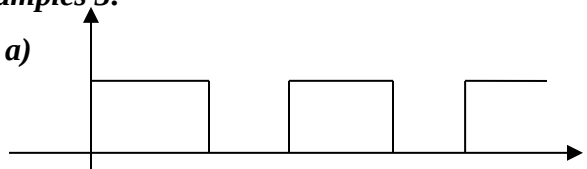
$y = A_n\ \sin nx$ is the nth harmonic, with amplitude A_n, and period $= \frac{360^0}{n} = \frac{2\pi}{n}$

1.1.3. Non-sinusoidal periodic functions

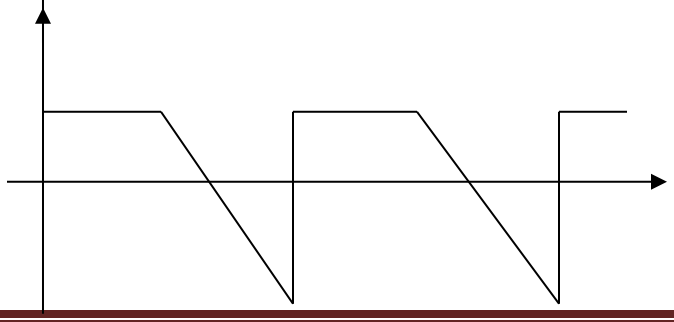
A function can be periodic without being obviously sinusoidal in appearance.

Examples 3:

a)



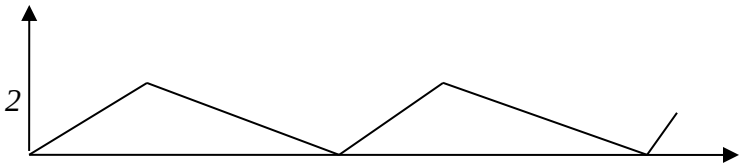
b)



1.1.4. Analytical description periodic function

Example 4:

$$f(x) = \begin{cases} x, & 0 < x < 2 \\ 3 - \dfrac{x}{2}, & 2 < x < 6 \end{cases}$$



$$f(x) = f(x+6)$$

EXERCISE 1.1

1. Sketch the graphs of the following, inserting relevant values

a) $f(x) = \begin{cases} 4, & 0 < x < 5 \\ 0, & 5 < x < 8 \\ f(x+8) \end{cases}$ *b)* $f(x) = \begin{cases} 3x - x^2, & 0 < x < 3 \\ f(x+3) \end{cases}$

c) $f(x) = \begin{cases} 2\sin x\,, & 0 < x < \pi \\ 0, & \pi < x < 2\pi \\ f(x+2\pi) \end{cases}$ *c)* $f(x) = \begin{cases} \frac{x^2}{4}, & 0 < x < 4 \\ 4, & 4 < x < 6 \\ 0, & 6 < x < 8 \\ f(x+8) \end{cases}$

1.1.5. Fourier 's theorem and the Fourier coefficients

The theorem states that: '*A periodic function that satisfies certain conditions which are Dirichlet conditions can be expressed as the sum of a number of sine functions of different amplitudes, phases and periods'.*

A Fourier series is an expansion of a periodic function $f(x)$ of period $T = 2\pi/k$ in which the base set is the set of sine functions, giving an expanded representation of the form

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \sin(nkx + \phi_n) \qquad (1)$$

Where A_0 is a constant term

$A_1, A_2, \cdots, A_n, \cdots$ denote the ***amplitudes*** of the compound sine terms

$\phi_1, \phi_2, \cdots, \phi_n, \cdots$ are constant auxiliary angles which are called ***phase angle*** *nw, which is*

The term $A_n \sin(nkx + \phi_n)$ is called the ***nth harmonic,*** and it has frequency, which is n times that of the fundamental.

Since, $A_n \sin(nkx + \phi_n) \equiv (A_n \cos\phi_n) \sin nkx + (A_n \sin\phi_n) \cos nkx$

$$\equiv a_n \cos nkx + b_n \sin nkx$$

Where $a_n = A_n \sin\phi_n$, $\quad b_n = A_n \cos\phi_n$

The expansion of a function in the form (1) had been used by Bernoulli, D' Alembert and Euler to solve problems associated with the vibration of strings. After Fourier postulated in 1807 that an arbitrary function could be represented by a trigonometric series as

Let $f(x)$ be defined in the interval $(-L, L)$; and outside of this interval by $f(x + 2L) = f(x)$, i.e., $f(x)$ is 2L-periodic. It is through this avenue that a new function on an infinite set of real numbers is created from the image on $(-L, L)$. The Fourier series or Fourier expansion corresponding to $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L} \right) \qquad (2)$$

where the Fourier coefficients a_n and b_n are

$$\begin{cases} a_n = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f(x) \cos\frac{n\pi x}{L}\, dx \\ b_n = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f(x) \sin\frac{n\pi x}{L}\, dx \end{cases} \quad \forall\, n = 0, 1, 2, \cdots$$

ORTHOGONALITY CONDITIONS FOR THE SINE AND COSINE FUNCTIONS

Notice that the Fourier coefficients are integrals. These are obtained by starting with the series, (2), and employing the following properties called orthogonality conditions:

a) $\int_{-L}^{L} \cos\frac{m\pi x}{L} \cos\frac{n\pi x}{L}\, dx = \begin{cases} 0, & if\ m \neq n \\ L, & if\ m = n \end{cases}$

b) $\int_{-L}^{L} \sin\frac{m\pi x}{L} \sin\frac{n\pi x}{L}\, dx = \begin{cases} 0, & if\ m \neq n \\ L, & if\ m = n \end{cases}$

c) $\int_{-L}^{L} \sin\frac{m\pi x}{L} \cos\frac{n\pi x}{L}\, dx = 0,$

Where m and n can assume any positive integer values

Examples 4: Determine the Fourier coefficient a_0 and a_1 ?

Solution:

a) To determine the Fourier coefficient a_0, integrate both sides of the Fourier series (2), i.e.,

$$\int_{-L}^{L} f(x)\,dx = \int_{-L}^{L}\left[\frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right)\right]dx$$

$$= \int_{-L}^{L}\frac{a_0}{2}\,dx + \int_{-L}^{L}\sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right)dx$$

$$= \int_{-L}^{L}\frac{a_0}{2}\,dx + \sum_{n=1}^{\infty}\left(a_n\int_{-L}^{L}\cos\frac{n\pi x}{L}\,dx + b_n\int_{-L}^{L}\sin\frac{n\pi x}{L}\,dx\right)$$

Since, $\int_{-L}^{L}\frac{a_0}{2}dx = a_0 L, \quad \int_{-L}^{L}\cos\frac{n\pi x}{L}dx = \int_{-L}^{L}\sin\frac{n\pi x}{L}dx = 0$

Thus, $\int_{-L}^{L} f(x)\,dx = a_0 L \rightarrow \boldsymbol{a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx}$

b) To determine a_1, multiply both sides of (2) by $\cos\frac{\pi x}{L}$ and then integrate.

$$\int_{-L}^{L} f(x)\cos\frac{\pi x}{L}dx = \int_{-L}^{L}\cos\frac{\pi x}{L}\left[\frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right)\right]dx$$

$$= \frac{a_0}{2}\int_{-L}^{L}\cos\frac{\pi x}{L}dx + \sum_{n=1}^{\infty}\left(a_n\int_{-L}^{L}\cos\frac{n\pi x}{L}\cos\frac{\pi x}{L}dx + b_n\int_{-L}^{L}\sin\frac{n\pi x}{L}\cos\frac{\pi x}{L}dx\right)$$

$$\int_{-L}^{L} f(x)\cos\frac{\pi x}{L}dx = a_1\int_{-L}^{L}\cos\frac{\pi x}{L}\cos\frac{\pi x}{L}dx$$

$$= a_1 L. \qquad After\ using\ above\ orthogonal\ conditions$$

Therefore, $\int_{-L}^{L} f(x)\cos\frac{\pi x}{L}dx = a_1 L \quad \rightarrow \boldsymbol{a_1 = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{\pi x}{L}dx}$

Remarks: *If $L = \pi$, the series (2) and the coefficients a_n and b_n are particularly simple. The function in this case has the period 2π*

Examples 5: Determine the Fourier series expansion of the following periodic functions of period 2π:

a) $f(x) = x, \quad 0 < x < 2\pi$

b) $f(x) = x^2 + x, \quad -\pi < x < \pi$

c) $f(x) = \begin{cases} x, & 0 < x < \frac{1}{2}\pi \\ \frac{1}{2}\pi, & \frac{1}{2}\pi < x < \pi \\ \pi - \frac{1}{2}x, & \pi < x < 2\pi \end{cases}$

Answers:

Fourier's coefficients determination:

a) $a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx = \frac{1}{\pi}\int_0^{2\pi} x\,dx = \frac{1}{\pi}\left[\frac{x^2}{2}\right]_0^{2\pi} = 2\pi$

$$\begin{cases} a_n = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx = \frac{1}{\pi}\int_0^{2\pi} x\cos nx\,dx = \frac{1}{\pi}\left[x\frac{\sin nx}{n} + \frac{\cos nx}{n^2}\right]_0^{2\pi} = 0 \\ b_n = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx = \frac{1}{\pi}\int_0^{2\pi} x\sin nx\,dx = \frac{1}{\pi}\left[-\frac{x}{n}\cos nx + \frac{\sin nx}{n^2}\right]_0^{2\pi} = -\frac{2}{n} \end{cases}$$

Hence,

$$f(x) = \pi - \sum_{n=1}^{\infty}\frac{2}{n}\sin nx$$

b) $a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi}(x^2+x)\,dx = \frac{2}{3}\pi^2$

$$\begin{cases} a_n = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi}(x^2+x)\cos nx\,dx = \frac{1}{\pi}\frac{4\pi}{n^2}\cos\pi x = \frac{4}{n^2}(-1)^n \\ b_n = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi}(x^2+x)\sin nx\,dx = \frac{2}{n}\cos\pi x = \frac{2}{n}(-1)^n \end{cases}$$

Hence,

$$f(x) = \frac{2}{3}\pi^2 + \left[\sum_{n=1}^{\infty}\frac{2}{n}(-1)^n\left[\frac{2}{n}\cos nx + \sin nx\right]\right]$$

c) $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\left[\int_0^{\pi/2} x\,dx + \int_{\pi/2}^{\pi}\frac{1}{2}\pi dx + \int_{\pi}^{2\pi}\left(\pi - \frac{1}{2}x\right)dx\right] = \frac{5}{8}\pi$

$$\begin{cases} a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}dx = \frac{1}{\pi}\left[\int_0^{\pi/2} x\cos\frac{n\pi x}{L}dx + \int_{\pi/2}^{\pi}\frac{1}{2}\pi\cos\frac{n\pi x}{L}dx + \int_{\pi}^{2\pi}\left(\pi-\frac{1}{2}x\right)\cos\frac{n\pi x}{L}dx\right] \\ a_n = \frac{1}{2\pi n^2}\left(2\cos\frac{1}{2}n\pi - 3 + \cos n\pi\right) = \begin{cases}\frac{1}{2\pi n^2}[(-1)^n - 1] & (even\ n)\\ -\frac{1}{2\pi n^2} & (odd\ n)\end{cases} \\ b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}dx = \frac{1}{\pi}\left[\int_0^{\pi/2} x\sin\frac{n\pi x}{L}dx + \int_{\pi/2}^{\pi}\frac{1}{2}\pi\sin\frac{n\pi x}{L}dx + \int_{\pi}^{2\pi}\left(\pi-\frac{1}{2}x\right)\sin\frac{n\pi x}{L}dx\right] \\ b_n = \frac{1}{\pi n^2}\sin\frac{1}{2}n\pi = \begin{cases}0 & (even\ n)\\ \frac{(-1)^{(n-1)/2}}{\pi n^2} & (odd\ n)\end{cases} \end{cases}$$

Hence,

$$f(x) = \frac{5}{16}\pi - \frac{2}{\pi}\left(\cos x + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \cdots\right) - \frac{2}{\pi}\left(\frac{\cos 2x}{2^2} + \frac{\cos 4x}{4^2} + \frac{\cos 6x}{6^2} + \cdots\right) + \frac{1}{\pi}\left(\sin x + \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} + \cdots\right)$$

Remark: *In the Fourier series corresponding to an odd function, only sine terms can be present. In the Fourier series corresponding to an even function, only cosine terms (and possibly a constant which we shall consider a cosine term) can be present*.

1.2.DIRICHLET'S CONDITIONS OF FOURIER'S SERIES CONVERGENCE

Suppose that

(1) $f(x)$ must be defined and single-valued on points in periodic interval $(-L, L)$;

(2) $f(x)$ must be continuous or have a finite number of finite discontinuities within a periodic interval $(-L, L)$ with period 2L;

(3) $f(x)$ and $f'(x)$ are piecewise continuous in periodic interval $(-L, L)$.

Then the series (2) with Fourier coefficients converges to

(a) $f(x)$ if x is a point of continuity

(b) $\frac{f(x+0)+f(x-0)}{2}$ if x is a point of discontinuity

Here $f(x+0)$ and $f(x-0)$ are the right- and left-hand limits of $f(x)$ at x and represent

$$\lim_{\in\to 0^+} f(x+\in) \quad and \quad \lim_{\in\to 0^-} f(x-\in) \quad \text{respectively.}$$

Examples 6: If the following functions are defined over the interval $-\pi < x < \pi$ and $(x) = f(x+2\pi)$, state whether or not each function can be represented by a Fourier series

a) $f(x) = x^3$ b) $f(x) = 4x - 5$ c) $f(x) = \frac{2}{x}$ d) $f(x) = \frac{1}{x-5}$ e) $f(x) = \tan x$

f) $f(x) = y$ where $x^2 + y^2 = 9$

Answers: A given function can be represented by a Fourier series sufficiently satisfying the above three conditions,

Nº	***Statement***	***Nº***	***Statement***
a)	*Yes*	*d)*	*yes*
b)	*Yes*	*e)*	*No*: because is infinitely discontinuity at $x = \frac{\pi}{2}$
c)	*No*: because is infinitely discontinuity at $x = 0$ i.e does not satisfy the	*f)*	*No*: because has two values

1.3. HALF RANGE FOURIER SINE OR COSINE SERIES

A half range Fourier sine or cosine series is a series in which only sine terms or only cosine terms are present, respectively. When a half range series corresponding to a given function is desired, the function is generally defined in the interval $(0, L)$ [which is half of the interval $(-L, L)$thus accounting for the name ***half range*** and then the function is specified as ***odd*** or ***even***, so that it is clearly defined in the other half of the interval, namely, $(-L, 0)$.

FOR A HALF RANGE COSINE SERIES

For a function $f(x)$ defined only over the finite interval $0 \leq x \leq L$ its **even periodic extension** $F(x)$ is the even periodic function defined by

$$\boldsymbol{F(x)} = \begin{cases} \boldsymbol{f(x)}; & \boldsymbol{0 < x < L} \\ \boldsymbol{f(-x)}; & \boldsymbol{-L < x < 0} \end{cases}$$

$$\boldsymbol{F(x + 2L) = F(x)}$$

If $f(x)$ satisfies Dirichlet's conditions in the interval $0 < x < L$, then since it is an even function of period $2L$, then even periodic extension $\boldsymbol{F(x)}$ will have a convergent Fourier series representation consisting of cosine terms only and given by

$$\boldsymbol{F(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}} \qquad (3)$$

Where

$$\boldsymbol{a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx}, \quad \boldsymbol{n = 0, 1, 2, 3, \cdots}$$

FOR A HALF RANGE SINE SERIES

For a function $f(x)$ defined only over the finite interval $0 \leq x \leq L$ **,** its **odd periodic extension** $G(x)$ is the even periodic function defined by

$$\boldsymbol{G(x)} = \begin{cases} \boldsymbol{f(x)}; & \boldsymbol{0 < x < L} \\ \boldsymbol{-f(-x)}; & \boldsymbol{-L < x < 0} \end{cases}$$

$$\boldsymbol{G(x + 2L) = G(x)}$$

If $f(x)$ satisfies Dirichlet's conditions in the interval $0 < x < L$, then since it is an odd function of period $2L$, then odd periodic extension $\boldsymbol{G(x)}$ will have a convergent Fourier series representation consisting of sine terms only and given by

$$\boldsymbol{G(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}} \qquad (4)$$

Where

$$\boldsymbol{b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx}, \quad \boldsymbol{n = 1, 2, 3, \cdots}$$

Example 7: Consider the following function defined in the interval $0 < x < 4$

$$f(x) = x,$$

Obtain:

a) a half-range cosine series expansion

b) a half-range sine series expansion

Answers:

a) ***Half-range cosine series expansion***. Define the periodic function

$$F(x) = \begin{cases} f(x) = x; & 0 < x < 4 \\ f(-x) = -x; & -4 < x < 0 \end{cases}$$

$$F(x+8) = F(x)$$

Then, since $F(x)$ is an even periodic function with period 8, it has a convergent Fourier series expansion given by (3). Taking $L = 4$, we have the Fourier's coefficients determination as:

$$a_0 = \frac{2}{4}\int_0^4 f(x)\,dx = \frac{1}{2}\int_0^4 x\,dx = 4$$

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx = \frac{2}{4}\int_0^4 x\cos\frac{n\pi x}{4}\,dx = \frac{1}{2}\left[\frac{4x}{n\pi}\frac{\sin n\pi x}{4} + \frac{16\cos n\pi x}{(n\pi)^2}\right]_0^4$$

$$= \frac{8}{(n\pi)^2}(\cos n\pi - 1) = \begin{cases} 0, & for\ \ n\ is\ even \\ -\dfrac{16}{(n\pi)^2}, & for\ \ n\ is\ odd \end{cases}$$

Hence,

$$F(x) = 2 - \frac{16}{\pi^2}\left(\cos\frac{1}{4}\pi x + \frac{1}{3^2}\cos\frac{3}{4}\pi x + \frac{1}{5^2}\cos\frac{5}{4}\pi x + \cdots\right)$$

or

$$F(x) = 2 - \frac{16}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}\cos\frac{1}{4}(2n-1)\pi x$$

Since $F(x) = f(x)$ for $0 < x < 4$, it follows that this Fourier series is representative of $f(x)$ within this interval. Thus the **half- range cosine series expansion of $\boldsymbol{f(x)}$** is

$$\boldsymbol{F(x) = x = 2 - \frac{16}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}\cos\frac{1}{4}(2n-1)\pi x,} \quad \boldsymbol{for\ 0} < x < 4$$

b) ***Half-range sine series expansion***. Define the periodic function

$$G(x) = \begin{cases} f(x) = x; & 0 < x < 4 \\ -f(-x) = x; & -4 < x < 0 \end{cases}$$

$$G(x+8) = G(x)$$

Then, since $G(x)$ is an even periodic function with period 8, it has a convergent Fourier series expansion given by (3). Taking $L = 4$, we have the Fourier's coefficients determination as:

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx = \frac{2}{4}\int_0^4 x\sin\frac{n\pi x}{4}\,dx = \frac{1}{2}\left[-\frac{4x}{n\pi}\cos\frac{1}{4}n\pi x + \frac{16}{(n\pi)^2}\sin\frac{1}{4}n\pi x\right]_0^4$$

$$= \frac{8}{n\pi}\cos n\pi = -\frac{8}{n\pi}(-1)^n$$

Hence,

$$G(x) = \frac{8}{\pi}\left(\sin\frac{1}{4}\pi x - \frac{1}{2}\sin\frac{1}{2}\pi x + \frac{1}{3}\sin\frac{3}{4}\pi x - \cdots\right)$$

or

$$G(x) = \frac{8}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\frac{1}{4}n\pi x$$

Since $G(x) = f(x)$ for $0 < x < 4$, it follows that this Fourier series is representative of $f(x)$ within this interval. Thus the **half- range sine series expansion of $f(x)$** is

$$\boldsymbol{f(x) = x = \frac{8}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\frac{1}{4}n\pi x, \quad for\ 0 < x < 4}$$

PARSEVAL'S IDENTITY

If a_n and b_n are the Fourier coefficients corresponding to $f(x)$ and if $f(x)$ satisfies the Dirichlet conditions, then

$$\boldsymbol{\frac{1}{L}\int_{-L}^{L}\{f(x)\}^2\,dx = \frac{{a_0}^2}{2} + \sum_{n=1}^{\infty}\left({a_n}^2 + {b_n}^2\right)}$$

Example 8: Consider the following function defined in the interval $0 < x < 4$

$$f(x) = x,$$

Verify the Parseval's identity

$$\frac{1}{2}\int_0^4 x^2\,dx = \frac{4^2}{2} + \sum_{n=1}^{\infty}\left(\left(\frac{-8}{n\pi^2}\right)^2 + 0^2\right),\ for\ \ n\ is\ odd$$

$$\frac{1}{2}\int_0^4 x^2\,dx = 8 + \frac{64}{\pi^4}\sum_{n=1}^{\infty}\frac{1}{n^2},\ for\ \ n\ is\ odd$$

$$\frac{32}{3} = 8 + \frac{64}{\pi^4}(1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots)$$

This imply that

$1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots = \sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = \frac{\pi^4}{24}$

1.4.FOURIER TRANSFORM

1.4.1. The Fourier integral

Let us assume the following conditions on $f(x)$*:*

(i) $f(x)$ *satisfies the Dirichlet conditions in every finite interval* $(-L, L)$*;*

(ii) $\int_{-\infty}^{\infty}|f(x)|dx$ *converges, i.e* $f(x)$ *is absolutely integrable in* $(-\infty, \infty)$*;*

Then Fourier's integral theorem states that the ***Fourier integral of a function*** $\boldsymbol{f}$ ***i****s*

$$\boldsymbol{\varphi(x) = \int_0^{\infty}\{A(\alpha)\cos\alpha x + B(\alpha)\sin\alpha x\}\ d\alpha} \qquad (\mathbf{5})$$

Where

$$\begin{cases} \boldsymbol{A(\alpha) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\cos\alpha x\,dx} \\ \boldsymbol{B(\alpha) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\sin\alpha x\,dx} \end{cases}$$

$A(\alpha)$ *and* $B(\alpha)$ *with* $-\infty < \alpha < \infty$ *are generalizations of the Fourier coefficients* a_n *and* b_n*. The right-hand side of (5)is also called a* ***Fourier integral expansion of*** $\boldsymbol{f}$ *.*

Remarks:

- *The result (5) holds if* x *is a point of continuity of* $f(x)$*.*

- *If x is a point of discontinuity, we must replace $f(x)$ by $\frac{f(x+0)+f(x-0)}{2}$ as in the case of Fourier series. Note that the above conditions are sufficient but not necessary.*
- *In the generalization of Fourier coefficients to Fourier integrals, a_0 may be neglected, since whenever*

$$\int_{-\infty}^{\infty} f(x)dx \ \ exists,$$

$$|a_0| = \left|\frac{1}{L}\int_{-L}^{L} f(x)dx\right| \to 0 \qquad as \ \ L \to \infty$$

EQUIVALENT FORMS OF FOURIER'S INTEGRAL THEOREM

Fourier's integral theorem can also be written in the forms

$$\varphi(x) = \frac{1}{\pi}\int_{\alpha=0}^{\infty}\int_{u=-\infty}^{\infty} f(u)\cos\alpha(x-u)\,du d\alpha$$

As there is no imaginary part, we can write $\cos\alpha(x-u) = e^{-i(\alpha x-\alpha u)}$, then the above expression becomes:

$$\varphi(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-i\alpha x}d\alpha \int_{-\infty}^{\infty} f(u)\, e^{i\alpha u}du \qquad (6)$$

$$\varphi(x) = \boldsymbol{\frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(u)\, e^{i\alpha(u-x)}d\alpha\, du}$$

Where it is understood that if $f(x)$ is not continuous at x the left side must be replaced by $\frac{f(x+0)+f(x-0)}{2}$.

These results can be simplified somewhat if $f(x)$ is either an odd or an even function, and we have:

$$\boldsymbol{\varphi(x)}$$
$$\boldsymbol{= \frac{2}{\pi}\int_0^{\infty}\left(\int_0^{\infty} f(u)\cos\alpha x\cos\alpha u\, du\right) d\alpha, \ \ if\ f(x) is\ an\ even\ function} \qquad (7)$$
$$\boldsymbol{\varphi(x)}$$
$$\boldsymbol{= \frac{2}{\pi}\int_0^{\infty}\left(\int_0^{\infty} f(u)\sin\alpha x\sin\alpha u\, du\right) d\alpha, \ \ if\ f(x) is\ an\ odd\ function} \qquad (8)$$

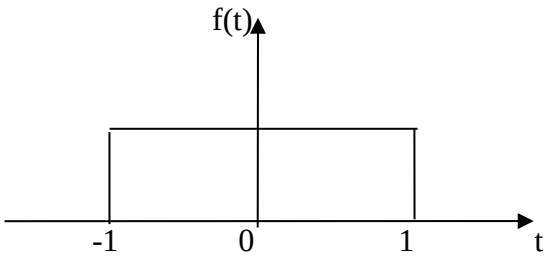
A (7) and (8) formulas are called the ***Fourier cosine integral***, and the ***Fourier sine integral*** respectively.

An entity of importance in evaluating integrals and solving differential and integral equations, then we can put (6) in the following form:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\alpha x} \left\{ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(u)\, e^{i\alpha u} du \right\} \; d\alpha$$

Example 9: Consider a rectangular pulse given by

$$f(t) = \begin{cases} 1, & |t| \le 1 \\ 0, & |t| \ge 1 \end{cases}$$



Calculate its Fourier integral

Answer: $f(t)$ is even function, then we using formula (7):

$$f(t) = \frac{2}{\pi} \int_0^{\infty} \left(\int_0^1 1 \cos \alpha t \cos \alpha u \, du \right) d\alpha = \frac{2}{\pi} \int_0^{\infty} \frac{\cos \alpha t \sin \alpha}{\alpha} \, d\alpha$$

As is an improper integral, then we consider frequencies $< \alpha_0$, when

$$f(t) = \frac{2}{\pi} \int_0^{\alpha_0} \frac{\cos \alpha t \sin \alpha}{\alpha} \, d\alpha$$

$$= \frac{1}{\pi} \int_0^{\alpha_0} \frac{\sin \alpha(t+1)}{\alpha} \, d\alpha$$

$$- \frac{1}{\pi} \int_0^{\alpha_0} \frac{\sin \alpha(t-1)}{\alpha} \, d\alpha \quad after\ trigonometric\ substitution$$

$$= \frac{1}{\pi} \int_0^{\alpha_0(t+1)} \frac{\sin u}{u} \, du$$

$$- \frac{1}{\pi} \int_0^{\alpha_0(t-1)} \frac{\sin u}{u} \, du \quad after\ changing\ \ variable\ of\ integration\ (*)$$

In infinite series, we have

$$\int_0^x \frac{\sin u}{u}\, du = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n+1)!} \qquad x \geq 0$$

Then (*) becomes:

$$f(t) = \frac{1}{\pi}\left[\sum_{n=0}^{\infty} \frac{(-1)^n[\alpha_0(t-1)]^{2n+1}}{(2n+1)(2n+1)!} - \sum_{n=0}^{\infty} \frac{(-1)^n[\alpha_0(t-1)]^{2n+1}}{(2n+1)(2n+1)!}\right], \alpha_0(t-1)$$
$$\geq 0 \;\; and \alpha_0(t+1) \geq 0$$

This expression is the Fourier integral of

$$f(t) = \begin{cases} 1, & |t| \leq 1 \\ 0, & |t| \geq 1 \end{cases}$$

1.4.2. The Fourier transform

From (6) it follows that

$$\boldsymbol{F(i\alpha) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, e^{-i\alpha x} dx} \qquad (\mathbf{9})$$

And

$$\boldsymbol{f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(i\alpha)\, e^{i\alpha x} d\alpha} \qquad (\mathbf{10})$$

Where the function $F(\alpha)$is called the ***Fourier transform*** of $f(x)$ and is sometimes written $F(i\alpha) = \mathcal{F}\{f(x)\}$.
The function $f(x)$ is the ***inverse Fourier transform*** of $F(i\alpha)$ and is written $f(x) = \mathcal{F}^{-1}\{f(x)\}$.

Example 10: Determine the Fourier transform of f if $f(x) = \begin{cases} e^{-x}, & for\ x \geq 0 \\ e^{2x}, & for\ x < 0 \end{cases}$

Answers:

$$F(i\alpha) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, e^{-i\alpha x} dx = \frac{1}{\sqrt{2\pi}} \left\{ \int_{-\infty}^{0} e^{2x}\, e^{-i\alpha x} dx + \int_0^{\infty} e^{-x}\, e^{-i\alpha x} dx \right\}$$
$$= \frac{1}{\sqrt{2\pi}} \left\{ \int_{-\infty}^{0} e^{(2-i\alpha)x}\, dx + \int_0^{\infty} e^{-(1+i\alpha)x}\, dx \right\}$$

$$= \frac{1}{\sqrt{2\pi}}\left\{\frac{e^{(2-i\alpha)x}}{2-i\alpha}\Big|_{x\to-\infty}^{x\to0^-} - \frac{e^{-(1+i\alpha)x}}{1+i\alpha}\Big|_{x\to0^+}^{x\to\infty}\right\}$$

$$= \frac{1}{\sqrt{2\pi}}\left\{\frac{e^{(2+i\alpha)x}}{2-i\alpha}\Big|_{x\to-\infty}^{x\to0^-} - \frac{e^{(1-i\alpha)(-x)}}{1+i\alpha}\Big|_{x\to0^+}^{x\to\infty}\right\}$$

$$= \frac{1}{\sqrt{2\pi}}\left\{\frac{1}{2+i\alpha} - \frac{1}{1+i\alpha}\right\} = \frac{1}{\sqrt{2\pi}}\left\{\frac{\alpha^2-2+i\alpha(5+2\alpha^2)}{(4+\alpha^2)(1+\alpha^2)}\right\}$$

Hence

$$F(i\alpha) = \frac{1}{\sqrt{2\pi}}\left\{\frac{\alpha^2-2+i\alpha(5+2\alpha^2)}{(4+\alpha^2)(1+\alpha^2)}\right\}$$

Remarks:

- If $f(x)$ is an even function, equation (7)yields

$$\begin{cases} \boldsymbol{F_c(i\alpha) = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cos\alpha x\, dx} \\ \boldsymbol{f(x) = \sqrt{\frac{2}{\pi}}\int_0^\infty F_c(i\alpha)\cos\alpha x\, d\alpha} \end{cases}$$

and we call $F_c(i\alpha)$ and $f(x)$***Fourier cosine transforms*** of each other.

- If $f(x)$ is an odd function, equation (8)yields

$$\begin{cases} \boldsymbol{F_s(i\alpha) = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\sin\alpha x\, dx} \\ \boldsymbol{f(x) = \sqrt{\frac{2}{\pi}}\int_0^\infty F_c(i\alpha)\sin\alpha x\, d\alpha} \end{cases}$$

and we call $F_s(i\alpha)$ and $f(x)$***Fourier sine transforms*** of each other.

- When the product of Fourier transforms is considered, a new concept called ***convolution*** comes into being, and in conjunction with it, a new pair (function and its Fourier transform) arises. In particular, if $F(i\alpha)$ and $G(i\alpha)$ are the Fourier transforms of f and g, respectively, and the convolution of f and g is defined to be

$$\boldsymbol{f * g = \frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty} f(u)\, g(x-u)du} \qquad \boldsymbol{(11)}$$

Then

$$\begin{cases} \boldsymbol{F(i\alpha)G(i\alpha) = \frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty} e^{-i\alpha u}\, f * g du} & \boldsymbol{(12)} \\ \boldsymbol{f * g = \frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty} e^{i\alpha x}\, F(\alpha)G(\alpha)d\alpha} & \boldsymbol{(13)} \end{cases}$$

where in both (11) and (13) the convolution $\boldsymbol{f * g}$ is a function of $\boldsymbol{x}$.
Now equate the representations of $\boldsymbol{f * g}$ expressed in (11) and (13), i.e.,

$$f * g = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(u)\, g(x-u) du = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{i\alpha x}\, F(\alpha) G(\alpha) d\alpha$$

and let the parameter x be zero, then

$$\int_{-\infty}^{\infty} \boldsymbol{f(u)\, g(-u) du} = \int_{-\infty}^{\infty} \boldsymbol{e^{i\alpha x}\, F(\alpha) G(\alpha) d\alpha} \qquad (\mathbf{14})$$

1.5.PROPERTIES OF TRANSFORMS OF SIMPLE FUNCTIONS.

1.5.1. The linearity property

Linearity property is a fundamental property of the Fourier transform, and may be stated as follows:

'*If* $f(x)$ *and* $g(x)$ *are functions having Fourier transforms* $F(i\alpha)$*and* $G(i\alpha)$ *respectively, and if* β *and* γ *are constants, then* s

$\boldsymbol{\mathcal{F}\{\beta f(x) + \gamma\, g(x)\} = \beta \mathcal{F}\{f(x)\} + \gamma \mathcal{F}\{g(x)\} = \beta F(i\alpha) + \gamma G(i\alpha)}$'

As a consequence of this, we say that the Fourier transform operator $\boldsymbol{\mathcal{F}}$ is a ***linear operator.***

Proof:

By definition we have:

$$\begin{aligned} \mathcal{F}\{\beta f(x) + \gamma\, g(x)\} &= \int_{-\infty}^{\infty} [\beta f(x) + \gamma\, g(x] e^{-i\alpha x} dx \\ &= \beta \textstyle\int_{-\infty}^{\infty} f(x) e^{-i\alpha x} dx + \gamma \int_{-\infty}^{\infty} g(x) e^{-i\alpha x}\, dx \\ &= \beta F(i\alpha) + \gamma G(i\alpha) \end{aligned}$$

Clearly the linearity property also applies to the inverse transform operator $\mathcal{F}^{-1}$

1.5.2. Differentiation property

If the function $f(x)$ has a Fourier transform $F(i\alpha)$, then by (10)

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(i\alpha) e^{i\alpha x}\, d\alpha$$

Differentiating with respect to x gives

$$\frac{df}{dx} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\partial}{\partial x} \left[F(i\alpha) e^{i\alpha x}\right] d\alpha = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (i\alpha) F(i\alpha) e^{i\alpha x}\, d\alpha$$

Imply that $\frac{df}{dx}$ is the inverse Fourier transform of $(i\alpha)F(i\alpha)$

In other words

$$\mathcal{F}\left\{\frac{df}{dx}\right\} = (i\alpha)F(i\alpha)$$

Repeating the argument n times, it follows that

$$\mathcal{F}\left\{\frac{d^n f}{dx^n}\right\} = (i\alpha)^n F(i\alpha) \qquad (15)$$

If $x = t \equiv time$ the above result (15) is referred to as the ***time-differentiation property***, and may be used to obtain ***frequency-domain representation of differential equations***.

Example 11: Show that if the time signals $y(t)$ and $u(t)$ have the Fourier transforms $Y(i\alpha)$ and $U(i\alpha)$ respectively, and if

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 7y(t) = 3\frac{du(t)}{dt} + 7u(t)$$

Then $Y(i\alpha) = G(i\alpha)U(i\alpha)$ for some function $G(i\alpha)$?

Answer: Taking Fourier transforms of above differential equation, then we have:

$$\mathcal{F}\left\{\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 7y(t)\right\} = \mathcal{F}\left\{3\frac{du(t)}{dt} + 7u(t)\right\}$$

Which, on using the linearity property, reduces to

$$\mathcal{F}\left\{\frac{d^2y(t)}{dt^2}\right\} + 3\mathcal{F}\left\{\frac{dy(t)}{dt}\right\} + 7\mathcal{F}\{y(t)\} = 3\mathcal{F}\left\{\frac{du(t)}{dt}\right\} + 7\mathcal{F}\{u(t)\}$$

Then, from (15) we get:

$$(i\alpha)^2Y(i\alpha) + 3(i\alpha)Y(i\alpha) + 7Y(i\alpha) = 3(i\alpha)U(i\alpha) + 2U(i\alpha)$$

That is,

$$(7 - \alpha^2 + 3i\alpha)Y(i\alpha) = (2 + 3i\alpha)U(i\alpha)$$

Giving $Y(i\alpha) = G(i\alpha)U(i\alpha)$

Where

$$G(i\alpha) = \frac{2 + 3i\alpha}{7 - \alpha^2 + 3i\alpha}$$

1.5.3. Time-shift property

If a function $f(t)$ has Fourier transform $F(i\alpha)$, then what is the Fourier transform of the shifted version of $f(\tau)$, defined by $g(t) = f(t - \tau)$?

Answer:

$\mathcal{F}\{g(t)\} = \int_{-\infty}^{\infty} g(t)e^{-i\alpha t}\,dt = \int_{-\infty}^{\infty} f(t - \tau)e^{-i\alpha t}\,dt$

Making the substitution $x = t - \tau$, we have

$$\mathcal{F}\{g(t)\} = \int_{-\infty}^{\infty} f(x)e^{-i\alpha(x+\tau)}\,dx = e^{-i\alpha\tau}\int_{-\infty}^{\infty} f(x)e^{-i\alpha x}\,dx = e^{-i\alpha\tau}F(i\alpha)$$

That is

$$\boldsymbol{\mathcal{F}\{f(t-\tau)\} = e^{-i\alpha\tau}F(i\alpha)} \qquad (\mathbf{16})$$

The result (16) is known as the ***time-shift property***, and implies that delaying a signal by a time $\boldsymbol{\tau}$ causes its Fourier transform to be multiplied by $e^{-i\alpha\tau}$.

Since

$$\left|e^{-i\alpha\tau}\right| = |\cos\alpha\tau - i\sin\alpha\tau| = 1$$

we have

$$\left|\boldsymbol{e^{-i\alpha\tau}F(i\alpha)}\right| = |\boldsymbol{F(i\alpha)}|$$

Indicating that the amplitude spectrum of $f(t-\tau)$ is identical with that of $f(t)$.

1.5.4. Frequency-shift property

Suppose that a function $f(t)$ has Fourier transform $F(i\alpha)$. Then, from the definition of Fourier transform we calculate the Fourier transform of $g(t) = e^{i\alpha_0 t}f(t)$ as

$$\mathcal{F}\{g(t)\} = \int_{-\infty}^{\infty} e^{i\alpha_0 t}f(t)e^{-i\alpha t}\,dt = \int_{-\infty}^{\infty} f(t)e^{-i(\alpha-\alpha_0)t}\,dt$$

$$= \int_{-\infty}^{\infty} f(t)e^{-i\tilde{\alpha}t}\,dt, \quad where\ \tilde{\alpha} = \alpha - \alpha_0$$

$$= F(i\tilde{\alpha})$$

Thus,

$$\boldsymbol{\mathcal{F}\{e^{i\alpha_0 t}f(t)\} = F[i(\alpha-\alpha_0)]} \qquad (\mathbf{17})$$

Example 12: Determine the frequency spectrum of the signal $g(t) = f(t)\cos\alpha_c t$

Answer*: As* $\cos\alpha_c t = \frac{1}{2}\left[e^{i\alpha_c t} + e^{-i\alpha_c t}\right]$

$$\mathcal{F}\{g(t)\} = \mathcal{F}\left\{\frac{1}{2}f(t)\left[e^{i\alpha_c t} + e^{-i\alpha_c t}\right]\right\} = \frac{1}{2}\left[\mathcal{F}\left\{e^{i\alpha_c t}f(t)\right\} + \mathcal{F}\left\{e^{-i\alpha_c t}f(t)\right\}\right]$$

Use property (17), then we get:

$$\frac{1}{2}\boldsymbol{F[i(\alpha-\alpha_c)]} + \frac{1}{2}\boldsymbol{F[i(\alpha+\alpha_c)]}$$

The effect of multiplying the signal $f(t)$ by the ***carrier signal*** $\cos\alpha_c t$ is thus to produce a signal whose spectrum consists of two (scaled) version of $F(i\alpha)$, the spectrum of $f(t)$;

one centred on $\boldsymbol{\alpha = \alpha_c}$ and the other on $\boldsymbol{\alpha = -\alpha_c}$**.** The ***carrier signal*** $\cos \alpha_c t$ is said to be modulated by the signal $f(t)$.

1.5.5. The symmetry property

We can establish the exact form of symmetric as:

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(i\alpha) e^{i\alpha t}\, d\alpha$$

Or, equivalently, by changing the dummy variable in the integration

$$\sqrt{2\pi} f(t) = \int_{-\infty}^{\infty} F(iy) e^{iyt}\, d\alpha$$

$$\sqrt{2\pi} f(-t) = \int_{-\infty}^{\infty} F(iy) e^{-iyt}\, d\alpha$$

Or on replacing t by w,

$$\sqrt{2\pi} f(-w) = \int_{-\infty}^{\infty} F(iy) e^{-iyw}\, d\alpha$$

$$\boldsymbol{\mathcal{F}\{F(iy)\} = \sqrt{2\pi} f(-w)}$$

Given that

$$\boldsymbol{\mathcal{F}\{f(t)\} = F(i\alpha)}$$

EXERCISES ON 1ST CHAPTER

1.1. Sketch the graphs of the following, inserting relevant values.

a) $f(x) = \begin{cases} 4, & 0 < x < 5 \\ 0, & 5 < x < 8 \\ f(x) = f(x+8) \end{cases}$ b) $f(x) = \begin{cases} 3, & 0 < x < 4 \\ 5, & 4 < x < 7 \\ f(x) = f(x+10) \end{cases}$ e

c) $f(x) = \begin{cases} 3x - x^2, & 0 < x < 3 \\ f(x) = f(x+3) \end{cases}$ d) $f(x) = \begin{cases} 2\sin x, & 0 < x < \pi \\ 0, & \pi < x < 2\pi \\ f(x) = f(x+2\pi) \end{cases}$

e) $f(x) = \begin{cases} \frac{x}{2}, & 0 < x < \pi \\ \pi - \frac{x}{2}, & \pi < x < 2\pi \\ f(x) = f(x+2\pi) \end{cases}$ f) $f(x) = \begin{cases} \frac{x^2}{4}, & 0 < x < 4 \\ 4, & 4 < x < 6 \\ 0, & 6 < x < 8 \\ f(x) = f(x+8) \end{cases}$

1.2.State whether each of the following products is odd, even, or neither

a) $x^2 \sin 2x$ b) $x^3 \cos x$ c) $\cos 2x \cos 3x$ d)$(2x+3)\ \sin 4x$ e) $x^3 e^x$ f) $\frac{1}{x+2} \cosh x$

1.3.If $f(x)$ is defined in the interval $-\pi < x < \pi$ and $f(x) = f(x + 2\pi)$, state whether or not each of the following functions can be represented by a Fourier series.

a)$f(x) = x^4$ b) $f(x) = 3 - 2x$ c) $f(x) = \frac{1}{x}$ d)$f(x) = e^{2x}$ e) $f(x) = \csc x$ f) $f(x) = \pm\sqrt{4x}$

1.4. Prove

a) $\int_{-L}^{L} \cos\frac{m\pi x}{L}\cos\frac{n\pi x}{L}dx = \int_{-L}^{L} \sin\frac{m\pi x}{L}\sin\frac{m\pi x}{L}dx = \begin{cases} 0, & m \neq n \\ L, & m = n \end{cases}$

b) $\int_{-L}^{L} \sin\frac{m\pi x}{L}\cos\frac{n\pi x}{L}dx = 0$, where m and n can be assume any of the values $1, 2, 3, \cdots$

2.1. Find the Fourier series of the following functions:

a) $f(x) = \begin{cases} 0, & -1 < x < 0 \\ 1, & 0 < x < 1 \end{cases}$ b) $f(x) = |x|,\ x \in [-1, 1]$ c) $f(x) = x,\ x \in [-1, 1]$

d) $f(x) = \begin{cases} -1, & -2 < x < -1 \\ 0, & -1 \leq x \leq 1 \\ 1, & 1 < x < 2 \end{cases}$ e) $f(x) = \begin{cases} a, & 0 < x < \frac{\pi}{3} \\ 0, & \frac{\pi}{3} < x < \frac{2\pi}{3} \\ -a, & \frac{2\pi}{3} < x < \pi \\ f(x) = f(x + \pi) \end{cases}$

2.2. a) Find the Fourier sine series of the following function: $f(x) = \begin{cases} -1, & -2 < x < 0 \\ 1, & 0 \leq x < 2 \end{cases}$

b) Show that the half-range Fourier sine series expansion of the function $f(x) = 1,\ 0 < x < \pi$

is $\frac{4}{\pi}\sum_{n=1}^{\infty} \frac{\sin(2n-1)x}{2n-1}$, $0 < x < \pi$.

2.3. a) Find the Fourier cosine series of the following functions: $f(x) = \begin{cases} 0, & -2 < x < -1 \\ 1, & -1 \leq x < 1 \\ 0, & 1 < x < 2 \end{cases}$

b) Determine the half-range Fourier cosine series expansion of the function

$$f(x) = 2x - 1, \qquad 0 < x < \pi$$

c) Determine the Fourier cosine series to represent the function $f(x)$ where

$$f(x) = \begin{cases} \cos x, & 0 < x < \dfrac{\pi}{2} \\ 0, & \dfrac{\pi}{2} < x < \pi \\ f(x) = f(x + 2\pi) \end{cases}$$

2.4. a) If $f(x)$ is defined by $f(x) = x(\pi - x)$, $0 < x < \pi$, express the function as

i) a half-range cosine series ii) a half-range sine series

b) If $f(x)$ is defined by $f(x) = 1 - x^2$, $0 < x < 1$, express the function as

i) a half-range cosine series ii) a half-range sine series

3.1. Calculate the Fourier transform of the two-sided exponential pulse given by

$$f(t) = \begin{cases} e^{at}, & t \leq 0 \\ e^{-at}, & t > 0 \end{cases}, \quad a > 0$$

3.2. Determine the Fourier transforms of a) $f(x) = \begin{cases} 2K, & |x| \leq 2 \\ 0, & |x| > 2 \end{cases}$, $K \text{ is a constant}$

b) $g(x) = \begin{cases} 2K, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$, $K \text{ is a constant}$

c) Sketch the function $h(x) = f(x) - g(x)$ and determine its Fourier transform

3.3. Calculate the Fourier transform of the 'off-on-off' pulse $f(t)$ defined by

$$f(t) = \begin{cases} 0, & t < -2 \\ -1, & -2 \leq t < -1 \\ 1, & -1 < t < 1 \\ -1, & 1 < t < 2 \\ 0, & t > 2 \end{cases}$$

3.4. Show that the Fourier transform of $f(x) = \begin{cases} \sin ax, & |x| \leq \frac{\pi}{a} \\ 0, & |x| > \frac{\pi}{a} \end{cases}$, $a \neq 0$ is

$\frac{i2a \sin(\pi\alpha/a)}{\alpha^2 - a^2}$

3.5. Show that the Fourier transform of $f(x) = \begin{cases} 0, & x < 0 \\ 1, & 0 \leq x \leq a \\ 0, & x > a \end{cases}$ are $\frac{1 - \cos xa}{x}$, $\frac{\sin xa}{x}$

3.6. Find the sine and cosine transforms of $f(x) = e^{-ax}H(x)$, $a > 0$

3.7. If $y(t)$ and $u(t)$ are signals with Fourier transforms $Y(i\alpha)$ and $U(i\alpha)$ respectively, and

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + y(t) = u(t)$$

Show that $Y(i\alpha) = H(i\alpha)U(i\alpha)$ for some function $H(i\alpha)$. What is $H(i\alpha)$?

3.8. Use the time- shift property to calculate the Fourier transform of the double pulse by

$$f(t) = \begin{cases} 1, & 1 \le |t| \le 2 \\ 0, & otherwise \end{cases}$$

Unit II PARTIAL DIFFERENTIAL EQUATIONS INCLUDING BOUNDARY VALUE PROBLEMS

2.0.INTRODUCTION

We will study functions $u = u(x_1, x_2, x_3, x_4, \cdots, x_n)$ and its partial derivatives. Here $(x_1, x_2, x_3, x_4, \cdots, x_n)$ are standard Cartesian coordinate on $\mathbb{R}^n$. We sometimes use the alternate notation $u(x, y), u(x, y, z)$ etc. We also $u(r, \theta, \Phi)$ for spherical coordinate on $\mathbb{R}^3$,etc. We sometimes also have a ***time*** coordinate t, in which case $t, x_1, x_2, x_3, x_4, \cdots, x_n$ denoted standard Cartesian coordinate on $\mathbb{R}^{1+n}$.

We use lots of different notation for partial derivatives:

$$\frac{\partial}{\partial x_i} u = \partial_i u; \qquad 1 \le i \le n \tag{0.1}$$

$$\frac{\partial^2}{\partial x_i \partial x_j} u = \frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} u = u_{x_i} u_{x_j}; \qquad 1 \le i, j \le n \tag{0.2}$$

If $i = j$, then we sometimes abbreviate $\partial_i \partial_j u \overset{\text{def}}{=} \partial_i^2 u$. If u is a function of (x, y), then we also write $u_x = \frac{\partial}{\partial x} u$, etc.

Definition 0.1: A partial differential equation (PDE) in a single unknown u is an equation involving u and its partial derivatives. All such equations can be written as

$$F\left(u, u_{x_1} \cdots, u_{x_n}, u_{x_1} u_{x_1}, \cdots, u_{x_{i_1} \cdots x_{i_N}}, x_1, x_2, x_3, x_4, \cdots, x_n\right) = 0, \;\; i_1, \cdots, i_N \in \{1, 2, \cdots, n\} \tag{0.3}$$

Here N is called the ***order of the PDE***. $N$is the maximum number of derivatives appearing in the equation.

Definition 0.2: The order of a PDE is the order of the highest derivatives appearing in the differentials.

Examples 0.1: *Consider $u = u(t, x)$ as a function of two variables*

1) $\partial_t^2 u + (1 + \cos u)\partial_x^3 u = 0$ *is a third-order PDE*
2) $\partial_t^2 u + 2\partial_x^2 u + u = 0$ *is a second-order PDE*

Definition 0.3: A PDE is termed ***linear PDE*** if and only if it is linear in the unknown function u and the partial derivatives of u. All other PDE are termed ***non-linear PDE.***

A linear PDE can be written as

$$\mathcal{L}u = f(x_1, x_2, x_3, x_4, \cdots, x_n)$$

For some linear operator $\mathcal{L}$ and some function f of the coordinates.

$\mathcal{L}$ is a linear operator iff $\mathcal{L}(au + bv) = a\mathcal{L}(u) + b\mathcal{L}(v)$ for $a, b \in \mathbb{R}$ and all function u, v.

Examples 0.2: *Consider* $= u(t,x)$, $u = u(x,y)$*and* $v = v(x,y)$ *as functions of two variables*

1) $\partial_t^2 u + (1 + \cos u)\partial_x^3 u = 0$ *is a third-order non-linear PDE*
2) $\partial_t^2 u + 2\partial_x^2 u + u = 0$ *is a second-order linear PDE*
3) ***Cauchy-Riemann equations***

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$
$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

Are first order linear PDE

Definition 0.4: If each term of a linear PDE contains the unknown function u or one of the partial derivatives of u, then a PDE is called ***Homogeneous PDE***, otherwise is a ***inhomogeneous PDE.***

Examples 0.3: *Consider* $= u(t,x)$, $u = u(x,y)$*and* $v = v(x,y)$ *as functions of two variables*

1) $\partial_t^2 u + (1 + \cos u)\partial_x^3 u = 0$ *is a third-order non-linear homogeneous PDE*
2) $\partial_t^2 u + 2\partial_x^2 u + u = x + 3t$ *is a second-order linear inhomogeneous PDE*

In general $\mathcal{L}u = f(x_1, x_2, x_3, x_4, \cdots, x_n)$ is an *homogeneous PDE iff* $f(x_1, x_2, x_3, x_4, \cdots, x_n) = 0$

Remarks:

- *We say that a given PDE is* ***constant coefficient linear PDE*** *iff* u *and its derivatives appear linearly(i.e. first power only) and are multiplied only by a constants;*

- *We say that a given PDE is **variable coefficient linear PDE** iff u and its derivatives appear linearly (i.e. first power only) and are multiplied only by a function of coordinates.*

Examples 0.4: *Consider $u = u(t,x)$ as a function of two variables*

1) $-\partial_t^2 u + 2\partial_x^2 u + u = 0$ *is **constant coefficient linear homogeneous PDE***
2) $\partial_t u + 2(1+x^2)\partial_x^3 u + u = x + 3t$ *is **variable coefficient linear inhomogeneous PDE***

Proposition 0.1: (Superposition principle). *If $u_1, u_2, u_3, \cdots, u_n$ are solution to the linear PDE*

$\mathcal{L}u = 0$, then its linear combination is also a solution

$$\sum_{i=0}^{n} c_i u_i \, , for \; c_1, c_2, \cdots, c_n \; \in \mathbb{R}$$

Is also a solution of linear homogeneous PDE.

Some important partial differential equations

The following are examples of important partial differential equations that commonly arise in problems of mathematical physics.

Benjamin-Bona-Mahony equation

$$u_t + u_x + u\,u_x - u_{xxt} = 0.$$

Biharmonic equation

$$\nabla^4 \phi = 0.$$

Boussinesq equation

$$u_{tt} - \alpha^2 u_{xx} = \beta^2 u_{xxtt}.$$

Cauchy-Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$
$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

Chaplygin's equation

$$u_{xx} + \frac{y^2}{1 - \frac{y^2}{c^2}} u_{yy} + y\,u_y = 0.$$

Euler-Darboux equation

$$u_{xy} + \frac{\alpha\,u_x - \beta\,u_y}{x - y} = 0.$$

Heat conduction equation

Helmholtz differential equation

$$\frac{\partial T}{\partial t} = \kappa \nabla^2 T.$$

$$\nabla^2 \psi + k^2 \psi = 0.$$

Klein-Gordon equation

$$\frac{1}{c^2}\frac{\partial^2 \psi}{\partial t^2} = \frac{\partial^2 \psi}{\partial x^2} - \mu^2 \psi.$$

Korteweg-de Vries-Burgers equation

$$u_t + 2u\,u_x - \nu\,u_{xx} + \mu\,u_{xxx} = 0.$$

Korteweg-de Vries equation

$$u_t + u_{xxx} - 6\,u\,u_x = 0.$$

Krichever-Novikov equation

$$\frac{u_t}{u_x} = \frac{1}{4}\frac{u_{xxx}}{u_x} - \frac{3}{8}\frac{u_{xx}^2}{u_x^2} + \frac{3}{2}\frac{p(u)}{u_x^2},$$

where $p(u) = \frac{1}{4}\left(4u^3 - g_2\,u - g_3\right).$

Laplace's equation

$$\nabla^2 \psi = 0.$$

Lin-Tsien equation

$$2\,u_{tx} + u_x\,u_{xx} - u_{yy} = 0.$$

Sine-Gordon equation

$$v_{tt} - v_{xx} + \sin v = 0.$$

Spherical harmonic differential equation

$$\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2} + l(l+1)\right]u = 0.$$

Tricomi equation

$$u_{yy} = y\,u_{xx}.$$

Wave equation

$$\nabla^2 \psi = \frac{1}{v^2}\frac{\partial^2 \psi}{\partial t^2}.$$

2.1.FORMATION AND SOLUTION OF STANDARD TYPES OF FIRST ORDER EQUATIONS

2.1.1. Formation of first order PDE

In the main we shall suppose that there are two independent variables x and y and that the dependent variable is denoted by z. If we write

$$p = \frac{\partial z}{\partial x}, \qquad q = \frac{\partial z}{\partial y}$$

, then the 1st order PDE is written as:

$$\boldsymbol{f(x,y,z,p,q)=0} \qquad (\mathbf{1.1})$$

Examples 1.1: If, for example, we take u to be the dependent variable and x, y and t to be independent variables, then the following equations:

1) $\left(\frac{\partial u}{\partial x}\right)^2+\frac{\partial u}{\partial t}=0$ is a first-order in two variables,
2) $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+\frac{\partial u}{\partial t}=0$ is a first-order in three variables.

2.1.2. Origins of first order PDE

Before discussing the solution of equation of the type (1.1), we shall examine the interesting question of how they arise.

Suppose that we consider the equation

$$x^2+y^2+(z-c)^2=a^2 \qquad (*)$$

In which the constants a and c are arbitrary. Then equation (*) represents the set of all spheres whose centers lie along the z axis. If we differentiate this equation with respect to x and with respect to y respectively, then we get:

$$x+p(z-c)=0 \quad and\ y+q(z-c)=0 \qquad (**)$$

By eliminating of arbitrary constant c from two equations of (**), then we obtain the PDE as:

$$\boldsymbol{yp-xq=0} \qquad (***)$$

The equation (***) is called the ***first-order PDE.*** In some sense, then, the set of all spheres with centers on the z axis is characterized by the PDE (***).

Problem 2.1

Eliminate the constants a and b from the following equations:

a) $z=(x+a)(y+b)$
b) $2z=(ax+y)^2+b$
c) $ax^2+by^2+z^2=1$

2.2. LAGRANGE'S LINEAR PDE

2.2.1. Formulation of Lagrange's linear PDE

By eliminating of arbitrary functions

Let u and v be any two given functions of x, y and z. Let u and v be connected by an arbitrary function φ by the relation

$$\boldsymbol{\varphi(u,v)=0} \qquad (****)$$

Now, we want to eliminate φ.

Differentiating partially, with respect $x$and y, we obtain

$$\begin{cases}\dfrac{\partial\varphi}{\partial x}=\dfrac{\partial\varphi}{\partial u}\dfrac{\partial u}{\partial x}+\dfrac{\partial\varphi}{\partial u}\dfrac{\partial u}{\partial z}\dfrac{\partial z}{\partial x}+\dfrac{\partial\varphi}{\partial v}\dfrac{\partial v}{\partial x}+\dfrac{\partial\varphi}{\partial v}\dfrac{\partial v}{\partial z}\dfrac{\partial z}{\partial x}=0\\ \dfrac{\partial\varphi}{\partial y}=\dfrac{\partial\varphi}{\partial u}\dfrac{\partial u}{\partial y}+\dfrac{\partial\varphi}{\partial v}\dfrac{\partial u}{\partial z}\dfrac{\partial z}{\partial y}+\dfrac{\partial\varphi}{\partial v}\dfrac{\partial v}{\partial y}+\dfrac{\partial\varphi}{\partial v}\dfrac{\partial v}{\partial z}\dfrac{\partial z}{\partial y}=0\end{cases}$$

$$\begin{cases}\dfrac{\partial\varphi}{\partial x}=\dfrac{\partial\varphi}{\partial u}\left(\dfrac{\partial u}{\partial x}+\dfrac{\partial u}{\partial z}p\right)+\dfrac{\partial\varphi}{\partial v}\left(\dfrac{\partial v}{\partial x}+\dfrac{\partial v}{\partial z}p\right)=0\\ \dfrac{\partial\varphi}{\partial y}=\dfrac{\partial\varphi}{\partial u}\left(\dfrac{\partial u}{\partial y}+\dfrac{\partial v}{\partial z}q\right)+\dfrac{\partial\varphi}{\partial v}\left(\dfrac{\partial v}{\partial y}+\dfrac{\partial v}{\partial z}q\right)=0\end{cases}$$

Eliminating $\frac{\partial\varphi}{\partial u}$ and $\frac{\partial\varphi}{\partial v}$, then we obtain

$$\begin{vmatrix}\dfrac{\partial u}{\partial x}+p\dfrac{\partial u}{\partial z} & \dfrac{\partial v}{\partial x}+p\dfrac{\partial v}{\partial z}\\ \dfrac{\partial u}{\partial y}+q\dfrac{\partial v}{\partial z} & \dfrac{\partial v}{\partial y}+q\dfrac{\partial v}{\partial z}\end{vmatrix}=0$$

Which simplifies to

$$\boldsymbol{Pp+Qq=R} \qquad (\mathbf{1.2})$$

Where $\begin{cases}P=\dfrac{\partial u}{\partial y}\dfrac{\partial v}{\partial z}-\dfrac{\partial u}{\partial z}\dfrac{\partial v}{\partial y}\equiv\dfrac{\partial(u,v)}{\partial(y,z)}\\ Q=\dfrac{\partial u}{\partial z}\dfrac{\partial v}{\partial x}-\dfrac{\partial u}{\partial x}\dfrac{\partial v}{\partial z}\equiv\dfrac{\partial(u,v)}{\partial(z,x)}\\ R=\dfrac{\partial u}{\partial x}\dfrac{\partial v}{\partial y}-\dfrac{\partial u}{\partial y}\dfrac{\partial v}{\partial x}\equiv\dfrac{\partial(u,v)}{\partial(x,y)}\end{cases}$

The equation (1.2) is called **Lagrange's linear PDE**

The relation $\varphi(u,v)=0$ is a solution of (1.2), whatever may the arbitrary function φ be.

Examples 1.2: *Form the PDE by eliminating the arbitrary function from*

i) $z=f(x^2+y^2)$

ii) $\varphi(x^2+y^2+y^2, lx+my+nz)=0$

Answer:

i) Differentiating partially, with respect $x$and y, we obtain

$$\begin{cases}\frac{\partial z}{\partial x} = p = 2xf'(x^2+y^2) \\ \frac{\partial z}{\partial y} = q = 2yf'(x^2+y^2)\end{cases}$$

Dividing, $\frac{p}{q} = \frac{x}{y} \rightarrow py - qx = 0$

$$\boldsymbol{y\frac{\partial z}{\partial x} - x\frac{\partial z}{\partial y} = 0}$$

ii) Now the given relation is of the form $\varphi(u,v) = 0$ where

$$\begin{cases}u = x^2 + y^2 + y^2 \\ v = lx + my + nz\end{cases}$$

Hence, the PDE is

$$Pp + Qq = R$$

Where $\begin{cases}P = \frac{\partial u}{\partial y}\frac{\partial v}{\partial z} - \frac{\partial u}{\partial z}\frac{\partial v}{\partial y} = 2ny - 2mz \\ Q = \frac{\partial u}{\partial z}\frac{\partial v}{\partial x} - \frac{\partial u}{\partial x}\frac{\partial v}{\partial z} = 2lz - 2nx \\ R = \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x} = 2mx - 2ly\end{cases}$

Therefore the required PDE is

$$(2ny - 2mz)p + (2lz - 2nx)q = (2mx - 2ly)$$

$$\boldsymbol{(ny - mz)\frac{\partial z}{\partial x} + (lz - nx)\frac{\partial z}{\partial y} = (mx - ly)}$$

Problem 2.2

1. Form the PDE by eliminating the arbitrary functions from:

a) $z = f(x^2 + y^2)$ b) $z = f(x + ct) + \Phi(x - ct)$ c) $z = f(ax + by) + g(\alpha x + \beta y)$

d) $z = xy + f(x^2 + y^2 + z^2)$ e) $z = f(x^2 + y^2 + z^2, x + y + z)$

f) $z = f(2x + y) + g(3x - y)$

2.2.2. Solution of Lagrange's linear PDE

Theorem 1.1: *The general solution of the linear PDE* $\boldsymbol{Pp + Qq = R}$ *is* $\varphi(u,v) = 0$ *Where* φ *is an arbitrary function and* $u(x,y,z) = c_1$ *and* $v(x,y,z) = c_2$*form a solution of the equations*

$$\boldsymbol{\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}} \qquad (\mathbf{1.3})$$

Procedure: *To solve the equation* $\boldsymbol{Pp + Qq = R}$ *we follow the following steps:*

STEP1: Form the auxiliary simultaneous equations

$$\frac{\boldsymbol{dx}}{\boldsymbol{P}} = \frac{\boldsymbol{dy}}{\boldsymbol{Q}} = \frac{\boldsymbol{dz}}{\boldsymbol{R}}$$

STEP2: Solve these auxiliary simultaneous equations giving two independent solutions $u(x, y, z) = c_1$ *and* $v(x, y, z) = c_2$*;*

STEP3: Then write down the solution as $\varphi(u, v) = 0$ *or* $u = f(v)$ *or* $v = F(u)$*, where the function is arbitrary.*

Examples 1.3: *Find the general integral of* $px + qy = z$

Answer: In term of comparison of $\boldsymbol{p + Qq = R}$ *, we get:*

$$P = x, Q = y, and\ R = z$$

STEP1: Form the auxiliary simultaneous equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \leftrightarrow \frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$$

STEP2: Solve these auxiliary simultaneous equations

$$\begin{cases} \dfrac{dx}{x} = \dfrac{dy}{y} \\ \dfrac{dy}{y} = \dfrac{dz}{z} \end{cases} \rightarrow \begin{cases} \ln x = \ln c_1 y \\ \ln y = \ln c_2 z \end{cases} \rightarrow \begin{cases} \dfrac{x}{y} = c_1 \\ \dfrac{y}{z} = c_2 \end{cases}$$

two independent solutions $u(x, y) = \frac{x}{y} = c_1$ *and* $v(y, z) = \frac{y}{z} = c_2$*;*

STEP3: Then write down the solution as $\varphi(u, v) = \varphi\left(\frac{x}{y}, \frac{y}{z}\right) = 0$

Solution of the subsidiary equation by the method of multipliers

The subsidiary equations $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$ can be solved as follows:

By algebra, we have, $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{ldx+mdy+ndz}{lP+mQ+nR} = \frac{l'dx+m'dy+n'dz}{l'P+m'Q+n'R}$

Where the set of multipliers $l, m, n; l', m', n'$ *by be constants or variables in* x, y, z*. Choosing* l, m, n *such that* $lP + mQ + nR = 0$*, we have*

$$\boldsymbol{ldx + mdy + ndz = 0} \qquad (\mathbf{1.4})$$

If $ldx + mdy + ndz$ *is a perfect differential of some function, say* $u(x, y, z)$ *then* $du = 0$ *, by(1.4). By integrating (1.4), we get* $u(x, y, z) = c_1$ *as one solution.*

Similarly, the other set of multipliers l', m', n' *can be found out so that* $l'P + m'Q + n'R = 0$

Hence $l'dx + m'dy + n'dz = 0$

This yield another solution $v(x, y, z) = c_2$

Therefore the general solution is $\varphi(u, v) = 0, or\ u = f(v)$.

Here, the set of multipliers l, m, n *and* l', m', n' *are called* ***Lagrangian multipliers.***

Examples 1.4: *Find the general solution of* $x(z^2 - y^2)p + y(x^2 - z^2)q = z(y^2 - x^2)$

The subsidiary equations are

$$\frac{dx}{x(z^2 - y^2)} = \frac{dy}{y(x^2 - z^2)} = \frac{dz}{z(y^2 - x^2)} \quad (*)$$

As $lP + mQ + nR = 0 \leftrightarrow lx(z^2 - y^2) + my(x^2 - z^2) + nz(y^2 - x^2) = 0$

Taking the two sets of multipliers multipliers as x, y, z *and* $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$*,each of ratio in* $(*)$*, we get*

$$\frac{xdx + ydy + zdz}{x^2(z^2 - y^2) + y^2(z^2 - y^2) + z^2(z^2 - y^2)} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{(z^2 - y^2) + (z^2 - y^2) + (z^2 - y^2)}$$

$$\frac{xdx + ydy + zdz}{0} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0}$$

Hence $xdx + ydy + zdz$ *and* $\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz$

By integrating we get, $\int xdx + \int ydy + \int z\,dz = x^2 + y^2 + z^2 = c_1$*and* $\int \frac{1}{x}dx + \int \frac{1}{y}dy + \int \frac{1}{z}dz = \ln x + \ln y + \ln z \equiv \ln xyz = c_2 \leftrightarrow xyz = e^{c_2} = c_3$

The general solution is $\varphi(x^2 + y^2 + z^2, xyz) = 0$

Examples 1.5: *Find the general solution of* $\frac{y^2z}{x}p + xzq = y^2$

The subsidiary equations are

$$\frac{xdx}{y^2z} = \frac{dy}{xz} = \frac{dz}{y^2} \quad (*)$$

$$\frac{xdx}{y^2z} = \frac{dy}{xz} \leftrightarrow x^2dx = y^2dy \rightarrow \int x^2\,dx = \int y^2\,dy \leftrightarrow x^3 - y^3 = c_1$$

$$\frac{xdx}{y^2z} = \frac{dz}{y^2} \leftrightarrow \frac{xdx}{z} = dz \leftrightarrow xdx = zdz \rightarrow \int x\,dx = \int z\,dz \leftrightarrow x^2 - y^2 = c_2$$

Hence the general solution is $\varphi(x^3 - y^3, x^2 - y^2) = 0$

2.3. METHODS OF SOLVING 1st ORDER PDES

The PDE of the 1st order can be written as $F(x, y, z, p, q) = 0$, where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$.

We shall see some standard forms of such equations and solve them by special methods.

2.3.1. Type 1. $F(p, q) = 0$

If the PDE contains p and q only, then suppose that $z = ax + by + c$ is a solution of the equation $F(p, q) = 0$

Then $p = \frac{\partial z}{\partial x} = a$ and $q = \frac{\partial z}{\partial y} = b$

After substituting these in a given PDE, then we obtain $F(a, b) = 0$

Hence the complete solution of a given PDE is $z = ax + by + c$, where $F(a, b) = 0$

Solving for b from $F(a, b) = 0$, we get $b = \varphi(a)$

Then $\boldsymbol{z = ax + \varphi(a)y + c}$ is a complete integral of a given PDE, since it contains two arbitrary constants.

Singular integral is got by eliminating a and c from $\boldsymbol{z = ax + \varphi(a)y + c}$ as

$$\frac{\partial z}{\partial a} = 0 = x$$

$$\frac{\partial z}{\partial c} = 0 = 1$$

This last equation being absurd, there is no singular integral for a given PDE.

For finding the general solution, put $c = f(a)$, f being arbitrary

Then $\begin{cases} \boldsymbol{z = ax + \varphi(a)y + f(a)} \\ \boldsymbol{and} \\ \boldsymbol{\frac{\partial z}{\partial a} = 0 = a + y\varphi'(a) + f'(a)} \end{cases}$

For eliminating a from this above system, then we obtain the **general solution.**

Examples 1.6: *Solve this PDE* $p^2 + q^2 = npq$

Answer: The solution of this PDE is $z = ax + by + c$ *subject to* $a^2 + b^2 = nab$

Solving for b*, we get* $b = \frac{na \pm \sqrt{n^2a^2 - 4a^2}}{2} = \frac{a}{2}\left[n \pm \sqrt{n^2 - 4}\right]$

Hence the complete solution is $z = ax + \frac{a}{2}\left[n \pm \sqrt{n^2 - 4}\right]y + c$

As $\frac{\partial z}{\partial c} = 0 = 1$ *which is absurd, then there is no singular solution*

For finding the general solution, put $c = f(a)$, f *being arbitrary*

$$\textit{Then} \begin{cases} z = ax + \frac{a}{2}\left[n \pm \sqrt{n^2 - 4}\right]y + f(a) \\ \textit{and} \\ \frac{\partial z}{\partial a} = 0 = a + \frac{1}{2}\left[n \pm \sqrt{n^2 - 4}\right]y) + f'(a) \end{cases}$$

$$\begin{cases} z = ax + \dfrac{a}{2}\left[n \pm \sqrt{n^2 - 4}\right]y + f(a) \\ \textit{and} \\ \dfrac{\partial z}{\partial a} = 0 = x + \dfrac{1}{2}\left[n \pm \sqrt{n^2 - 4}\right]y) + f'(a) \end{cases}$$

Eliminating a between above system, then we obtain the general solution of a given PDE.

2.3.2. Type 2. Clairaut'form $z = px + qy + f(p, q)$

Suppose that the given PDE is of the form $z = px + qy + f(p, q)$ (1)

Its complete solution is $z = ax + by + f(a, b)$ (2)

Where a and b are arbitrary constants.

Differential (2) partially with respect to a and b, we get

$$\begin{cases} \dfrac{\partial z}{\partial a} = x + \dfrac{\partial f}{\partial a} = 0 \\ \textit{and} \\ \dfrac{\partial z}{\partial b} = y + \dfrac{\partial f}{\partial b} = 0 \end{cases} \quad (3)$$

By eliminating a and b from (2), and (3), we get the singular solution of (1).

Taking $b = \varphi(a)$, (2) becomes

$z = ax + \varphi(a)y + f(a, \varphi(a))$ (4)

Differential (4) partially with respect to a, we get

$$\frac{\partial z}{\partial a} = 0 = x + \varphi'(a), y + f'\big(a, \varphi(a)\big) \quad (5)$$

Eliminating a between (4) and (5), then we obtain the general solution of a given PDE.

Examples 1.7: *Solve this PDE* $z = px + qy + p^2q^2$

Answer: This is Clairaut's form

The complete solution of this PDE is $z = ax + by + a^2b^2$

Differential $z = ax + by + a^2b^2$ partially with respect to a and b, we get

$$\begin{cases} \dfrac{\partial z}{\partial a} = 0 = x + 2ab^2 \\ \quad and \\ \dfrac{\partial z}{\partial b} = 0 = y + 2ab^2 \end{cases} \leftrightarrow \begin{cases} x = -2ab^2 \\ \quad and \\ y = -2a^2 b \end{cases}$$

$$\frac{x}{b} = \frac{y}{a} = -2ab = \frac{1}{k} \rightarrow a = ky \ \ and \ b = kx$$

$$x = -2ab^2 = -2k^3 yx^2 \rightarrow k^3 = -\frac{1}{2xy}$$

$$z = ax + by + a^2 b^2 \leftrightarrow z = kxy + kxy + k^4 x^2 y^2$$

$$z = 2kxy + kx^2 y^2 \left(-\frac{1}{2xy}\right) = 2kxy - \frac{k}{2}xy = \frac{3}{2}kxy$$

$$z^3 = \frac{27}{8}k^3 x^3 y^3 = \frac{27}{8}\left(-\frac{1}{2xy}\right)x^3 y^3 = -\frac{27}{16}x^2 y^2$$

$16z^3 + 27x^2 y^2 = 0$ *is a singular solution*

Taking $b = \varphi(a)$, (2) becomes

$z = ax + \varphi(a), y + a^2[\varphi(a)]^2,$ (*)

Differential (*) partially with respect to a, we get

$$\frac{\partial z}{\partial a} = 0 = x + \varphi'(a)\, y + 2a[\varphi(a)]^2 + 2a^2\varphi'(a) \qquad (**)$$

Eliminating a between (*) and (**), then we obtain the general solution of a given PDE

2.3.3. **Type 3.**

Case1. $\boldsymbol{F(z, p, q) = 0}$

This PDE form not containing x and $y$explicitily. As a trial solution, assume that z is a function of $u = x + ay$, where a is an arbitrary constant.

$z = f(u) = f(x + ay)$

$$p = \frac{\partial z}{\partial x} = \frac{dz}{du}.\frac{\partial u}{\partial x} = \frac{dz}{du}.1 = \frac{dz}{du}$$

$$q = \frac{\partial z}{\partial y} = \frac{dz}{du}.\frac{\partial u}{\partial y} = \frac{dz}{du}.a = a\frac{dz}{du}$$

Substituting these values of p and $q$in $F(z,p,q) = 0$**,** we obtain $F\left(z, \frac{dz}{du}, a\frac{dz}{du}\right) = 0$ which is an ordinary differential equation of the 1st order. Solving for $\frac{dz}{du}$, we obtain $\frac{dz}{du} = \varphi(z,a)$.

$$\frac{dz}{\varphi(z,a)} = du \leftrightarrow \int \frac{dz}{\varphi(z,a)} = u + c \rightarrow f(z,a) = u + c = x + ay + c$$

$$\boldsymbol{f(z,a) = x + ay + c}$$

This is the complete integral.

The singular and general integrals are found out as usual.

Case2. $\boldsymbol{F(x,p,q) = 0}$

Since z is a function of x and y, then

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy = pdx + qdy$$

Assume that $q = a$, then the equation becomes $\boldsymbol{F(x,p,a) = 0}$

Solving for p, we obtain $p = \Phi(x,a) \rightarrow$

$$dz = \Phi(x,a)dx + ady \rightarrow \int dz = \int \Phi(x,a)\,dx + \int a\,dy \leftrightarrow z = f(x,a) + ay + c$$

$$\boldsymbol{z = f(x,a) + ay + c}$$

Is a complete integral of a given PDE since it contains two arbitrary constants a and c.

Case3. $\boldsymbol{F(y,p,q) = 0}$

Since z is a function of x and y, then

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy = pdx + qdy$$

Assume that $p = a$, then the equation becomes $\boldsymbol{F(x, a, q) = 0}$

Solving for q, we obtain $q = \Phi(y, a) \rightarrow$

$$dz = pdx + \Phi(y,a)dy \rightarrow \int dz = \int a\,dx + \int \Phi(y,a)\,dy \leftrightarrow z = ax + f(y,a) + c$$

$$\boldsymbol{z = ax + f(y, a) + c}$$

Is a complete integral of a given PDE since it contains two arbitrary constants a and c.

Examples 1.8: *Solve the following PDEs*

a) $p(1 + q) = qz$

b) $q = px + p^2$

c) $pq = y$

Answers:

a) $p(1 + q) = qz$

Assume $u = x + ay, p = \frac{dz}{du} and\ q = a\frac{dz}{du}$

Substituting these values of p and $q$in a given PDE**,** we obtain $\frac{dz}{du}(1 + a\frac{dz}{du}) = a\frac{dz}{du}z$

$$\frac{dz}{du}\left(1 + a\frac{dz}{du}\right) = a\frac{dz}{du}z \leftrightarrow 1 + a\frac{dz}{du} = az \rightarrow adz = (az - 1)du$$

$$\rightarrow \int \frac{adz}{(az-1)} = \int du \leftrightarrow \ln(az - 1) = \frac{1}{a}u + c$$

$$\ln(az - 1) = \frac{1}{a}(x + ay) + c = \frac{1}{a}x + y + c$$

$\ln(az - 1) = \frac{1}{a}x + ay + c$.***This is the complete integral.***

The singular and general integrals are found out as usual.

b) $q = px + p^2$

Assume that $q = a = constant$, then the equation becomes $a = px + p^2$

$$\leftrightarrow p^2 + px + a = 0 \leftrightarrow p = \frac{-x \pm \sqrt{x^2 + 4a}}{2}$$

Since $dz = pdx + qdy = \left(\frac{-x \pm \sqrt{x^2+4a}}{2}\right) dx + ady = 0$

$$\int dz = \int \left(\frac{-x \pm \sqrt{x^2+4a}}{2}\right) dx + \int a\, dy \rightarrow z = -\frac{x^2}{4} \pm \frac{1}{2}\int \sqrt{x^2 + 4a}\, dx + ay + b$$

$$z = -\frac{x^2}{4} \pm \frac{1}{2}\left\{2a \sinh^{-1}\left(\frac{x}{2\sqrt{a}}\right) + \frac{x}{2}\sqrt{x^2 + 4a}\right\} + ay + b$$

This is the complete integral. The singular and general integrals are found out as usual.

c) $pq = y$

Assume that $p = a = constant$, then the equation becomes $aq = y$

$$\leftrightarrow q = \frac{y}{a}$$

Since $dz = pdx + qdy = adx + \frac{y}{a}dy$

$$\int dz = \int a\, dx + \int \frac{y}{a} dy \rightarrow z = ax + \frac{y^2}{a} + b$$

$$z = ax + \frac{y^2}{a} + b$$

This is the complete integral. The singular and general integrals are found out as usual.

2.3.4. **Type 4. Separable equations**

We say that a 1st order PDE is ***separable*** if it can be written as $\boldsymbol{f(x,p) = \varphi(y,q)}$

If these two functions are equally then we assume that are constants, means that $f(x,p) = \varphi(y,q) = a$

Solving for p and q, we get $p = f_1(x,a)$ and $q = f_2(y,a)$

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

Hence $dz = pdx + qdy = f_1(x,a)dx + f_2(y,a)dy$

$$\int dz = \int f_1(x,a)\, dx + \int f_2(y,a)\, dy$$

$$\boldsymbol{z = \int f_1(x,a)\, dx + \int f_2(y,a)\, dy + b}$$

This expression contains two arbitrary constants and hence it is the complete integral. The singular and general integrals are found out as usual.

Examples 1.9: *Solve the following PDEs*

$$p^2y(1+x^2) = qx^2$$

Answers:

This equation is separable PDE.

$$p^2\frac{(1+x^2)}{x^2} = \frac{q}{y} = a$$

$$p^2\frac{(1+x^2)}{x^2} = a \rightarrow p = \frac{x\sqrt{a}}{\sqrt{1+x^2}}$$

$$\frac{q}{y} = a \rightarrow q = ay$$

Hence $dz = pdx + qdy = \frac{x\sqrt{a}}{\sqrt{1+x^2}}dx + aydy$

$$\int dz = \sqrt{a}\int \frac{x}{\sqrt{1+x^2}}dx + a\int y\,dy$$

$$\boldsymbol{z = \sqrt{a(1+x^2)} + \frac{1}{2}ay^2 + b}$$

This is the complete integral.

Differentiating partially w.r.t b, we find that there is no singular integral.

2.3.5. Type 5. Equations reducible to standard forms

Many non-linear PDEs of 1st order do not fall under any of the four standard types discussed so far. However, in some cases, it is possible to transform the given PDE into one of the standard types by change the variables.

Case1. $\boldsymbol{F(x^m p,\ y^m q) = 0, \quad \forall m,n\ are\ constants}$

This type of PDE can be transform into an equation of the 1st type.

By putting $x^{1-m} = X$ and $y^{1-n} = Y$,where $m \neq 1$ and $n \neq 1$, we get

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial X}\frac{dX}{dx} = (1-m)x^{-m}\frac{\partial z}{\partial X} = (1-m)x^{-m}P$$

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial Y}\frac{dY}{dy} = (1-n)y^{-n}\frac{\partial z}{\partial Y} = (1-n)y^{-n}Q$$

Where $P = \frac{\partial z}{\partial X}$ and $Q = \frac{\partial z}{\partial Y}$

Hence the equation reduces to $\boldsymbol{F\big((1-m)P,\ (1-n)Q\big) = 0,}$ which is of the form $\boldsymbol{f(P,Q) = 0}$

Case2. $\boldsymbol{F(x^m p,\ y^m q, z) = 0, \quad \forall m,n\ are\ constants}$

This type of PDE can be transform into standard form.

By putting $x^{1-m} = X$ and $y^{1-n} = Y$, where $m \neq 1$ and $n \neq 1$, we get $\boldsymbol{f(P, Q, z) = 0}$

Case3. For 1st and 2nd cases $\forall \boldsymbol{m = n = 1}$

If $= 1$, put and $X = \ln x$ and $n = 1$, put and $Y = \ln y$, we get

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial X}\frac{dX}{dx} = \frac{1}{x}\frac{\partial z}{\partial X} = P\frac{1}{x} \rightarrow px = P$$

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial Y}\frac{dY}{dy} = \frac{1}{y}\frac{\partial z}{\partial Y} = Q\frac{1}{y} \rightarrow qy = Q$$

Case4. $\boldsymbol{F(z^k p,\ z^k q) = 0}$, $\forall \boldsymbol{k}$ *is constant*

This type of PDE can be transform into an equation of the 1st type by proper substitution

If $k \neq -1$,put $Z = z^{k+1}$, then $P = \frac{\partial Z}{\partial x} = (k+1)z^k\frac{\partial z}{\partial x} = (k+1)z^k p \rightarrow \frac{P}{k+1} = z^k p$ and $Q = \frac{\partial Z}{\partial y} = (k+1)z^k\frac{\partial z}{\partial y} = (k+1)z^k q \rightarrow \frac{Q}{k+1} = z^k q$

Hence the equation reduces to $\boldsymbol{F\left(\frac{1}{k+1}P,\ \frac{1}{k+1}Q\right) = 0}$**,** which is of the form $\boldsymbol{f(P, Q) = 0}$

If $k = -1$,put $Z = \ln z$, then $P = \frac{\partial Z}{\partial x} = \frac{1}{z}\frac{\partial z}{\partial x} = \frac{1}{z}p$ and $Q = \frac{\partial Z}{\partial y} = \frac{1}{z}\frac{\partial z}{\partial y} = \frac{1}{z}q$ Hence the equation reduces to $\boldsymbol{F(P,\ Q) = 0}$**,** which is of the form $\boldsymbol{f(P, Q) = 0}$

Case5. $\boldsymbol{F(x^m z^k p,\ y^n z^k q) = 0}$, $\forall \boldsymbol{k}$ *is constant*

It may be transformed into the standard type $\boldsymbol{f(P, Q) = 0}$ by putting $x^{1-m} = X$,$y^{1-n} = Y$ and $Z = z^{k+1}$, if $m \neq 1, n \neq 1$ and $k \neq 1$ or by putting $X = \ln x$, $Y = \ln y$, $Z = \ln z$ if $m = 1, n = 1$ and $k = -1$

Examples 1.10: *Solve the following PDEs*

a) $x^2p^2 + y^2q^2 = z^2$

b) $z^2(p^2 + q^2) = x^2 + y^2$

Answers:

a) $x^2p^2 + y^2q^2 = z^2$

This equation is not in any of the four standard types. But this is reducible to one of the standard types by proper substitution of the variables.

$$x^2p^2 + y^2q^2 = z^2 \leftrightarrow \left(\frac{xp}{z}\right)^2 + \left(\frac{yq}{z}\right)^2 = 1$$

This is of the form explained in case 5, where $m = 1, n = 1$ and $k = -1$ .Then put $= \ln x$, $Y = \ln y$, $Z = \ln z$

Then $P = \frac{\partial Z}{\partial X} = \frac{\partial Z}{\partial z} \cdot \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial X} = \frac{1}{z} \cdot p \cdot x$ and $Q = \frac{\partial Z}{\partial Y} = \frac{qy}{z}$

The equation reduces to $P^2 + Q^2 = 1$

The complete solution is $Z = aX + bY + c,$ where $a^2 + b^2 = 1 \rightarrow b = \sqrt{1 - a^2}$

This means that $\ln z = a \ln x + \sqrt{1 - a^2} \ln y + c \rightarrow z = e^{\{a \ln x + \sqrt{1-a^2} \ln y + c\}}$

b) $z^2(p^2 + q^2) = x^2 + y^2 \leftrightarrow (zp)^2 + (zq)^2 = x^2 + y^2$

Put $Z = z^2$, then $P = \frac{\partial Z}{\partial x} = \frac{\partial Z}{\partial z} \cdot \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial x} = 2z \cdot p$ and $Q = \frac{\partial Z}{\partial y} = \frac{\partial Z}{\partial z} \cdot \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial y} = 2z \cdot q$

The equation reduces to $P^2 + Q^2 = 4(x^2 + y^2)$

$$P^2 - 4x^2 = 4y^2 - Q^2 = 4a \rightarrow \begin{cases} P = \sqrt{4a + 4x^2} \\ Q = \sqrt{4y^2 - 4a} \end{cases}$$

$$dZ = Pdx + Qdy = 2\sqrt{a + x^2}dx + 2\sqrt{y^2 - a}dx$$

$$Z = \int dZ = \int Pdx + \int Qdy = \int 2\sqrt{a + x^2}dx + \int 2\sqrt{y^2 - a}dx$$

$$Z = 2\left[\frac{x}{2}\sqrt{a + x^2} + \frac{a}{2}\sinh^{-1}\frac{x}{\sqrt{a}} + \frac{y}{2}\sqrt{y^2 - a} - \frac{a}{2}\cosh^{-1}\frac{y}{\sqrt{a}}\right] + b$$

$$Z = x\sqrt{a + x^2} + a\sinh^{-1}\frac{x}{\sqrt{a}} + y\sqrt{y^2 - a} - a\cosh^{-1}\frac{y}{\sqrt{a}} + b$$

EXERCISES 2.1

1. Solve the following equations:

a) $p^2 + q^2 = npq$ b) $z = px + qy + \sqrt{1 + p^2 + q^2}$ c) $z = px + qy + p^2q^2$ d) $z = px + qy + p^2 - q^2$ e) $p = 2qx$ f) $9(p^2z + q^2) = 4$ g) $z = p^2 + q^2$ h) $p^2 + q^2 = x + y$ i) $p - x^2 = q + y^2$ j) $2x^4p^2 - yzq - 3z^2 = 0$

2. Find the general solution of the following PDEs

a) $px^2 + qy^2 = (x + y)z$ b) $p\sqrt{x} + q\sqrt{y} = \sqrt{z}$ c) $px^2 - qy^2 = z^2$ d) $px + qy = nz$

3. Form the PDE by eliminating the arbitrary constants: a) $z = ax^n + by^n$ b) $z = a(x + \ln y) - \frac{x^2}{2} - b$ c) $z = ax + by + \frac{a}{b} - b$

4. Obtain PDEs by eliminating the arbitrary functions:

a) $xyz = f(x + y + z)$ b) $z = f(2x + y) + g(3x - y)$ c) $z = e^y f(x + y)$

5.Show that the PDE $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = {2u}/{x}$ is satisfied by $u = \frac{1}{x} f(y - x) + f'(y - x)$ where f is an arbitrary function.

6. If $z = f(x + iy) + F(x - iy$, prove that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$, where $f\ and\ F$ are arbitrary function.

7. If $u = f(x^2 + y) + F(x^2 - y)$, show that $\frac{\partial^2 u}{\partial x^2} - \frac{1}{x} \cdot \frac{\partial u}{\partial x} - 4x^2 \frac{\partial^2 u}{\partial y^2} = 0$

2.4.LINEAR HOMOGENEOUS PARTIAL DIFFERENTIAL EQUATIONS OF SECOND ORDER WITH CONSTANT COEFFICIENTS.

Homogeneous Equations

Let $D_x = \frac{\partial}{\partial x}$, $D_y = \frac{\partial}{\partial y}$, $D_x^i = \frac{\partial^i}{\partial x^i}$, $D_y^i = \frac{\partial^i}{\partial y^i}$,

We are looking for solving equations of the type

$$\frac{\partial^2 u}{\partial x^2} + k_1 \frac{\partial^2 u}{\partial x \partial y} + k_2 \frac{\partial^2 u}{\partial y^2} = 0 \qquad (2.4.1)$$

where k_1 and k_2 are constants. Then (2.4.1) can be written as

$$\left(D_x^2 + k_1 D_x D_y + k_2 D_y^2\right) u = 0$$

or $F(D_x, D_y)\, u = 0$ (2.4.2)

The auxiliary equation of (24.1.) is

$$D_x^2 + k_1 D_x D_y + k_2 D_y^2 = 0$$

Let the roots of this equation be m_1 and m_2, that is, $D_x=m_1Dy$, $Dx=m_2D_y$

$$(D_x-m_1Dy)\,(D_x-m_2D_y)u=0- \qquad (2.4.3)$$

This implies $(D_x-m_2D_y)\,u=0$ or $p-m_2q=0$

The auxiliary system of equations for $p-m_2q=0$ is of the type $\frac{dx}{1}=\frac{dy}{-m_2}=\frac{du}{0}$

This gives us $-m_2dx=dy$ or $y+m_2x=c$ and $u=c_1=\varphi(c)$

Thus, $u=\varphi(y+m_2x)$ is a solution of (2.4.1). From (2.4.2) we also have $(D_x-m_1D_y)\,u=0$

or $p-m_1q=0$

Its auxiliary system of equations is $\frac{dx}{1}=\frac{dy}{-m_1}=\frac{du}{0}$

This gives $-m_1dx=dy$ or $m_1x+y=c_1$ and $u=c_2$ and so $u=\psi(y+m_1x)$ is a solution of (2.4.1).

Therefore $u=\varphi(y+m_2x)+\psi(y+m_1x)$ is the complete solution of (2.4.1).

If the roots are equal ($m_1 = m_2$) then Equation 2.4.1 is equivalent to

$$(D_x-m_1D_y)^2\,u = 0$$

Putting $(D_x-m_1D_y)\,u = z$, we get $(D_x-m_1D_y)\,z=0$ which gives $z=\varphi(y+m_1x)$

Substituting z in $(D_x-m_1D_y)\,u=z$ gives $(D_x-m_1D_y)\,u = \varphi(y+m_1x)$ or $p-m_1q = \varphi(y+m_1x)$

Its auxiliary system of equations is $\frac{dx}{1} = \frac{dy}{-m_1} = \frac{du}{\varphi(y+m_1x)}$ which gives $y+m_1x = a$

and $u + \varphi(a)\,x+b$

The complete solution in this case is $u= x\,\varphi(y+m_1x)+\psi(y+m_1x)$

Example 2.4.1. Find the solution of the equation

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$$

Solution: In the terminology introduced above this equation can be written as

$(D_x^2 - D_y^2)\, u = 0$. or $(D_x - D_y)(D_x + D_y)u = 0$

Its auxiliary equation is $(D_x - D_y)(D_x + D_y) = 0$, that is, $D_x - D_y = 0$ or $D_x = -D_y$. that is,

$p = q$ or $p = -q$, $p - q = 0$ or $p + q = 0$

Auxiliary system of equations for $p - q = 0$ is $\frac{dx}{1} = \frac{dy}{-1} = \frac{du}{0}$

This gives $x + y = c$. The auxiliary system for $p + q = 0$ is $\frac{dx}{1} = \frac{dy}{1} = \frac{du}{0}$

This gives $x - y = c_1$ .The complete solution is

$u = \varphi(x+y) + \psi(x-y)$ where φ and ψ are arbitrary functions.

Non-homogeneous Partial Differential Equations of the second-order

Equations of the type

$$\frac{\partial^2 u}{\partial x^2} + k_1 \frac{\partial^2 u}{\partial x \partial y} + k_2 \frac{\partial^2 u}{\partial y^2} = f(x,y) \qquad (2.4.4)$$

are called non-homogeneous partial differential equations of the second-order with constant coefficients. Let u_c be the general solution of

$$\frac{\partial^2 u}{\partial x^2} + k_1 \frac{\partial^2 u}{\partial x \partial y} + k_2 \frac{\partial^2 u}{\partial y^2} = 0 \qquad (2.4.5)$$

and let u_p be a particular solution of (2.4.4)

Then $u_c + u_p$ is the solution of (2.4.4).

We have discussed the method for finding the general solution (complementary function) of (2.4.4).The method of undetermined coefficients for ordinary differential equations is applicable in finding particular solution of partial differential equations of the type (2.4.3).

Let $f(D_x,D_y)$ be ***a linear partial differential operator with constant coefficients, then the corresponding inverse operator is defined*** as $\frac{1}{f(D_x,D_y)}$

The following results hold $f(D_x,D_y)\left[\frac{1}{f(D_x,D_y)}\varphi(x,y)\right]=\varphi(x,y)$ (2.4.6)

$$\frac{1}{f_1(D_x,D_y)f_2(D_x,D_y)}\varphi(x,y)=\frac{1}{f_1(D_x,D_y)}\left[\frac{1}{f_2 D_x,D_y)}\varphi(x,y)\right] \quad (2.4.7)$$

$$=\frac{1}{f_2(D_x,D_y)}\left[\frac{1}{f_1(D_x,D_y)}\varphi(x,y)\right] \quad (2.4.8)$$

$$\frac{1}{f(D_x,D_y)}\left[\alpha\varphi_1(x,y)+\beta\varphi_2(x,y)\right]=\alpha\frac{1}{f(D_x,D_y)}\varphi_1(x,y)+\beta\frac{1}{f(D_x,D_y)}\varphi_2(x,y) \quad (2.4.9)$$

$$\frac{1}{f(D_x,D_y)}e^{ax+by}=\frac{1}{f(a,b)}=e^{ax+by},f(a,b)\neq 0 \quad (2.4.10)$$

$f(D_x,D_y)\ \varphi\ (x,y)\ e^{ax+by}=e^{ax+by}\ f(D_x+a,\ D_y+b)\ \varphi(x,y)$

$$\frac{1}{f(D_x,D_y)}\varphi(x,y)e^{ax+by}=e^{ax+by}\frac{1}{f(D_x+a,D_y+b)}\varphi(x,y) \quad (2.4.11)$$

$$= e^{ax} \frac{1}{f(D_x + a, D_y)} e^{by} \varphi(x,y) = e^{by} \frac{1}{f(D_x, D_y + b)} e^{ax} \varphi(x,y)$$

(2.4.12)

$$f(D_x^2, D_y^2) \cos(ax+by) = f(-a^2, -b^2) \cos(ax+by)$$

$$\frac{1}{f(D_x^2, D_y^2)} \cos(ax+by) = \frac{1}{f(-a^2, -b^2)} \cos(ax+by)$$

(2.4.13)

$$f(D_x^2, D_y^2) \sin(ax+by) = f(-a^2, -b^2) \sin(ax+by)$$

$$\frac{1}{f(D_x^2, D_y^2)} \sin(ax+by) = \frac{1}{f(-a^2, -b^2)} \sin(ax+by) \qquad (2.4.14)$$

When $\varphi(x,y)$ is any function of x and y, we resolve $\frac{1}{f(D_x, D_y)}$ into partial fractions treating $f(D_x, D_y)$ as a function of D_x alone and operate each partial fraction on $\varphi(x,y)$, remembering that

$$\frac{1}{D_x - mD_y} \varphi(x,y) = \int \varphi(x, c - mx) dx$$

where c is replaced by y+mx after integration.

Example 2.4.2. Find the particular solution of the following partial differential equations

(i) $$3\frac{\partial^2 u}{\partial x^2} + 4\frac{\partial^2 u}{\partial x \partial y} - \frac{\partial u}{\partial y} = e^{x-3y}$$

(ii) $$3\frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial y} = e^x \sin(x+y)$$

Solution: (i) The equation can be written as

$$(3D_x^2 + 4D_xD_y - D_y)u = e^{x-3y}$$

$$u_p = \frac{1}{3D_x^2 + 4D_x D_y - D_y} e^{x-3y}$$

$$= \frac{1}{3 + 4(-3) - (-3)} e^{x-3y} \text{ by (2.4.9)}$$

$$= -\frac{1}{6} e^{x-3y}$$

(ii) The equation can be written as

$(3D^2_x - D_y)u = e^x \sin(x+y)$

$$u_p = \frac{1}{3D_x^2 - D_y} e^x \sin(x+y)$$

$$= e^x \frac{1}{(3(D_x+1)^2 - D_y)} \sin(x+y)$$

$$= e^x \frac{1}{(3D_x^2 + 6D_x + 3 - D_y)} \sin(x+y)$$

$$= e^x \frac{1}{(3(-1) + 6D_x + 3 - D_y)} \sin(x+y)$$

$$= e^x \left(\frac{1}{6D_x - D_y}\right) \sin(x+y) \qquad = e^x \frac{(6D_x + D_y)}{36D_x^2 - D^2{}_y} \sin(x+y)$$

$$= e^x \frac{7\cos(x+y)}{-35}$$

$$= -\frac{1}{5} e^x \cos(x+y).$$

Example 2.4.3. Solve the partial differential equation

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = e^{-x} \sin t$$

Solution: The equation can be written as

$(D_t^2 - c^2 D_x^2)\, u = e^{-x} \sin t$

The particular solution is

$$u_p = \frac{1}{D_t^2 - c^2 D_x^2} e^{-x} \sin t$$

$$= e^{-x} \frac{1}{D_t^2 - (c(D_x - 1)} \sin t = e^{-x} \frac{1}{-1 - c^2} \sin t$$

$$= -\frac{1}{c^2 + 1} e^{-x} \sin t$$

By proceeding on the lines of the solution of Example 2.4.1 we get

$u_c = \varphi(x-ct) + \psi(x+ct)$

$$u(x,t) = \varphi(x-ct) + \psi(x+ct) - \frac{1}{c^2 + 1} e^{-x} \sin t$$

The solution u_c is known as the d' **Alembert's solution** of the wave equation

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0.$$

2.5. CLASSIFICATION OF SECOND ORDER LINEAR PARTIAL DIFFERENTIAL EQUATIONS

A partial differential equation is said to be **linear** if the unknown function u(.,.) and all its partial derivatives appear in an algebraically linear form, 'that is, of the first degree. For example the equation

$$A u_{xx} + 2B u_{xy} + C u_{yy} + D u_x + E u_y + F u = f \qquad (2.5.1)$$

where the coefficients A,B,C,D.E and F and the function f are functions of x and y, is a second-order linear partial differential equation in the unknown u(x,y).

Left hand side of (2.5.1) can be abbreviated by Lu, where u has continuous partial derivatives of upto second order.

If u is a function having continuous partial derivatives of appropriate order, say n then a partial derivative can be written as Lu=f where L is a differential operator, that is, L carries u to the sum of scalar multiplications of its partial derivatives of different order. An operator L is called linear differential operator if $\boldsymbol{L(\alpha u+\beta v)= \alpha Lu+\beta v}$ where α and β are scalars and u and v are any functions with continuous partial derivatives of appropriate order. A partial differential equation is called **homogeneous** if Lu=0, that is, f on the right hand side of a partial differential equation is zero, say f=0 in (2.5.1). The partial differential equation is called **non-homogeneous** if $f\neq 0$.

Examples 2.1:

a) $(x+2y)\, u_x + x^2 u_y = \sin(x^2+y^2)$ is a non-homogeneous partial differential equation of first-order.

b) $(x+2y)\, u_x + x^2 u_y = 0$ is a homogeneous linear partial differential equation of first-order

c) $xu_{xx} + yu_{xy} + u_{yy} = 0$ is a homogeneous linear partial differential equation of second-order.

d) $xu_{xx} + y\, u_{xy} + u_{yy} = \sin x$ is a non-homogeneous linear partial differential equation of second-order.

For f=0 in Equation (2.5.1), the most general form of a second-order homogeneous equation $\mathbf{A\, u_{xx} + 2B\, u_{xy} + C\, u_{yy} + D\, u_x + E\, u_y + Fu = 0}$ (2.5.2)

For a correspondence of this equation with an algebraic quadratic equation, we replace u_x by α, u_y by β, u_{xx} by α^2, u_{xy} by $\alpha\beta$, and u_{yy} by β^2. The left hand side of Equation (2.2) reduces to a second degree polynomial in α and β:

$$\mathbf{P(\alpha,\beta)=A\alpha^2+2B\alpha\beta+C\beta^2+D\alpha+E\beta+F=0} \qquad \mathbf{(2.5.3)}$$

It is known from analytical geometry and algebra that the polynomial equation $P(\alpha,\beta)=0$ represents a hyperbola, parabola, or ellipse according as its discriminant. B^2-AC is positive, zero, or negative. Thus, the partial differential equation (2.5.3) is classified as **hyperbolic**, **parabolic,** or **elliptic** according as the quantity

B^2-AC>0, B^2-AC=0, or B^2-AC<0.

The equation

$$\mathbf{A\,u^2_x+2B\,u_{xy}+C\,u^2_y=0} \qquad (2.5.4)$$

is called the **characteristic equation** of the partial differential equation (2.5.3). Solutions of (2.5.4) are called the characteristics

Example 2.2: Examine whether the following partial differential equations are hyperbolic, parabolic, or elliptic.

(i) $\dfrac{\partial^2 u}{\partial x^2} + x\dfrac{\partial^2 u}{\partial y^2} + 4 = 0$ (ii) $\dfrac{\partial^2 u}{\partial x^2} + y\dfrac{\partial^2 u}{\partial y^2} = 0$ (iii) $y^2\dfrac{\partial^2 u}{\partial x^2} - \dfrac{\partial^2 u}{\partial y^2} = 0$

(iv) $u_{xx} + x^2 u_{yy} = 0$ (v) $x\,u_{xx} + 2x\,u_{xy} + y\,u_{yy} = 0$

Solutions: **(i)** A = 1, C = x, B = 0

B^2-AC = 0 –x <0 for x>0

Thus the equation is elliptic if x > 0, is hyperbolic if x < 0 and it is parabolic if x = 0.

(ii) A=1, B=0, C=y

B^2-AC=0-y >0 if y<0 and so the equation is hyperbolic if y<0. It is parabolic if y=0 and it is elliptic if y>0.

(iii) $A=y^2$, B=0, C = -1.

B^2-AC=y^2>0 for all y. Therefore the equation is hyperbolic.

(iv) A=1, B=0, C=x^2

B^2-AC=0-x^2<0 for all x. The equation is elliptic

(v) A=x, B=x, C=y

B^2-AC=x^2-xy=x(x-y)>0 for x>0 x>y

In this case the equation is hyperbolic B^2-AC=o if x=y. For this the equation is parabolic.

B^2-AC <0 if x>y and x<0 or if x<y and x>0

In this case the equation is elliptic.

Exercises2.2

Write down the order and degree of partial differential equations in problems 1-5.

1. $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = u^2$

2. $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$

3. $\left(\frac{\partial u}{\partial x}\right)^3 + \frac{\partial u}{\partial y} = 0$

4. $\frac{\partial u}{\partial t} + 100\ \frac{\partial u}{\partial x} = 0$

5. $\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^3 = 0$

6. Verify that the functions $u(x,y)=x^2-y^2$ and $u(x,y) = e^x \sin y$ are solutions of the equation

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y}=0$$

7. Let u=f(x,y), where f is an arbitrary differentiable function. Show that u satisfies the equation

$x\, u_x - y\, u_y = 0$

Examine whether cos (xy), e^{xy} and $(xy)^3$ are solutions of this partial differential equation.

Classify the partial differential equations as hyperbolic, parabolic, or elliptic.

8. $4\, u_{xx}-7\, u_{xy} + 3\, u_{yy}= 0$

9. $4\, u_{xx}-8\, u_{xy} + 4\, u_{yy}= 0$

10. $a^2\, u_{xx}+2a\, u_{xy} +u_{yy} = 0,\ a\neq 0$

11. $$4\,\frac{\partial^2 u}{\partial t^2} - 12\,\frac{\partial^2 u}{\partial x \partial t}+9\frac{\partial^2 u}{\partial x^2}=0$$

12. $$8\,\frac{\partial^2 u}{\partial x^2} - 2\,\frac{\partial^2 u}{\partial x \partial y} - 3\frac{\partial^2 u}{\partial y^2}=0$$

For what values of x and y are the following partial differential equations hyperbolic, parabolic, or elliptic?

13. $u_{xx}+2xu_{xy}+(1-y^2)\, u_{yy}=0$

14. $(1+y^2)\, u_{xx}+(1+x^2)\, u_{yy}=0$

15. $u_{xx} + x^2\, u_{yy} = 0$

16. $u_{xx} - 2 \sin x\, u_{xy} - \cos^2 x\, u_y = 0$

17. Find the general solution of $2\, u_x - 3\, u_y = \cos x$

18. Solve $u_x+e^x u_y=y,\ u(0,y) = 1+y$

Find the complete solutions of the equations in problem 19-25

19. $p=(u+qy)^2$

20. $2(u+xp+yq)=yp^2$

21. $u^2=pqxy$

22. $xp+3yq=2(u-x^2q^2)$

23. $pq=1$

24. $p^2y(1+x^2)=qx^2$

25. $u=p^2-q^2$

26. $p^2q^2+x^2y^2=x^2q^2(x^2-y^2)$

Solve partial differential equations of problems 27to 31.

27. $\frac{\partial^2 u}{\partial x^2}+12\frac{\partial u}{\partial x}+2=0$

28. $4\frac{\partial^2 u}{\partial x^2}-16\frac{\partial^2 u}{\partial y^2}+15\frac{\partial^2 u}{\partial y^2}=0$

29. $3\frac{\partial^2 u}{\partial x^2}+4\frac{\partial^2 u}{\partial x\partial y}-\frac{\partial u}{\partial y}=0$

30. $3\frac{\partial^2 u}{\partial x^2}-\frac{\partial u}{\partial y}=\sin(ax+by)$

31. $3\frac{\partial^2 u}{\partial x^2}-2\frac{\partial u}{\partial x}\frac{\partial u}{\partial y}-5\frac{\partial^2 u}{\partial y^2}=3x+y+e^{x-y}$

2.6.SOLUTIONS OF ONE–DIMENSIONAL WAVE EQUATION, ONE-DIMENSIONAL HEAT EQUATION

2.6.1. The Heat Equation

For a material of constant density ρ, constant specific heat μ and constant thermal conductivity K, the partial differential equation governing the temperature u at any location (x, y, z) and any time t is

$$\frac{\partial u}{\partial t} = k \nabla^2 u \ , \quad \text{where} \quad k = \frac{K}{\mu \rho}$$

Example 2.6.1

Heat is conducted along a thin homogeneous bar extending from $x = 0$ to $x = L$. There is no heat loss from the sides of the bar. The two ends of the bar are maintained at temperatures T_1 (at $x = 0$) and T_2 (at $x = L$). The initial temperature throughout the bar at the cross-section x is $f(x)$.

Find the temperature at any point in the bar at any subsequent time.

The partial differential equation governing the temperature $u(x, t)$ in the bar is

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} \qquad \text{[Parabolic]}$$

together with the boundary conditions

$$u(0, t) = T_1 \quad \text{and} \quad u(L, t) = T_2$$

and the initial condition

$$u(x, 0) = f(x)$$

[Note that if an end of the bar is insulated, instead of being maintained at a constant temperature, then the boundary condition changes to $\frac{\partial u}{\partial t}(0,t) = 0$ or $\frac{\partial u}{\partial t}(L,t) = 0$.]

Attempt a solution by the method of separation of variables.

$$u(x, t) = X(x)\, T(t)$$

$$\Rightarrow \quad X T' = k X'' T \qquad \Rightarrow \quad \frac{T'}{T} = k \frac{X''}{X} = c$$

Again, when a function of t only equals a function of x only, both functions must equal the same absolute constant. Unfortunately, the two boundary conditions cannot both be satisfied unless $T_1 = T_2 = 0$. Therefore we need to treat this more general case as a perturbation of the simpler ($T_1 = T_2 = 0$) case.

Let $u(x, t) = v(x, t) + g(x)$
Substitute this into the PDE:

$$\frac{\partial}{\partial t}\big(v(x,t)+g(x)\big) = k\,\frac{\partial^2}{\partial x^2}\big(v(x,t)+g(x)\big) \quad \Rightarrow \quad \frac{\partial v}{\partial t} = k\left(\frac{\partial^2 v}{\partial x^2} + g''(x)\right)$$

This is the standard heat PDE for v if we choose g such that $g''(x) = 0$.
$g(x)$ must therefore be a linear function of x.

We want the perturbation function $g(x)$ to be such that

$$u(0, t) = T_1 \ , \quad u(L, t) = T_2$$

and

$$v(0, t) = v(L, t) = 0$$

Therefore $g(x)$ must be the linear function for which $g(0) = T_1$ and $g(L) = T_2$.
It follows that

$$g(x) = \left(\frac{T_2 - T_1}{L}\right)x + T_1$$

and we now have the simpler problem

$$\frac{\partial v}{\partial t} = k\,\frac{\partial^2 v}{\partial x^2}$$

together with the boundary conditions

$$v(0, t) = v(L, t) = 0$$

and the initial condition

$$v(x, 0) = f(x) - g(x)$$

Now try separation of variables on $v(x, t)$:
$v(x, t) = X(x)\,T(t)$

$$\Rightarrow \quad X\,T' = k\,X''T \qquad \Rightarrow \quad \frac{1}{k}\frac{T'}{T} = \frac{X''}{X} = c$$

But $v(0, t) = v(L, t) = 0 \quad \Rightarrow \quad X(0) = X(L) = 0$

This requires c to be a negative constant, say $-\lambda^2$.
The solution is very similar to that for the wave equation on a finite string with fixed ends (section 4.3). The eigenvalues are $\lambda = \dfrac{n\pi}{L}$ and the corresponding eigenfunctions are any non-zero constant multiples of

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

Example 2.6.1 (continued)

The ODE for $T(t)$ becomes

$$T' + \left(\frac{n\pi}{L}\right)^2 kT = 0$$

whose general solution is

$$T_n(t) = c_n e^{-n^2\pi^2 kt/L^2}$$

Therefore

$$v_n(x,t) = X_n(x)T_n(t) = c_n \sin\left(\frac{n\pi x}{L}\right)\exp\left(-\frac{n^2\pi^2 kt}{L^2}\right)$$

If the initial temperature distribution $f(x) - g(x)$ is a simple multiple of $\sin\left(\frac{n\pi x}{L}\right)$ for some integer n, then the solution for v is just $v(x,t) = c_n \sin\left(\frac{n\pi x}{L}\right)\exp\left(-\frac{n^2\pi^2 kt}{L^2}\right)$.

Otherwise, we must attempt a superposition of solutions.

$$v(x,t) = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right)\exp\left(-\frac{n^2\pi^2 kt}{L^2}\right)$$

such that $v(x,0) = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right) = f(x) - g(x)$.

The Fourier sine series coefficients are $c_n = \frac{2}{L}\int_0^L \left(f(z) - g(z)\right)\sin\left(\frac{n\pi z}{L}\right)dz$

so that the complete solution for $v(x, t)$ is

$$v(x,t) = \frac{2}{L}\sum_{n=1}^{\infty}\left(\int_0^L \left(f(z) - \frac{T_2 - T_1}{L}z - T_1\right)\sin\left(\frac{n\pi z}{L}\right)dz\right)\sin\left(\frac{n\pi x}{L}\right)\exp\left(-\frac{n^2\pi^2 kt}{L^2}\right)$$

and the complete solution for $u(x, t)$ is

$$u(x,t) = v(x,t) + \left(\frac{T_2 - T_1}{L}\right)x + T_1$$

Note how this solution can be partitioned into a transient part $v(x, t)$ (which decays to zero as t increases) and a steady-state part $g(x)$ which is the limiting value that the temperature distribution approaches.

Example 2.6.1 (continued)

As a specific example, let $k = 9$, $T_1 = 100$, $T_2 = 200$, $L = 2$ and

$f(x) = 145x^2 - 240x + 100$, (for which $f(0) = 100$, $f(2) = 200$ and $f(x) > 0 \ \forall x$).

Then $g(x) = \frac{200-100}{2} x + 100 = 50x + 100$

The Fourier sine series coefficients are

$$c_n = \int_0^2 \left(\left(145z^2 - 240z + 100\right) - \left(50z + 100\right) \right) \sin\left(\frac{n\pi z}{2}\right) dz$$

$$\Rightarrow \quad c_n = 145 \int_0^2 \left(z^2 - 2z\right) \sin\left(\frac{n\pi z}{2}\right) dz$$

D	I
$z^2 - 2z$	$\sin\left(\frac{n\pi z}{2}\right)$
$2z - 2$	$-\left(\frac{2}{n\pi}\right)\cos\left(\frac{n\pi z}{2}\right)$
2	$-\left(\frac{2}{n\pi}\right)^2 \sin\left(\frac{n\pi z}{2}\right)$
0	$\left(\frac{2}{n\pi}\right)^3 \cos\left(\frac{n\pi z}{2}\right)$

$$\Rightarrow \quad c_n = 145 \left[\left(\left(-z^2 + 2z\right)\frac{2}{n\pi} + \frac{16}{(n\pi)^3} \right) \cos\left(\frac{n\pi z}{2}\right) + \frac{8(z-1)}{(n\pi)^2} \sin\left(\frac{n\pi z}{2}\right) \right]_{z=0}^{z=2}$$

$$\Rightarrow \quad c_n = 145 \left[\left(z(2-z)\frac{2}{n\pi} + \frac{16}{(n\pi)^3} \right) \cos\left(\frac{n\pi z}{2}\right) + \frac{8(z-1)}{(n\pi)^2} \sin\left(\frac{n\pi z}{2}\right) \right]_{z=0}^{z=2}$$

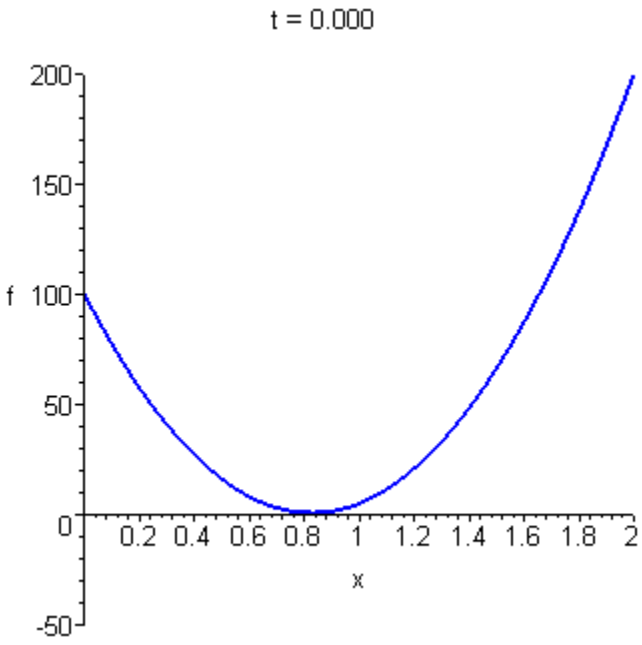
$$\Rightarrow \quad c_n = \frac{2320}{(n\pi)^3}\left((-1)^n - 1\right)$$

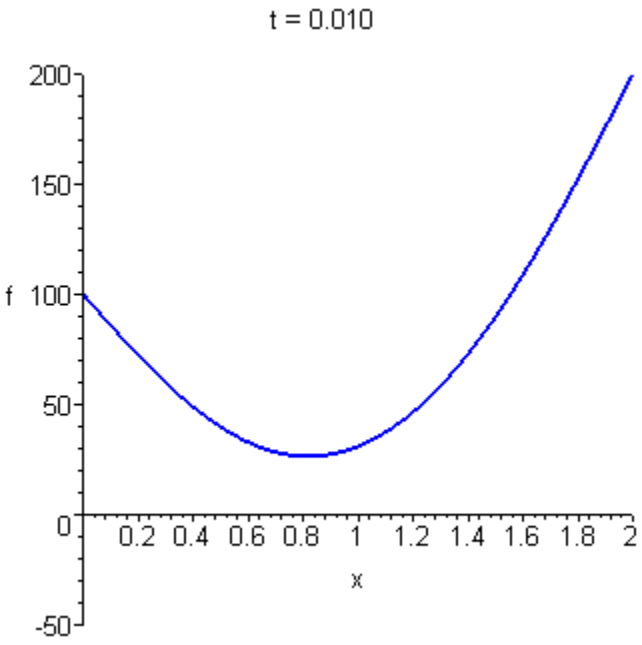
The complete solution is

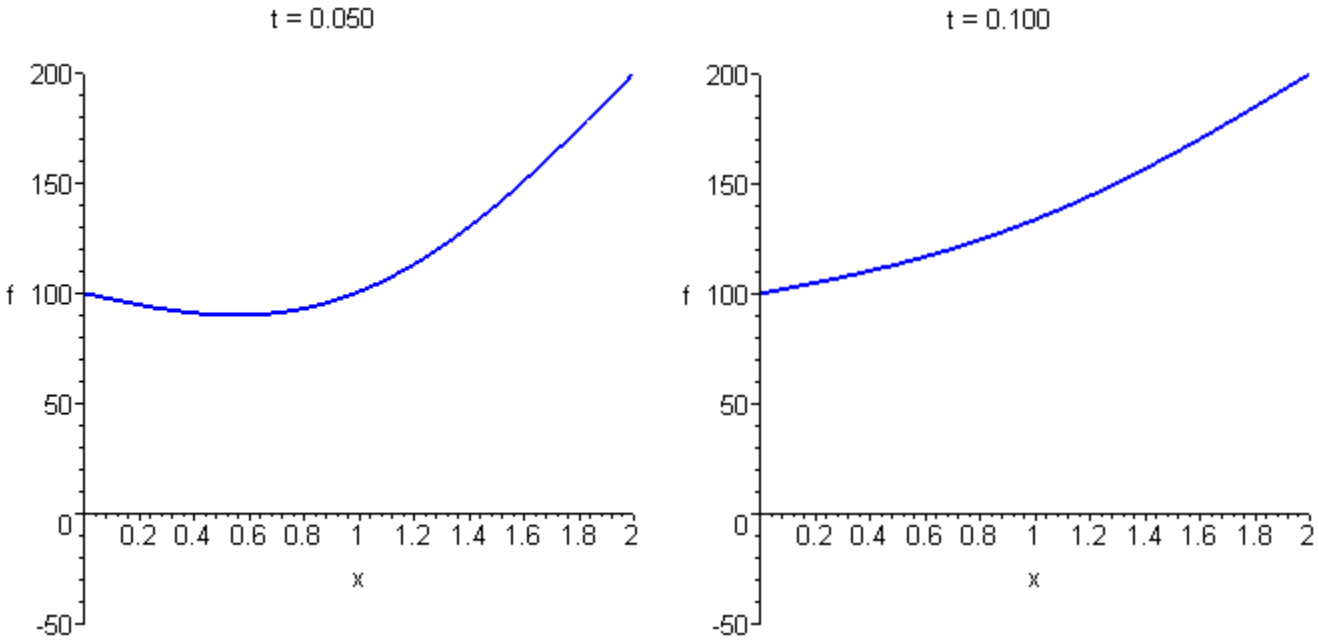
$$u(x,t) = 50x + 100 - \frac{2320}{\pi^3}\sum_{n=1}^{\infty}\left(\frac{1-(-1)^n}{n^3}\right)\sin\left(\frac{n\pi x}{2}\right)\exp\left(-\frac{9n^2\pi^2 t}{4}\right)$$

Some snapshots of the temperature distribution (from the tenth partial sum) from the Maple file at "www.engr.mun.ca/~ggeorge/5432/demos/ex451.mws" are shown on the next page.

Example 2.6.1 (continued)







The steady state distribution is nearly attained in much less than a second!

2.6.2. Wave equation

The **wave equation**: $\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$

or its one-dimensional special case $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ [which is hyperbolic everywhere]

(where u is the displacement and c is the speed of the wave);

The **heat** (or **diffusion**) **equation**: $\mu\rho \frac{\partial u}{\partial t} = K \nabla^2 u + \vec{\nabla} K \bullet \vec{\nabla} u$

a one-dimensional special case of which is

$$\frac{\partial u}{\partial t} = \frac{K}{\mu\rho}\frac{\partial^2 u}{\partial x^2}$$ [which is parabolic everywhere]

(where u is the temperature, μ is the specific heat of the medium, ρ is the density and K is the thermal conductivity);

The potential (or **Laplace's**) **equation**: $\nabla^2 u = 0$

a special case of which is $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ [which is elliptic everywhere]

The complete solution of a PDE requires additional information, in the form of initial conditions (values of the dependent variable and its first partial derivatives at $t = 0$), boundary conditions (values of the dependent variable on the boundary of the domain) or some combination of these condition

d'Alembert Solution

Example 2.6.2

Show that

$$y(x,t) = \frac{f(x+ct) + f(x-ct)}{2}$$

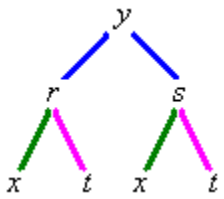
is a solution to the wave equation

$$\frac{\partial^2 y}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 y}{\partial t^2} = 0$$

with initial conditions $y(x, 0) = f(x)$ and $\left.\frac{\partial}{\partial t} y(x,t)\right|_{t=0} = 0$

for ***any*** twice differentiable function $f(x)$.

Let $r = x + ct$ and $s = x - ct$, then $y(r,s) = \frac{f(r) + f(s)}{2}$



and

$$\frac{\partial y}{\partial x} = \frac{\partial y}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial y}{\partial s}\frac{\partial s}{\partial x} = \frac{1}{2}\Big(\big(f'(r)+0\big)\times 1 + \big(0+f'(s)\big)\times 1\Big),$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial y}{\partial x}\right) = \frac{\partial}{\partial r}\left(\frac{\partial y}{\partial x}\right)\frac{\partial r}{\partial x} + \frac{\partial}{\partial s}\left(\frac{\partial y}{\partial x}\right)\frac{\partial s}{\partial x} = \frac{1}{2}\big(f''(r)\times 1 + f''(s)\times 1\big),$$

$$\frac{\partial y}{\partial t} = \frac{\partial y}{\partial r}\frac{\partial r}{\partial t} + \frac{\partial y}{\partial s}\frac{\partial s}{\partial t} = \frac{1}{2}\Big(\big(f'(r)+0\big)\times c + \big(0+f'(s)\big)\times(-c)\Big),$$

$$\frac{\partial^2 y}{\partial t^2} = \frac{\partial}{\partial r}\left(\frac{\partial y}{\partial t}\right)\frac{\partial r}{\partial t} + \frac{\partial}{\partial s}\left(\frac{\partial y}{\partial t}\right)\frac{\partial s}{\partial t} = \frac{1}{2}\big(c\,f''(r)\times c - c\,f''(s)\times(-c)\big),$$

$$\Rightarrow \quad \frac{\partial^2 y}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 y}{\partial t^2} = \frac{1}{2}\big(f''(r) + f''(s)\big) - \frac{1}{2c^2}\big(c^2 f''(r) + c^2 f''(s)\big) = 0,$$

Therefore $y(x,t) = \dfrac{f(x+ct) + f(x-ct)}{2}$ is a solution to the wave equation for all twice differentiable functions $f(x)$. This is part of the d'Alembert solution.

Example 2.6.2 (continued)

This d'Alembert solution satisfies the initial displacement condition:

$$y(x,0) = \frac{f(x+0) + f(x-0)}{2} = f(x)$$

Also $\left.\dfrac{\partial}{\partial t} y(x,t)\right|_{t=0} = \left.\dfrac{c\,f'(x+ct) - c\,f'(x-ct)}{2}\right|_{t=0} = \dfrac{c\,f'(x) - c\,f'(x)}{2} = 0$

The d'Alembert solution therefore satisfies both initial conditions.

A more general d'Alembert solution to the wave equation for an infinitely long string is

$$\boxed{y(x,t) = \frac{f(x+ct) + f(x-ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(u)\,du}$$

This satisfies the wave equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \qquad \text{for } -\infty < x < \infty \quad \text{and} \quad t > 0$$

and

Initial configuration of string: $y(x, 0) = f(x)$ for $x \in \mathbb{R}$

and

Initial speed of string: $\left.\dfrac{\partial y}{\partial t}\right|_{(x,0)} = g(x)$ for $x \in \mathbb{R}$

for *any* twice differentiable functions $f(x)$ and $g(x)$.

Physically, this represents two identical waves, moving with speed c in opposite directions along the string.

Proof that $y(x,t) = \frac{1}{2c}\int_{x-ct}^{x+ct} g(u)\,du$ satisfies both initial conditions:

$$y(x,t) = \frac{1}{2c}\int_{x-ct}^{x+ct} g(u)\,du \quad \Rightarrow \quad y(x,0) = \frac{1}{2c}\int_{x}^{x} g(u)\,du = 0$$

Using a Leibnitz differentiation of the integral:

$$\frac{\partial y}{\partial t} = \frac{1}{2c}\left(g(x+ct)\cdot\frac{\partial}{\partial t}(x+ct) - g(x-ct)\cdot\frac{\partial}{\partial t}(x-ct) + \int_{x-ct}^{x+ct}\frac{\partial}{\partial t}g(u)\,du \right)$$

$$= \frac{1}{2c}\left(c\,g(x+ct) + c\,g(x-ct) + 0\right) = \frac{g(x+ct) + g(x-ct)}{2}$$

$$\Rightarrow \quad \left.\frac{\partial y}{\partial t}\right|_{t=0} = \frac{g(x+0) + g(x-0)}{2} = g(x)$$

Example 2.6.3

An elastic string of infinite length is displaced into the form $y = \cos \pi x/2$ on [–1, 1] only (and $y = 0$ elsewhere) and is released from rest. Find the displacement $y(x, t)$ at all locations on the string $x \in \mathbb{R}$ and at all subsequent times ($t > 0$).

For this solution to the wave equation we have initial conditions

$$y(x,0) = f(x) = \begin{cases} \cos\left(\frac{\pi x}{2}\right) & (-1 \le x \le 1) \\ 0 & (\text{otherwise}) \end{cases}$$

and

$$\frac{\partial y}{\partial t}(x,0) = g(x) = 0$$

The d'Alembert solution is

$$y(x,t) = \frac{f(x+ct) + f(x-ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(u)\,du = \frac{f(x+ct) + f(x-ct)}{2} + 0$$

where $f(x+ct) = \begin{cases} \cos\left(\dfrac{\pi(x+ct)}{2}\right) & (-1-ct \le x \le 1-ct) \\ 0 & (\text{otherwise}) \end{cases}$

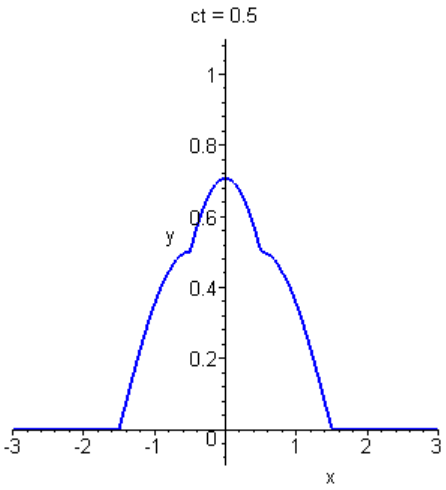
and $f(x-ct) = \begin{cases} \cos\left(\dfrac{\pi(x-ct)}{2}\right) & (-1+ct \le x \le 1+ct) \\ 0 & (\text{otherwise}) \end{cases}$

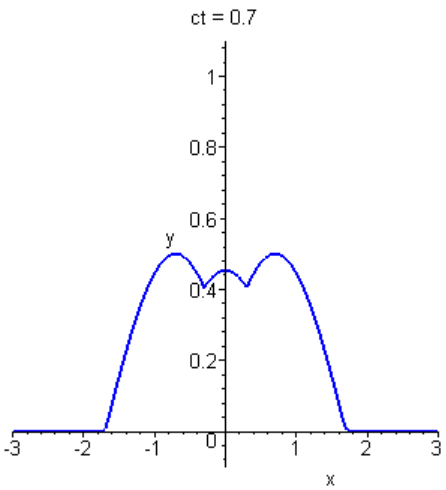
We therefore obtain two waves, each of the form of a single half-period of a cosine function, moving apart from a superposed state at $x = 0$ at speed c in opposite directions.

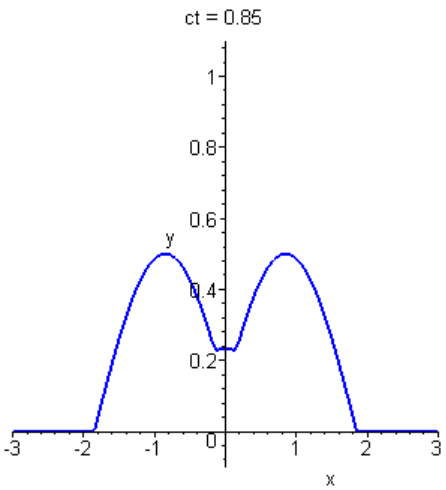
See the web page "www.engr.mun.ca/~ggeorge/5432/demos/ex422.html" for an animation of this solution.

Example 2.6.3.(continued)

Some snapshots of the solution are shown here







A more general case of a d'Alembert solution arises for the homogeneous PDE with constant coefficients

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x\,\partial y} + C\frac{\partial^2 u}{\partial y^2} = 0$$

The characteristic (or auxiliary) equation for this PDE is

$$A\lambda^2 + B\lambda + C = 0$$

This leads to the complementary function (which is also the general solution for this homogeneous PDE)

$$\boxed{u(x, y) = f_1(y + \lambda_1 x) + f_2(y + \lambda_2 x),}$$

where

$$\boxed{\lambda_1 = \frac{-B - \sqrt{D}}{2A} \quad \text{and} \quad \lambda_2 = \frac{-B + \sqrt{D}}{2A}}$$

and $D = B^2 - 4AC$

and f_1, f_2 are **arbitrary** twice-differentiable functions of their arguments.

λ_1 and λ_2 are the roots (or eigenvalues) of the characteristic equation.

In the event of equal roots, the solution changes to

$$u(x, y) = f_1(y + \lambda x) + h(x, y)\, f_2(y + \lambda x)$$

where $h(x, y)$ is any non-trivial linear function of x and/or y (except $y + \lambda x$).

The wave equation is a special case with $y = t$, $A = 1$, $B = 0$, $C = -1/c^2$ and $\lambda = \pm 1/c$.

Example 2.6.4

$$\frac{\partial^2 u}{\partial x^2} - 3\frac{\partial^2 u}{\partial x\,\partial y} + 2\frac{\partial^2 u}{\partial y^2} = 0$$

$u(x, 0) = -x^2$

$u_y(x, 0) = 0$

(a) Classify the partial differential equation.

(b) Find the value of u at $(x, y) = (0, 1)$.

(a) Compare this PDE to the standard form

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x\,\partial y} + C\frac{\partial^2 u}{\partial y^2} = 0$$

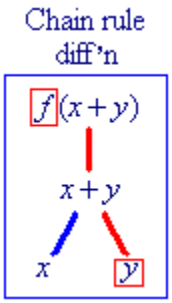
$A = 1$, $B = -3$, $C = 2$ $\Rightarrow$ $D = 9 - 4\times 2 = 1 > 0$

Therefore the PDE is **hyperbolic** everywhere.

(b) $\lambda = \dfrac{+3 \pm \sqrt{1}}{2} = 1$ or 2

The complementary function (and general solution) is

$u(x, y) = f(y + x) + g(y + 2x)$



$\Rightarrow$ $u_y(x, y) = f'(y + x) + g'(y + 2x)$

Initial conditions:

$u(x, 0) = f(x) + g(2x) = -x^2$ **(1)**

and

$u_y(x, 0) = f'(x) + g'(2x) = 0$ **(2)**

$\dfrac{d}{dx}(1) = f'(x) + 2g'(2x) = -2x$ **(3)**

(3) – (2) $\Rightarrow$ $g'(2x) = -2x$ $\Rightarrow$ $g'(x) = -x$

$$\Rightarrow \quad g(x) = -\tfrac{1}{2}x^2 + k \qquad \Rightarrow \quad g(y+2x) = -\tfrac{1}{2}(y+2x)^2 + k$$

Example 2.6.4 (continued)

Also **(1)** $\Rightarrow$ $f(x) = -x^2 - g(2x) = -x^2 + \tfrac{1}{2}(2x)^2 - k = x^2 - k$

$\Rightarrow$ $f(y + x) = (y + x)^2 - k$

Therefore $u(x, y) = f(y + x) + g(y + 2x)$

$= (y + x)^2 - k - (y + 2x)^2 / 2 + k$

$$= \frac{1}{2}\left(2y^2 + 4xy + 2x^2 - y^2 - 4xy - 4x^2\right) = \frac{1}{2}\left(y^2 - 2x^2\right)$$

The complete solution is therefore $u(x, y) = \frac{1}{2}\left(y^2 - 2x^2\right)$

$$\Rightarrow \quad u(0,1) = \frac{1}{2}\left(1^2 - 0^2\right) = \underline{\underline{\mathbf{\frac{1}{2}}}}$$

[It is easy (though tedious) to confirm that $u(x, y) = \frac{1}{2}\left(y^2 - 2x^2\right)$ satisfies the partial differential equation $\frac{\partial^2 u}{\partial x^2} - 3\frac{\partial^2 u}{\partial x\,\partial y} + 2\frac{\partial^2 u}{\partial y^2} = 0$ together with both initial conditions $u(x, 0) = -x^2$ and $u_y(x, 0) = 0$.]

[Also note that the arbitrary constants of integration for f and g cancelled each other out. This cancellation happens generally for this method of d'Alembert solution.]

Example 2.6.5

Find the complete solution to

$$6\frac{\partial^2 u}{\partial x^2} - 5\frac{\partial^2 u}{\partial x\,\partial y} + \frac{\partial^2 u}{\partial y^2} = 14,$$

$$u(x, 0) = 2x + 1,$$

$$u_y(x, 0) = 4 - 6x.$$

This PDE is non-homogeneous.

For the particular solution, we require a function such that the combination of second partial derivatives resolves to the constant 14. It is reasonable to try a quadratic function of x and y as our particular solution.

Try $u_P = ax^2 + bxy + cy^2$

$$\Rightarrow \quad \frac{\partial u_P}{\partial x} = 2ax + by \quad \text{and} \quad \frac{\partial u_P}{\partial y} = bx + 2cy$$

$$\Rightarrow \quad \frac{\partial^2 u_P}{\partial x^2} = 2a, \quad \frac{\partial^2 u_P}{\partial x\,\partial y} = b \quad \text{and} \quad \frac{\partial^2 u_P}{\partial y^2} = 2c$$

$$\Rightarrow \quad 6\frac{\partial^2 u_P}{\partial x^2} - 5\frac{\partial^2 u_P}{\partial x\,\partial y} + \frac{\partial^2 u_P}{\partial y^2} = 12a - 5b + 2c = 14$$

We have one condition on three constants, two of which are therefore a free choice.

Choose $b = 0$ and $c = a$, then $14a = 14 \Rightarrow c = a = 1$

Therefore a particular solution is $u = x^2 + y^2$

Complementary function:

$A = 6$, $B = -5$, $C = 1 \Rightarrow D = 25 - 4\times 6 = 1 > 0$

Therefore the PDE is **hyperbolic** everywhere.

$$\lambda = \frac{+5 \pm \sqrt{1}}{12} = \frac{1}{3} \text{ or } \frac{1}{2}$$

The complementary function is

$$u_C(x,y) = f\left(y+\tfrac{1}{3}x\right) + g\left(y+\tfrac{1}{2}x\right)$$

and the general solution is

$$u(x,y) = f\left(y+\tfrac{1}{3}x\right) + g\left(y+\tfrac{1}{2}x\right) + x^2 + y^2$$

Example 2.6.5 (continued)

$$u(x,y) = f\left(y+\tfrac{1}{3}x\right) + g\left(y+\tfrac{1}{2}x\right) + x^2+y^2$$

$$\Rightarrow \quad \frac{\partial u}{\partial y} = f'\left(y+\tfrac{1}{3}x\right) + g'\left(y+\tfrac{1}{2}x\right) + 2y$$

Imposing the two boundary conditions:

$$u(x,0) = f\left(\tfrac{1}{3}x\right) + g\left(\tfrac{1}{2}x\right) + x^2 = 2x+1 \qquad \textbf{(1)}$$

and

$$u_y(x,0) = f'\left(\tfrac{1}{3}x\right) + g'\left(\tfrac{1}{2}x\right) + 0 = 4-6x \qquad \textbf{(2)}$$

$$\frac{d}{dx}\textbf{(1)} = \frac{1}{3}f'\left(\tfrac{1}{3}x\right) + \frac{1}{2}g'\left(\tfrac{1}{2}x\right) + 2x = 2 \qquad \textbf{(3)}$$

(2) – 2×(3) $\Rightarrow \quad \frac{1}{3}f'\left(\tfrac{1}{3}x\right) - 4x = 4 - 6x - 4$

$$\Rightarrow \quad f'\left(\tfrac{1}{3}x\right) = -6x = -18\left(\tfrac{1}{3}x\right) \quad \Rightarrow \quad f'(x) = -18x$$

$$\Rightarrow \quad f(x) = -9x^2+k$$

(1) $\Rightarrow \quad g\left(\tfrac{1}{2}x\right) = 2x+1-x^2 - f\left(\tfrac{1}{3}x\right) = 2x+1-x^2 + 9\left(\frac{x^2}{9}\right) - k$

$$\Rightarrow \quad g(x) = 4x + 1 - k$$

But

$$u(x,y) = f\left(y+\tfrac{1}{3}x\right) + g\left(y+\tfrac{1}{2}x\right) + x^2+y^2$$

$$\Rightarrow \quad u(x,y) = -9\left(y+\tfrac{1}{3}x\right)^2 + k + 4\left(y+\tfrac{1}{2}x\right) + 1 - k + x^2+y^2$$

[again the arbitrary constants cancel - they can be omitted safely.]

$$= -9y^2-6xy-x^2 + 4y + 2x + 1 + x^2+y^2$$

Therefore the complete solution is

$$\boxed{u(x,y) = 1 + 2x + 4y - 6xy - 8y^2}$$

Example 2.6.6

Find the complete solution to

$$\frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x\,\partial y} + \frac{\partial^2 u}{\partial y^2} = 0,$$

$u = 0$ on $x = 0$,

$u = x^2$ on $y = 1$.

$A = 1,\quad B = 2,\quad C = 1 \quad \Rightarrow \quad D = 4 - 4\times 1 = 0$

Therefore the PDE is **parabolic** everywhere.

$$\lambda = \frac{-2 \pm \sqrt{0}}{2} = -1 \text{ or } -1$$

The complementary function (and general solution) is

$$u(x,y) = f(y-x) + h(x,y)\,g(y-x)$$

where $h(x, y)$ is any convenient non-trivial linear function of (x, y) except a multiple of $(y - x)$.

Choosing, arbitrarily, $h(x, y) = x$,

$$u(x,y) = f(y-x) + x\,g(y-x)$$

Imposing the boundary conditions:

$u(0, y) = 0 \quad \Rightarrow \quad f(y) + 0 = 0$

Therefore the function f is identically zero, for any argument including $(y - x)$.

We now have $u(x, y) = x\, g(y - x)$.

$u(x, 1) = x^2 \Rightarrow x\, g(1 - x) = x^2 \Rightarrow g(1 - x) = x$

Let $z = 1 - x$, then $x = 1 - z$ and $g(z) = 1 - z \Rightarrow g(x) = 1 - x$

Therefore

$u(x, y) = x\, g(y - x) = x\,(1 - (y - x))$

The complete solution is

$$\boxed{u(x, y) = x(x - y + 1)}$$

Unit III INTRODUCTION TO PROBABILITY AND STATISTICS

3.1. Descriptive Statistics: Measures of Central tendency, Measures of Dispersion and Measures of Forms.

3.1.1. Measure of central tendency and Location

Descriptive statistical measures have two functions: they provide a mental image of a data distribution, and they are an essential component of inferential statistics, the basis of both estimation and hypothesis testing.

The clustering of the measurements near the centre of a distribution is called *central tendency*, and the statistical measures that describe aspects of the "centre" of a distribution are called *measures of central tendency*. Because of the usual concentration of measurements in the centre of a distribution, the various measures of central tendency are generally also called *measures of average value*.

Measures of location show where the characteristics of a distribution are located in relation to the measurement scale. The measures of location we will present are $x_{\min}$ and $x_{\max}$, the minimum and maximum values in the data set; x_{Median} , the median, which is the boundary point to the left of which (and to the right of which) are 50% of the data, and x_{Mode} , the mode, which is the most frequently occurring value in a set of data.

Arithmetic Mean

Let $x_1, x_2, \cdots, x_n$ be an array of n measurements of a variable X. Suppose that these n measurements can be arranged into k categories, and let $R_i, i = 1, 2, \cdots, k$ and $f_i, i = 1, 2, \cdots, k$ represent the frequency and relative frequency in each of the k categories, respectively. Then, the arithmetic mean of the measurements is given by

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{\sum_{i=1}^{k} R_i x_i}{n} = \sum_{i=1}^{k} f_i x_i$$

Geometric Mean

For a set of positive numbers $x_1, x_2, \cdots, x_n$, the geometric mean is the principal n^{th} root of the product of the n numbers.

$$\bar{x}_G = \sqrt[n]{\prod_{i=1}^{n} x_i} = \sqrt[n]{\prod_{i=1}^{k} x_i^{R_i}}$$

Harmonic Mean

The harmonic mean of a set of data $x_1, x_2, \cdots, x_n$ is the reciprocal of the arithmetic mean of the reciprocals of the data.

$$\bar{x}_H = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}} = \frac{n}{\sum_{i=1}^{k} \frac{R_i}{x_i}}$$

It can be shown that

$$\bar{x}_H \le \bar{x}_G \le \bar{x}$$

Example 3.1

Consider the height measurements of 25 students given in below frequency table. Calculate the arithmetic mean, harmonic mean and geometric mean, and compare the results.

Solution

Table 3.1 gives basic calculations to be done

Table 31: Basic calculations

i_i	x_i	R_i	$R_i \times x_i$	$x_i^{R_i}$	$\frac{R_i}{x_i}$
1	152	1	152	152	0.0066
2	154	1	154	154	0.0065
3	155	1	155	155	0.0065
4	159	2	318	25281	0.0126
5	160	6	960	1.67772E+13	0.0375
6	161	2	322	25921	0.0124
7	162	1	162	162	0.0062
8	167	1	167	167	0.0060
9	170	4	680	835210000	0.0235
10	171	1	171	171	0.0058
11	172	4	688	875213056	0.0233
12	173	1	173	173	0.0058
			4102	**2.3337E+55**	**0.1526**

Arithmetic mean $\bar{x} = \frac{\sum_{i=1}^{12} R_i x_i}{25} = \frac{4102}{25} = 164.08$.

Geometric mean $\bar{x}_G = \sqrt[25]{\prod_{i=1}^{12} x_i^{R_i}} = \sqrt[25]{2.3337 \times 10^{55}} = 163.95$

Harmonic mean $\bar{x}_H = \frac{n}{\sum_{i=1}^{12} \frac{R_i}{x_i}} = \frac{25}{0.1526} = 163.8270$

The comparison $\bar{x}_H \le \bar{x}_G \le \bar{x}$ is verified. In fact, $\bar{x}_H = 163.8270 \le \bar{x}_G = 163.95 \le \bar{x} = 164.08$.

Median

Suppose that the n observations $x_1, x_2, \ldots, x_n$ have been sorted according to size as $x_{(1)}, x_{(2)}, \ldots, x_{(n)}$. We define the median denoted by x_{median} or x, as the middle observation or the arithmetic mean of the two middle observations. There are as many data values below the median as above it. If there is an odd number of values in an array, then the median is the middle value of the array; if there is an even number of values in an array, then the median is the arithmetic mean of the two middle values. More precisely,

$$x = \begin{cases} \dfrac{x_{n+1}}{2} & \text{if } n \text{ is odd} \\ \dfrac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2} & \text{if } n \text{ is even} \end{cases}$$

Quartiles

The median is one of many possible quartiles that can be calculated from a data set organized into ascending array. Some of them are quartiles, deciles and percentiles. There are three quartiles: first quartile (Q_1), second quartile (Q_2) and third quartile (Q_3). They divide arrays into four equal parts.

The first quartile Q_1 is given by

$$Q_1 = \frac{x_{\left[\frac{n}{4}\right]} + x_{\left[\frac{n}{4}\right]+1}}{2}$$

where $\left[\frac{n}{4}\right]$ is the integer part of $\frac{n}{4}$.

The third quartile Q_3 is given by

$$Q_3 = \frac{x_{\left[\frac{3n}{4}\right]} + x_{\left[\frac{3n}{4}\right]+1}}{2}$$

where $\left[\frac{3n}{4}\right]$ is the integer part of $\frac{3n}{4}$.

The interquartile range Δ is the difference between the third and first quartiles, and is thus given by

$$\Delta = Q_3 - Q_1$$

The second quartile is equal to the median, this is

$$Q_2 = x$$

Mode

The mode of a set of numbers is that value which occurs with the greatest frequency; that is, it is the most common value. The mode may not exist, and even if it does exist it may not be unique.

Example 3.2

Consider the height measurements of 25 students given in ascending order are

$$\begin{array}{cccccccccccccc} 152 & 154 & 155 & 159 & 159 & 160 & 160 & 160 & 160 & 160 & 160 & 161 & 161 \\ 162 & 167 & 170 & 170 & 170 & 170 & 171 & 172 & 172 & 172 & 172 & 173. \end{array}$$

- The median is

$$x = x_{\frac{n+1}{2}} = x_{\frac{25+1}{2}} = x_{13} = 161.$$

- The mode is

$$x_{mode} = 160.$$

- The first quartile is

$$Q_1 = \frac{x_{\left[\frac{n}{4}\right]} + x_{\left[\frac{n}{4}\right]+1}}{2} = \frac{x_{\left[\frac{25}{4}\right]} + x_{\left[\frac{25}{4}\right]+1}}{2} = \frac{x_6 + x_{6+1}}{2} = \frac{x_6 + x_7}{2} = \frac{160+160}{2} = 160.$$

- The second quartile is

$$Q_2 = x = 161.$$

- The third quartile is

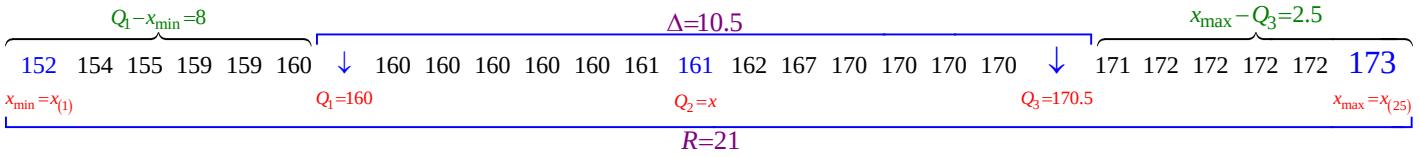
$$Q_3 = \frac{x_{\left[\frac{3n}{4}\right]} + x_{\left[\frac{3n}{4}\right]+1}}{2} = \frac{x_{\left[\frac{75}{4}\right]} + x_{\left[\frac{75}{4}\right]+1}}{2} = \frac{x_{19} + x_{19+1}}{2} = \frac{x_{19} + x_{20}}{2} = \frac{170+171}{2} = 170.5.$$

- The interquartile range is

$$\Delta = Q_3 - Q_1 = 170.5 - 160 = 10.5$$

- The minimum and maximum values are respectively

$$x_{\min} = x_{(1)} = 152 \text{ and } x_{\max} = x_{(25)} = 173.$$



3.1.2. Measures of Dispersion

The degree to which numerical data tend to spread about an average value is called the *dispersion*, or *variation*, of the data. The most common measures of the dispersion are the range, mean deviation, and standard deviation.

Range

The *range* is the largest value in a data set minus the smallest value.

$$R = x_{\max} - x_{\min} = x_{(n)} - x_{(1)}$$

The range has the drawback of involving only two values and does not involve the remaining numbers in the data set.

Mean Deviation

The mean deviation (or average deviation) of a set of n observations $x_1,\ldots,x_n$ is defined as

$$x_{MD}=\frac{\sum_{i=1}^{n}\left|x_i-\bar{x}\right|}{n}=\frac{\sum_{i=1}^{k}R_i\left|x_i-\bar{x}\right|}{n}=\sum_{i=1}^{k}f_i\left|x_i-\bar{x}\right|$$

Standard Deviation

The standard deviation of a set of n numbers $x_1,\ldots,x_n$, denoted by s , is the root mean square of the deviations from the arithmetic mean. More precisely,

$$s=\sqrt{\frac{\sum_{i=1}^{n}\left(x_i-\bar{x}\right)^2}{n}}=\sqrt{\frac{\sum_{i=1}^{k}R_i\left(x_i-\bar{x}\right)^2}{n}}=\sqrt{\sum_{i=1}^{k}f_i\left(x_i-\bar{x}\right)^2}$$

The variance of a set of data is defined as the square of the standard deviation and is thus given by s^2.

Coefficient of Variation

The coefficient of variation (also called the coefficient of variability, the coefficient of dispersion, or the relative standard deviation) is defined by

$$CV=\frac{s}{\bar{x}}\quad\text{or}\quad CV=\frac{s}{\bar{x}}\times 100$$

Example 3.5

Consider the height measurements of 25 students given in table 3.1. Calculate the mean deviation, the standard deviation and the coefficient of variation.

Solution

The mean deviation is $x_{MD}=\dfrac{\sum_{i=1}^{12}R_i\left|x_i-164.08\right|}{25}=\dfrac{148.24}{25}\cong 5.93$.

The standard deviation is $s=\sqrt{\dfrac{\sum_{i=1}^{12}R_i\left(x_i-164.08\right)^2}{25}}=\sqrt{\dfrac{1031.84}{25}}\cong 6.42$.

The coefficient of variation is $CV=\dfrac{s}{\bar{x}}\times 100=\dfrac{6.42}{164.08}\times 100=3.91\%$.

Table 3.2. Shows basic calculations to be done.

Table 3.2: Basic calculations

i	x_i	R_i	$x_i - \bar{x}$	$R_i \lvert x_i - \bar{x} \rvert$	$R_i \left(x_i - \bar{x} \right)^2$
1	152	1	-12.08	12.08	145.93
2	154	1	-10.08	10.08	101.61
3	155	1	-9.08	9.08	82.45
4	159	2	-5.08	10.16	51.61
5	160	6	-4.08	24.48	99.88
6	161	2	-3.08	6.16	18.97
7	162	1	-2.08	2.08	4.33
8	167	1	2.92	2.92	8.53
9	170	4	5.92	23.68	140.19
10	171	1	6.92	6.92	47.89
11	172	4	7.92	31.68	250.91
12	173	1	8.92	8.92	79.57
				148.24	**1031.84**

3.1.3. Measures of the shape

Skewness

This is a concept which is commonly used in statistical decision making. It refers to the degree in which a given frequency curve is deviating away from the normal distribution

There are 2 types of skewness namely:

i. Positive skewness
ii. Negative skewness

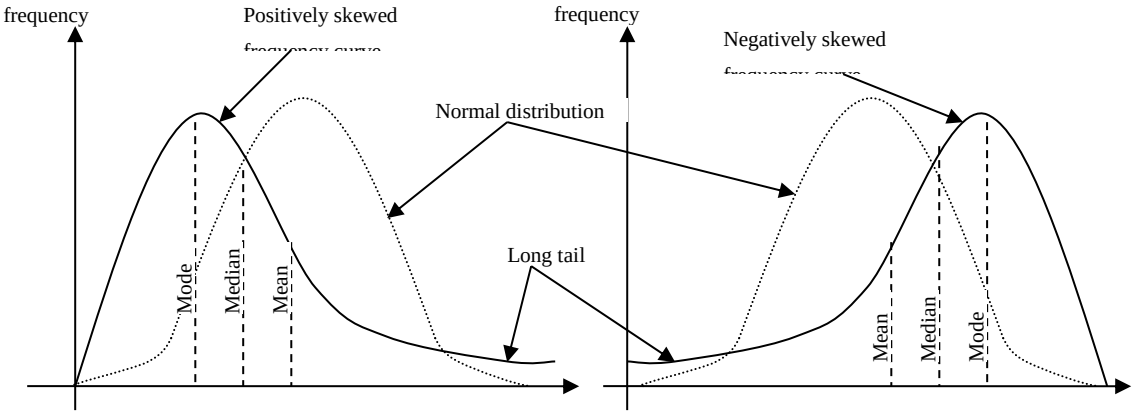
1. Positive Skewness

This is the tendency of a given frequency curve leaning towards the left. In a positively skewed distribution, the long tail extended to the right.

In this distribution one should note the following

i. The mean is usually bigger than the mode and median
ii. The median always occurs between the mode and mean
iii. There are more observations below the mean than above the mean

This frequency distribution as represented in the skewed distribution curve is characteristic of the age distributions in the developing countries



2. Negative Skewness

This is an asymmetrical curve in which the long tail extends to the left

NB: This frequency curve for the age distribution is characteristic of the age distribution in developed countries

- The mode is usually bigger than the mean and median
- The median usually occurs in between the mean and mode
- The no. of observations above the mean are usually more than those below the mean (see the shaded region)

Measures of skewness

1. Coefficient Skewness = $3 \times \frac{(\text{mean - median})}{\text{Standard deviation}}$
2. Coefficient of skewness = $\frac{\text{mean - mode}}{\text{Standard deviation}}$

NB: These 2 coefficients above are also known as Pearsonian measures of skewness.

3. Quartile Coefficient of skewness = $\frac{Q_3 + Q_1 - 2Q_2}{Q_3 + Q_1}$

Where Q_1 = 1st quartile

Q_2 = 2nd quartile

Q_3 = 3rd quartile

Example

The following information was obtained from an NGO which was giving small loans to some small scale business enterprises in 1998. The loans are in the form of thousands of RWf.

Loans	46-50	51-55	56-60	61-65	66-70	71-75	76-80	81-85	86-90	91-95	**Total**
Units	32	62	97	120	92	83	52	40	21	11	**610**

Required

Using the Pearsonian measure of skewness, calculate the coefficients of skewness and hence comment briefly on the nature of the distribution of the loans.

Arithmetic mean = 66.51

The standard deviation = 10.68

The Median = 65.27

Therefore the Pearsonian coefficient = $3\frac{(66.51 - 65.27)}{10.68} = 0.348$

Comment

The coefficient of skewness obtained suggests that the frequency distribution of the loans given was positively skewed .

This is because the coefficient itself is positive. But the skewness is not very high implying the degree of deviation of the frequency distribution from the normal distribution is small

Example 2

Using the above data calculate the quartile coefficient of skewness

Quartile coefficient of skewness = $\frac{Q_3 + Q_1 - 2Q_2}{Q_3 + Q_1}$

$Q1 = 55.5 + \frac{(152.75 - 94)}{97} \times 5 = 58.53$

$Q_3 = 70.55 + \frac{(458.25 - 403)}{83} \times 5 = 73.83 \times 5$

$Q_2 = 60.5 + \frac{(305.5 - 191)}{120} \times 5 = 65.27$

The required coefficient of skewness = $\frac{73.83 + 58.53 - 2(65.27)}{73.83 + 58.53} = 0.013$

Kurtosis

This is a concept, which refers to the degree of peakedness of a given frequency distribution. The degree is normally measured with reference to normal distribution.

The concept of kurtosis is very useful in decision making processes i.e. if is a frequency distribution happens to have either a higher peak or a lower peak, then it should not be used to make statistical inferences.

Generally there are 3 types of kurtosis namely:

a) A frequency distribution which is lepkurtic has generally a higher peak than that of the normal distribution. The coefficient of kurtosis when determined will be found to be more than 3. Thus frequency distributions with a value of more than 3 are definitely **leptokurtic**

b) Some frequency distributions when plotted may produce a curve similar to that of the normal distribution. Such frequency distributions are referred to as **mesokurtic**. The degree of kurtosis is usually equal to 3

c) When the frequency curve contacted produces a peak which is lower that that of a normal distribution when such a curve is said to be **platykurtic**. The coefficient of such is usually less than 3

It is necessary to calculate the numerical measure of kurtosis. The commonly used measure of kurtosis is the percentile coefficient of kurtosis. This coefficient is normally determined using the following equation

Percentile measure of kurtosis, K (Kappa) = $\frac{1}{2}\frac{(Q3 - Q1)}{P90 - P10}$

Example

Refer to the table above for loans to small business firms/units

Required

Calculate the percentile coefficient of Kurtosis

$P90 = \frac{90}{100}(n+1) = 0.9(610+1)$

$= 0.9\ (611)$

$= 549.9$

The actual loan for a firm in this position

$(549.9) = 80.5 + \frac{(549.9 - 538)}{40} \times 5 = 81.99$

$P10 = \frac{10}{100}\ (n + 1) = 0.1\ (611) = 61.1$

The actual loan value given to the firm on this position is

$50.5 + \frac{(61.1 - 32)}{62} \times 5 = 52.85$

$= 0.9\ (611)$

$= 549.9$

$\therefore$ Percentile measure of kurtosis K (Kappa) $= \frac{1}{2}\frac{(Q3 - Q1)}{P90 - P10} = \frac{1}{2}\frac{(73.83 - 58.53)}{81.99 - 52.85} = 0.26$

Since 0.26 < 3, it can be concluded that the frequency distribution exhibited by the distribution of loans is platykurtic.

3.2. Probability: Basic concepts and definition of probability, Conditional probability.

3.2.1. Basic concepts and definition of probability

Random experiment

In the study of probability, any process of observation is referred to as an *experiment*. The results of an observation are called the *outcomes* of the experiment.

When different outcomes are obtained in repeated trials, the experiment is called a *random experiment*. More precisely, an experiment is called a random experiment if its outcome cannot be predicted.

Sample space

The set of all possible outcomes of a random experiment is called the sample space (or universal set), and it is denoted by Ω. An element in Ω is called a sample point. Each outcome of a random experiment corresponds to a sample point.

Example

If we toss a coin, there are two possible outcomes, heads (H) or tails (T).

Thus the sample space of the experiment of tossing a coin is

$\Omega = \{H, T\}$.

Example

If we toss a coin twice, the sample space of this experiment is

$\Omega = \{HH, HT, TH, TT\}$.

Example

If we toss a die, the set of all possible outcomes is given by

$\Omega = \{1, 2, 3, 4, 5, 6\}$.

Example

If an experiment consists of measuring "lifetimes" of electric light bulbs produced by a company, then the result of the experiment is a time t in hours that lies in some interval $\Omega = \{t : 0 \le t \le 4000\}$ where we assume that no bulb lasts more than 4000 hours.

Events

An event is any subset of the sample space Ω. A sample point of Ω is often referred to as a *simple* or *elementary event*. The sample space Ω is the subset of itself, that is $\Omega \subset \Omega$. Since Ω is the set of all possible outcomes, it is often called the *sure* or *certain event*. The empty set $\varnothing$, which is called the *impossible event* because an element of $\varnothing$ cannot occur.

Example

If we toss a coin twice, the event that only one head comes up is the subset $A = \{HT, TH\}$ of the sample space $\Omega = \{HH, HT, TH, TT\}$.

If the sets corresponding to events A and B are disjoint, i.e., $A \cap B = \varnothing$, the events are said to be *mutually exclusive*. This means that they cannot both occur. We say that a collection of events $A_1, \ldots, A_n$ is mutually exclusive if every pair in the collection is mutually exclusive.

Set operations on events

Let A and B be the events in the sample space Ω. Then

- $A \cup B$ is the event "either A or B or both". $A \cup B$ is called the *union* of A and B.
- $A \cap B$ is the event "both A and B". $A \cap B$ is called the *intersection* of A and B.
- $\overline{A}$ is the event "not A". $\overline{A}$ is called the *complement* of A.
- $A - B = A \cap \overline{B}$ is the event "A but not B". In particular, $\overline{A} = \Omega - A$.

Example

Referring to the experiment of tossing a coin twice, let A be the event "at least one head occurs" and B the event "the second toss results is a tail".
Then $A = \{HT, TH, HH\}$, $B = \{HT, TT\}$, and so we have

$A \cup B = \{HT, TH, HH, TT\}$, $A \cap B = \{HT\}$, $\overline{A} = \{TT\}$, $A - B = \{TH, HH\}$

The Concept of Probability

In any random experiment there is always uncertainty as to whether a particular event will or will not occur. As a measure of the *chance*, or *probability*, with which we can expect the event to

occur, it is convenient to assign a number between 0 and 1. If we are sure or certain that an event will occur, we say that its probability is 100% or 1. If we are sure that the event will not occur, we say that its probability is zero.

If, for example, the probability is $\frac{1}{4}$, we would say that there is a 25% chance it will occur and a 75% chance that it will not occur.

There are two important procedures by means of which we can estimate the probability of an event.

Classical approach: If an event A can occur in h different ways out of a total of n possible ways, all of which are equally likely, then the probability of the event is

$P(A) = \frac{h}{n}$.

Example

Suppose we want to know the probability that a head will turn up in a single toss of a coin. Since there are two equally likely ways in which the coin can come up (heads and tails) assuming it does not roll away or stand on its edge, and of these two ways a head can arise in only one way, we reason that the required probability is $\frac{1}{2}$.

Frequency approach: If after n repetitions of an experiment, where n is very large, an event A is observed to occur in h of these, then the probability of the event is

$P(A) = \frac{h}{n}$.

This is also called the *empirical probability* of the event A.

Example

If we toss a coin 1000 times and find that it comes up heads 532 times, we estimate the probability of a head coming up to be $\frac{532}{1000} = 0.532$.

The Axioms of Probability

Suppose we have a sample space Ω. To each event A in the class C of events, we associate a real number $P(A)$. Then P is called a *probability function*, and $P(A)$ the *probability* of the event A, if the following axioms are satisfied.

Axioms

For every event A in the class C,

$$P(A) \geq 0$$

For the sure or certain event Ω in the class C,

$$P(\Omega) = 1$$

For any number of mutually exclusive events $A_1, A_2, \ldots$ in the class C,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

In particular, for two mutually exclusive events A_1 and A_2,

$$P(A_1 \cup A_2) = P(A_1) + P(A_2)$$

Some Important Theorems on Probability

Theorem 3.1

If $A_1 \subset A_2$, then

$$P(A_1) \leq P(A_2) \text{ and } P(A_2 - A_1) = P(A_2) - P(A_1)$$

Proof

If $A_1 \subset A_2$, then $A_2 = A_1 \cup (\overline{A}_1 \cap A_2)$ and $P(A_2) = P(A_1) + P(\overline{A}_1 \cap A_2) \geq P(A_1)$, since

$P(\overline{A}_1 \cap A_2) = P(A_2 - A_1)$.

Theorem 3.2

For every event A,

$$0 \leq P(A) \leq 1$$

i.e., a probability is between 0 and 1.

Theorem 3.3

For $\varnothing$, the empty set,

$$P(\varnothing) = 0$$

i.e., the impossible event has probability zero.

Proof

Note that $\Omega = \Omega \cup \varnothing$ and Ω and $\varnothing$ are mutually exclusive. Then $P(\Omega) = P(\Omega) + P(\varnothing)$, and $P(\varnothing) = 0$ since $P(\Omega) = 1$.

Theorem 3.4

If $\overline{A}$ is the complement of A, then,

$$P(\overline{A}) = 1 - P(A)$$

Proof

Note that $\Omega = A \cup \overline{A}$ and A and $\overline{A}$ are mutually exclusive. Then $P(\Omega) = P(A) + P(\overline{A})$, and $P(\overline{A}) = 1 - P(A)$ since $P(\Omega) = 1$.

Theorem 3.5

If $A = A_1 \cup A_2 \cup \ldots \cup A_n$, where $A_1, A_2, \ldots, A_n$ are mutually exclusive event, then,

$$P(A) = P(A_1) + P(A_2) + \ldots + P(A_n)$$

In particular, if $A = \Omega$, then

$$P(A) = P(A_1) + P(A_2) + \ldots + P(A_n) = 1$$

Theorem 2.6

If A and B are two events, then,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof

Since $A \cup B = A \cup (B \cap \overline{A})$, where A and $B \cap \overline{A}$ are mutually exclusive, and

$B=(A\cap B)\cup\left(B\cap\overline{A}\right)$, where $A\cap B$ and $B\cap\overline{A}$ are mutually exclusive, then

$P(A\cup B)=P(A)+P\left(B\cap\overline{A}\right)$ and $P(B)=P(A\cap B)+P\left(B\cap\overline{A}\right)$. Substracting,

$P(A\cup B)-P(B)=P(A)-P(A\cap B)$, and thus $P(A\cup B)=P(A)+P(B)-P(A\cap B)$.

More generally, if A_1, A_2, A_3 are any three events, then

$$P(A_1\cup A_2\cup A_3)=P(A_1)+P(A_2)+P(A_3)-P(A_1\cap A_2)-P(A_1\cap A_3)$$
$$-P(A_2\cap A_3)+P(A_1\cap A_2\cap A_3)$$

Generalisations to n events can also be made:

If $A_1, A_2, \cdots, A_n$ are n arbitrary events in Ω, then

$$P\left(\bigcup_{i=1}^{n}A_i\right)=\sum_{i=1}^{n}P(A_i)-\sum_{i\neq j}P(A_i\cap A_j)+\sum_{i\neq j\neq k}(A_i\cap A_j\cap A_k)+$$
$$-\cdots+(-1)^{n-1}P(A_1\cap A_2\cap\cdots\cap A_n)$$

Theorem 3.7

For any events A and B,

$$P(A)=P(A\cap B)+P\left(A\cap\overline{B}\right)$$

Theorem 3.8

If an event must result in the occurrence of one of the mutually exclusive events $A_1, A_2, \cdots, A_n$, then,

$$P(A)=P(A\cap A_1)+P(A\cap A_2)+\cdots+P(A\cap A_n)$$

Example

A single die is tossed once. The sample space is $\Omega=\{1,2,3,4,5,6\}$.

If we assign the equal probability to the sample points, i.e., if we assume that the die is fair, then

$P(1)=P(2)=\cdots=P(6)=\frac{1}{6}$.

The event that either a 2 or 5 turns up is indicated by $A=2\cup5$. Therefore, the probability of a 2 or 5 turning up is

$$P(A) = P(2 \cup 5) = P(2) + P(5) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

Example

Suppose events A and B are not mutually exclusive and we know that $P(A) = 0.20$, $P(B) = 0.30$ and $P(A \cap B) = 0.10$. Then

- $P(\overline{A}) = 1 - P(A) = 0.80$
- $P(\overline{B}) = 1 - P(B) = 0.70$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.20 + 0.30 - 0.10 = 0.40$

$$P(\overline{A} \cap \overline{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - \left[P(A) + P(B) - P(A \cap B)\right] = 1 - 0.40 = 0.60$$

3.2.2. Conditional probability

Let A and B be two events such that $P(A) > 0$. Denote by $P(B|A)$ the probability of B given that A has occurred. Since A is known to have occurred, it becomes the new sample space replacing the original Ω. From this we are led to the definition

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

or

$$P(A \cap B) = P(A) \times P(B|A)$$

where $P(A \cap B)$ is the joint probability of A and B.

Similarly, for $P(B) > 0$, the probability of A given that B has occurred is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

or

$$P(A \cap B) = P(B) \times P(A|B)$$

From above equations, we have

$$P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$$

Bayes´ rule

$$P(A|B)=\frac{P(A)\times P(B|A)}{P(B)}$$

Example

Find the probability that a single toss of a die will result in a number less than 4 if

(a) no other information is given and

(b) it is given that the toss resulted in an odd number.

Solution

(a) Let B denote the event {less than 4}, i.e., $B=\{1,2,3\}$. Since B *is* the union of the events 1, 2, or 3 turning up, we see by **Theorem 2. 5** that

$$P(B)=P(1\cup 2\cup 3)=P(1)+P(2)+P(3)=\frac{1}{6}+\frac{1}{6}+\frac{1}{6}=\frac{1}{2}$$

We assume equal probabilities for the sample points.

(b) Letting A be the event {odd number}, i.e. $A=\{1,3,5\}$. We see that

$$P(A)=P(1\cup 3\cup 5)=P(1)+P(3)+P(5)=\frac{1}{6}+\frac{1}{6}+\frac{1}{6}=\frac{1}{2}.$$

As $A\cap B=\{1,3\}$, we see that

$$P(A\cap B)=\frac{2}{6}=\frac{1}{3}.$$

Then $P(B|A)=\dfrac{P(A\cap B)}{P(A)}=\dfrac{1/3}{1/2}=\dfrac{2}{3}$. Hence, the added knowledge that the toss results in an odd number raises the probability from $\frac{1}{2}$ to $\frac{2}{3}$.

Theorems on Conditional Probability

Theorem3.9

For any three events A_1, A_2, A_3 we have

$$P(A_1\cap A_2\cap A_3)=P(A_1)\times P(A_2|A_1)\times P(A_3|A_1\cap A_2)$$

The result above is easily generalized to n events.

Theorem 3.10

If an event A must result in one of the mutually exclusive events $A_1, A_2, \ldots, A_n$, then

$$P(A) = \sum_{i=1}^{n} P(A_i) \times P(A|A_i)$$

Total Probability

- **Definition (Mutually exclusive and exhaustive events)**

The events $A_1, A_2, \ldots, A_n$ are mutually exclusive and exhaustive if

$$\bigcup_{i=1}^{n} A_i = \Omega \text{ and } A_i \cap A_j = \varnothing, \forall i \neq j$$

- **Total Probability**

Let B be any event in Ω. Then

$$P(B) = \sum_{i=1}^{n} P(B \cap A_i) = \sum_{i=1}^{n} P(A_i) \times P(B|A_i)$$

Eq. (2.25) is known as the total probability of event B.

Proof

Let us consider the Venn diagram of **Fig. 3.1**.

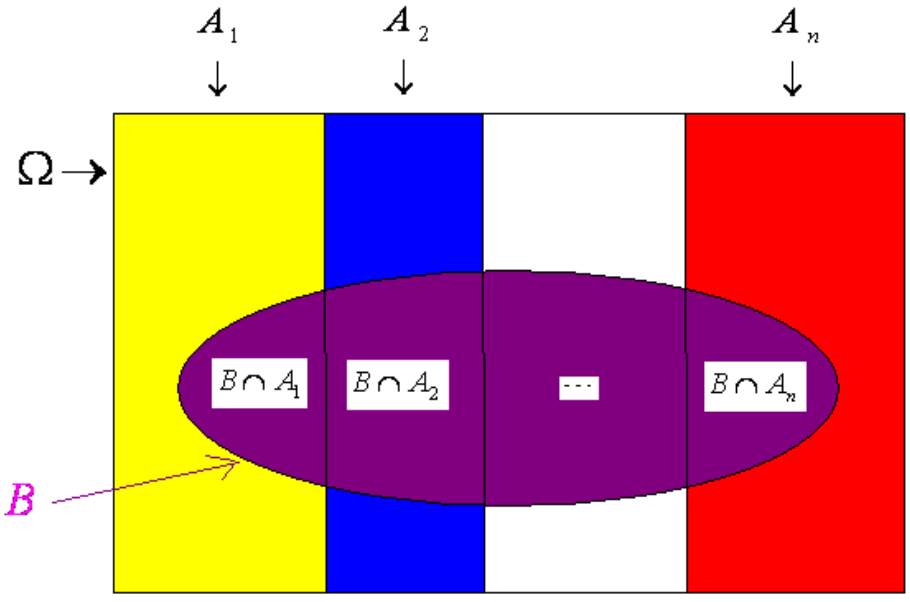


Fig. 3.1: Total Probability

Since $B = B \cap \Omega$, we have

$$B = B \cap \Omega = B \cap (A_1 \cup \ldots \cup A_n)$$
$$= (B \cap A_1) \cup \cdots \cup (B \cap A_n).$$

The events $(B \cap A_1), i = 1, 2, \cdots, n$ are mutually exclusive. We obtain

$$P(B) = P(B \cap \Omega) = P\left[(B \cap A_1) \cup \cdots \cup (B \cap A_n)\right]$$
$$= \sum_{i=1}^{n} P(B \cap A_i) = \sum_{i=1}^{n} P(A_i) \times P(B|A_i).$$

Bayes´ Theorem

Let $A_1, A_2, \ldots, A_n$ be mutually exclusive and exhaustive events. Then if B is any event in Ω, we obtain

$$P(A_i|B) = \frac{P(A_i) \times P(B|A_i)}{\sum_{j=1}^{n} (A_j) \times P(B|A_j)}$$

Example

Three facilities supply microprocessors to a manufacturer of elementary equipment. All are supposedly made to the same specifications. However, the manufacturer has for several years tested the microprocessors, and records indicate the following information (Table 3.1).

Table 3.1: Test results

Supply Facility	Fraction Defective	Fraction Supplied
1	0.02	0.15
2	0.01	0.80
3	0.02	0.05

The manufacturer has stopped testing because of the costs involved, and it may be reasonably assumed that the fractions that are defective and the inventory mix are the same as during the period of record keeping. The director of manufacturing randomly selects a microprocessor, takes it to the test department, and finds that it is defective. If we let A be the event that an item is defective, and B_i be the event that the item came from facility i ($i = 1, 2, 3$), then we can evaluate $P(B_i|A)$. Suppose, for instance, that we are interested in determining $P(B_3|A)$. Then

$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{P(B_1)P(A|B_1)+P(B_2)P(A|B_2)+P(B_3)P(A|B_3)}$$

$$= \frac{0.05\times 0.03}{0.15\times 0.02+0.80\times 0.01+0.05\times 0.03} = \frac{3}{25}.$$

Independents Events

Two events A and B are said to be independent if and only if

$$P(A\cap B) = P(A)\times P(B)$$

It follows immediately that if A and B are independent, then

$P(A|B) = P(A)$ and $P(B|A) = P(B)$

If two events A and B are independent, then it can be shown that A and $\overline{B}$ are also independent, that is

$$P(A\cap \overline{B}) = P(A)\times P(\overline{B})$$

Then $$P(A|\overline{B}) = \frac{P(A\cap \overline{B})}{P(\overline{B})} = P(A)$$

Three events A, B, C are independent if and only if

1. $P(A\cap B\cap C) = P(A)\times P(B)\times P(C)$
2. $P(A\cap B) = P(A)\times P(B)$
3. $P(A\cap C) = P(A)\times P(C)$
4. $P(B\cap C) = P(B)\times P(C)$

To distinguish between the mutual exclusiveness (or disjointness) and independence of a collection of events we summarize as follows:

- If $\{A_i, i = 1,2,\cdots,n\}$ is a sequence of mutually exclusive events, then

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i)$$

- If $\{A_i, i = 1,2,\cdots,n\}$ is a sequence of independent events, then

$$P\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} P(A_i)$$

3.3. Probability distributions including Discrete distributions e.g. binomial and Poisson distributions and Continuous distribution e.g. Normal Distribution.

a) Introduction

Random variable: the measurements of a random variable vary in a seemingly random and unpredictable manner. A random variable assumes a unique numerical value for each of the outcomes in the sample space of the probability experiment.

If we toss a coin twice, the number of Heads obtained could be 0, 1 or 2. The probabilities of these occurrences are:

P (No heads) =P (TT) = (1/2)* (1/2) =1/4

P (1 head) =P (TH) +P (HT) = (0.5*0.5) + (0.5*0.5) =0.5

P (2 heads) =P (HH) =0.5*0.5=0.25

The probability distribution is often written as follow:

x	0	1	2
P(X=x)	0.25	0.5	0.25

Random Variable: a variable whose values are determined by chance.

A random variable X is a function defined as follow:

$$X : \Omega \to X(\Omega) = IR = \{x_1, x_2, ..., x_n\}$$

$$\underbrace{\omega}_{Event} \to \underbrace{X(\omega)}_{Real\ number}$$

b) Discrete random variables

(i) Definition

X can take only values $x_1, x_2, ..., x_n$ .The probability associated with these values are

$P_1, P_2, ..., P_n$

Where

$P(X = x_1) = P_1$
$P(X = x_2) = P_2$
.....
.....
$P(X = x_n) = P_n$

Then X is a **discrete random variable** if $\sum_{i=1}^{n} P_i = 1$ *or* $\sum_{all\ x} P(X = x) = 1$

We always denote a random variable by a capital letter X,Y,Z….and the particular values by lower case letter x,y,z….

(ii) Probability density function (pdf)

This is the function that is responsible for allocating probabilities. It is written as

$f(x_i) = P(X = x_i)$ *with* $1 \le i \le n$.

x_1	$f(x_1) = P_1$
x_2	$f(x_2) = P_2$
x_3	$f(x_3) = P_3$
……	……..
x_n	$f(x_n) = P_n$

$0 \le f(x_i) \le 1$ *and* $\sum_{i=1}^{n} f(x_i) = 1$

Example: The following table gives the pdf of X

i	1	2	3	4
x_i	-3	-2	1	4
$P(X = x_i)$	0.1	0.2	0.3	0.4

(iii) Characteristics of a random variable.

- **Expectation:** $\underbrace{E(X)}_{Expected\ value\ or\ Mean}$

Let X be a r.v with pdf f(x)=P(X=x) and x is discrete. The expectation of X written $E(X)$ is given by

$$E(X)=\sum x\,P(X=x)\ or\ E(X)=\sum_{i=1}^{n} x_i\,P_i\ or\ E(X)=\sum_{i=1}^{n} x_i\,f(x_i)$$

It is the average of the numbers giving the gains and the losses weighted by their probabilities.
The game permits to expect an average gain or an average loss (by part) of E(X).
One says that the game is favourable if E(X) is positive and it is not favourable if E(X) is negative.
If E(X) =0, the game is balanced.

Theorems:

1. Let consider an r.v X and a real number k:

$$E(kX)=k\,E(X)$$
$$E(X+k)=E(X)+k$$

2. Let consider X and Y, two r.v:

$$E(X+Y)=E(X)+E(Y)$$

Example: A random variable has pdf as shown. Find E(X).

X	-2	-1	0	1	2
P(X=x)	0.3	0.1	0.15	0.4	0.05

Solution

$$E(X)=\sum_{all\ x} x\,P(X=x)=(-2*0.3)+(-1*0.1)+(0*0.15)+(1*0.4)+(2*0.05)=-0.2$$

- **The expectation of any function of X, E[g(x)]**

Let X be a r.v and pdf f(x) =P(X=x). Let also g(x) be a function of the r.v. X. Then , the expected value of g(x) written E[g(x)] is given by: $E[g(x)]=\sum_{all\ x} g(x)P(X=x)$

Example:

The r.v X has pdf P(X=x) for x=1, 2, 3.

X	1	2	3

P(X=x)	0.1	0.6	0.3

Compute

a) E(3) b) E(X) c) E(5X) d) E(5X+3)

Solution

a) $E(X)=\sum_{all\ x} x\,P(X=x)$

But since x=3, we have

$$E(x)=\sum_{all\ x}3P(X=x)=3\sum_{all\ x}P(X=x)=3*(0.1+0.6+0.3)=3*1=3$$

b) $E(X)=\sum_{all\ x} x\,P(X=x)=(1*0.1)+(2*0.6)+(3*0.3)=2.2$

c) $E(5X)=\sum_{all\ x}5x\,P(X=x)=(5*1*0.1)+(5*2*0.6)+(3*5*0.3)=11$

d) $E(5X+3)=\sum_{all\ x}(5x+3)\,P(X=x)=8*0.1+13*0.6+18*0.3=14$

- **Variance (Var X) and standard deviation** σ_X

Let X be a r.v having the pdf $f(x)=P(X=x), x\in IR$ and the $E(X)=\mu$ where μ is constant. The variance of X, written Var (X) is given by: $Var\,X=E\left[(X-\mu)^2\right]$.Alternatively,

$$
\begin{aligned}
Var\ X &= E\left[(X-\mu)^2\right] \\
&= E\left(X^2 - 2\mu X + \mu^2\right) \\
&= E\left(X^2\right) - 2\mu * E(X) + E\left(\mu^2\right) \\
&= E\left(X^2\right) - 2 * \mu * \mu + \mu^2 \\
&= E\left(X^2\right) - 2\mu^2 + \mu^2 \\
&= E\left(X^2\right) - \mu^2 \\
Var\ X &= E\left(X^2\right) - \left[E(X)\right]^2 \\
&= \sum_{i=1}^{n} x_i^2 * P_i - E^2(X) \\
\sigma_X &= \sqrt{Var\ X}
\end{aligned}
$$

The variance is defined as the average of the sum of squared deviationsfrom different values of gains (losses) around the expected value (=mean) of gain (or loss).

The standard deviation is a parameter that indicates the spread of the gains or the losses.

Properties

$Var(k) = 0$

$Var(kX) = k^2 Var(X)$

$Var(X + k) = Var(X)$

$Var(kX + +b) = k^2 Var(X)$

$\sigma(kX) = \sigma_{kX} = |k| * \sigma_X$

Example:

An r.v X has the following distribution:

x	1	2	3	4	5
P(X=x)	0.1	0.3	0.2	0.3	0.1

Find E(X), Var X and σ_X

Solution

a) $E(X) = \sum_{all\ x} xP(X = x) = (1*0.1) + (2*0.3) + (3*0.2) + (4*0.3) + (5*0.1) = 3$

b) $Var\,X = E\left(X^2\right) - \left[E(X)\right]^2$

$E\left(X^2\right) = \sum_{all\ x} x^2 P(X = x) = (1*0.1) + (4*0.3) + (9*0.2) + (16*0.3) + (25*0.1) = 10.4$

$So\ that\ Var\ X = 10.4 - 3^2 = 1.4$

$\sigma_X = \sqrt{1.4} = 1.1832$

EXERCISE

FIND THE MEAN, VARIANCE, AND STANDARD DEVIATION FOR THE FOLLOWING PROBABILITY DISTRIBUTION.

x	1	2	3	4	5	6	**sum**
p(x)	1/6	1/6	1/6	1/6	1/6	1/6	**6/6=1**

Hint:

x	1	2	3	4	5	6	**sum**
p(x)	1/6	1/6	1/6	1/6	1/6	1/6	**6/6 = 1**
x p(x)	1/6	2/6	3/6	4/6	5/6	6/6	**21/6 = 3.5**
x^2 p(x)	1/6	4/6	9/6	16/6	25/6	36/6	**91/6 = 15.1667**

The joint probability distribution of X and Y.

X \ Y	y_1	y_2	….. y_j …..	y_m	Total
x_1	$h(x_1, y_1)$	$h(x_1, y_2)$	…….	$h(x_1, y_m)$	$f(x_1)$
x_2	$h(x_2, y_1)$	$h(x_2, y_2)$	……..	$h(x_2, y_m)$	$f(x_2)$
…. x_i …	…….	……	…….	……	$f(x_i)$
x_n	$h(x_n, y_1)$	$h(x_n, y_2)$	……..	$h(x_n, y_m)$	$f(x_n)$
Total	$g(y_1)$	$g(y_2)$	$g(y_j)$	$g(y_m)$	1

The functions $f(x_i)$ and $g(y_j)$ are called marginal pdf. of X and Y respectively.

Example: Let consider two independent r.v X and Y with the following corresponding pdf:

x_i	1	2
$f(x_i)$	0.3	0.7

y_j	2	3	4
$g(y_j)$	0.1	0.5	0.4

Then the joint probability distribution is stated as follow:

X \ Y	**2**	**3**	**4**	Total
1	0.3*0.1=**0.03**	0.3*0.5=**0.15**	0.3*0.4=**0.12**	**0.3**
2	0.7*0.1=**0.07**	0.7*0.5=**0.35**	0.7*0.4=**0.28**	**0.7**
Total	**0.1**	**0.5**	**0.4**	1

Note:

Two variables X and Y are independent iff:

1) $E(X,Y)=E(X)*E(Y)$
2) $Var(X+Y)=Var\,X+Var\,Y$
3) $Cov(X,Y)=0$

- **Covariance and correlation coefficient**

Let consider two r.v X and Y:

$Cov(X,Y)=E(X\cdot Y)-\mu_X*\mu_Y$ *where* $\mu_X=E(X)$ *and* $\mu_Y=E(Y)$

The Pearson correlation coefficient of X and Y is $r(X,Y)=\dfrac{Cov(X,Y)}{\sigma_X*\sigma_Y}$.

The Pearson correlation coefficient is a dimensionless quantity with following properties:

1) $r(X,Y)=r(Y,X)$
2) $r(X,X)=1$ *and* $r(X,-X)=-1$
3) $r(aX+b,cY+d)=r(X,Y)$ *with* $a,c\neq 0$

Exercises

1. Three girls Aileen, Barbara and Cathy pack biscuits in a factory. From the batch allotted to them Aileen packs 55%, Barbara 30% and Cathy 15%. The probability that Aileen breaks some biscuits in a pack is 0.7 and the respective probability for Barbara and Cathy are 0.2 and 0.1.

What is the probability that a packet with broken biscuits found by the checker was packed by Aileen.

2. Calculate the expected value , the variance and the standard deviation for the following pdf:

x_i	-5	-4	1	2
$f(x_i)$	$1/4$	$1/8$	$1/2$	$1/8$

3. A perfect coin is tossed until one obtains either Head or 5 Tails. Compute
$E(X)$, $Var(X)$ *and* σ_X
4. Let consider two pdf of X and Y:

X	1	2	3	4
P(X=x)	$1/4$	$1/4$	$1/4$	$1/4$

Y	0	1	2
P(Y=y)	$1/4$	$1/2$	$1/4$

a) Obtain the probability distribution of X and Y.
b) Find $E(X)$, $E(Y)$, $Var(X)$ *and* $Var(Y)$.
c) Obtain the probability distribution of X + Y.
d) Find $E(X+Y)$ *and* $Var(X+Y)$.

(iv) Special discrete probability distributions.

- **Binomial distribution**

Binomial experiment: an experiment with a fixed number of independent trials. Each trial can only have two outcomes, or outcomes which can be reduced to two outcomes. The probability of each outcome must remain constant from trial to trial.

Binomial distribution: the outcomes of a binomial experiment with their corresponding probabilities.

Let consider an experiment with n independent trials and only two possible outcomes. We call one of the outcomes successful outcome and the probability of its occurring is P(s) =p. The other outcome is called fail outcome and the probability of its occurring is P (f) =q=1-p.

The probability for getting exactly k successes in n trials is

$$P_k = B(k,\ n,\ p) = B(n,\ p) = \binom{n}{k} p^k q^{n-k} \ \ and\ k = 0,1,2,...,n.$$

Then the pdf of X is given by $f(x) = P(X = x) = \binom{n}{k} p^k q^{n-k} \ \ and\ k = 0,1,2,...,n$

We write $X \sim Bin(n,\ p)$ and read "X is a random variable following the binomial distribution with parameter n and p".

Example:

The probability that a person supports party A is 0.6.Find the probability that a randomly selected sample of 8 voters, there are:

a) Exactly 3 who support party A.
b) More than 5 who support party A

Solution

Consider "support of party A to be success". Then p=0.6 and q=1-0.6=0.4.

Let X be the r.v." the number of party A supporters"

$X \sim Bin(n,\ p)$

$X \sim Bin(8, 0.6)$ and its pdf is given by

$$f(x) = P(X = x) = \binom{n}{k} p^k q^{n-k} \quad and \ k = 0,1,2,...,n$$

$$f(x) = P(X = x) = \binom{8}{k} (0.6)^k (0.4)^{8-k} \quad and \ k = 0,1,2,...,8$$

Thus

a) k=3

P(X=3)= $\binom{8}{3} (0.6)^3 (0.4)^{8-3} = 0.124$

b) P(X>5)=P(X=6)+P(X=7)+P(X=8)

$$= \binom{8}{6} (0.6)^6 (0.4)^{8-6} + \binom{8}{7} (0.6)^7 (0.4)^{8-7} + \binom{8}{8} (0.6)^8 (0.4)^{8-8}$$

$$= \binom{8}{6} (0.6)^6 (0.4)^2 + \binom{8}{7} (0.6)^7 (0.4)^1 + \binom{8}{8} (0.6)^8 (0.4)^0$$

=0.315

Expectation value and variance.

If the r.v X is such that $X \sim Bin(n, p)$, then $E(X) = np \ and \ Var \ X = npq \ where \ q = 1 - p$

Example: If the probability that it will be a fine day is 0.4.Find the expected number of fine days in a week end and the standard deviation.

Solution

Let "fine day" be "success". Then, p=0.4 and q=0.6.

Let X be an r.v. "number of fine days in a week ".Then $X \sim Bin(n, p)$, where n=7 and p=0.4.

Hence $X \sim Bin(7, 0.4)$

a) $E(X) = np = 7 * (0.4) = 2.8$

b) $Var \ X = npq = 7 * (0.4) * (0.6) = 1.68$

$\Rightarrow \sigma = \sqrt{Var \ X} = \sqrt{1.68} = 1.2961$

Mode

Let the r.v. X be such that $X \ Bin(n,\ p)$. To compute the mode of the probability distribution of X, we only consider the values of X close to E(X).

Example:

If $X \sim Bin(10,\ 0.45)$, find the mode of the probability distribution of X.

Solution

Here X has pdf as follow: $f(x) = P(X = x) = \binom{n}{k} p^k q^{n-k} \ and \ k = 0,1,2,...,n$

So that $P(X = x) = \binom{10}{k} (0.45)^k (0.55)^{n-k} \ and \ k = 0,1,2,...,10$

But E(x) =np=10*0.45=4.5

Now $P(X = 3) = \binom{10}{3} (0.45)^3 (0.55)^{10-3} = 0.1664$

$P(X = 4) = \binom{10}{4} (0.45)^4 (0.55)^{10-4} = 0.2383$

$P(X = 5) = \binom{10}{5} (0.45)^5 (0.55)^{10-5} = 0.2340$

$P(X = 6) = \binom{10}{6} (0.45)^6 (0.55)^{10-6} = 0.1595$

Here X with the highest probability is 4. Therefore the mode of X is 4.

- **Poisson distribution**

Poisson distribution: a probability distribution used when a density of items is distributed over a period of time. The sample size needs to be large and the probability of success to be small.
Let X be a discrete r.v. It is said to follow the Poisson distribution if its pdf is of the form:

$P(X=x)=\dfrac{e^{-\lambda}\,\lambda^{x}}{x!}$ *for* $x=0,\,1,\,2,\ldots..$ *to* inf *inity where* λ *is the parameter of the distribution.*

We write : $X \sim \mathrm{P}_0(\lambda)$

Then the Poisson distribution is defined by

$p(k,\lambda)=p_0(\lambda)=\dfrac{\lambda^k e^{-\lambda}}{k!}$, $k=0,\,1,\,2,\ldots..$ *where* λ *is any cons* tan *t.* This distribution is an infinitely countable distribution.

Example If $X \sim \mathrm{P}_0(3.5)$, find:

a) $P(X=4)$
b) $P(X\le 2)$
c) $P(X>1)$

Solution

$P(X=x)=\dfrac{e^{-3.5}(3.5)^{x}}{x!}$, $x=0,\,1,\,2,\ldots..$

a) $P(X=4)=\dfrac{e^{-3.5}(3.5)^{4}}{4!}=0.1888\ldots..$

b)

$$P(X\le 2)=P(X=0)+P(X=1)+P(X=2)$$
$$=\frac{e^{-3.5}(3.5)^{0}}{0!}+\frac{e^{-3.5}(3.5)^{1}}{1!}+\frac{e^{-3.5}(3.5)^{2}}{2!}$$
$$=e^{-3.5}\left(1+3.5+\frac{(3.5)^{2}}{2!}\right)=0.3208$$

c)

$$P(X>1)=1-P(X\le 1)$$
$$=1-\left[P(X=0)+P(X=1)\right]$$
$$=1-\left(e^{-3.5}+e^{-3.5}(3.5)\right)=0.8641$$

Expectation value and variance If X is a discrete r.v such that $X \sim \mathrm{P}_0(\lambda)$, then

$E(X)=\lambda$ *and Var* $X=\lambda$

Example

If $X \sim \mathrm{P}_0(\lambda)$ with standard deviation 3. Find

a) E(X)
b) P(X<4)

Solution

a) Since standard deviation $\sigma = \sqrt{Var\ X}$, we have $\sqrt{Var\ X} = 3 \Rightarrow Var\ X = 9 = \lambda$
Consequently, since $X \sim \mathrm{P}_0(\lambda)$, $E(X) = Var\ X = 9$

Thus the pdf of X is given by $P(X = x) = \frac{e^{-9}\, 9^x}{x!}, x = 0,1,\ 2,\ .....$

b) $P(X < 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$

$= e^{-9} + e^{-9} * 9 + \frac{e^{-9} * 9^2}{2!} + \frac{e^{-9} * 9^3}{3!} = 0.02122...$

Uses of Poisson distribution

There are two main practical uses of Poisson distribution:

a) When we consider the distribution of random events

When an event is randomly scattered in time (or in space) and has mean number of occurrences λ in a given interval and if X is a random variable "the number of occurrences in a given interval, then $X \sim \mathrm{P}_0(\lambda)$.

Examples:-Car accidents on a particular street of road in one day

-Accidents in a factory per week

-Telephone calls made to a switchboard in a given minute.

Exercise

The mean number of bacteria per millimetre of a liquid is known to be 4. Assuming that the number of bacteria follows a Poisson distribution, find the probability that 1 ml of liquid there will be:

- ❖ No bacteria
- ❖ 4 bacteria
- ❖ Less than 3 bacteria

Solution

$X \sim \mathrm{P}_0(4)$

$$P(X = x) = \frac{e^{-4}\,4^x}{x!}$$

Where X is the r.v. "the number of bacteria in 1 ml of liquid"

- $P(X = 0) = P(\text{no bacteria}) = e^{-4} = 0.0183$
- $P(X = 4) = P(4 \text{ bacteria in } 1 \text{ ml}) = \frac{e^{-4}4^4}{4!} = 0.195$
- $P(X < 3) = P(\text{less than } 3 \text{ bacteria in } 1 \text{ ml})$
  $= P(X = 0) + P(X = 1) + P(X = 2)$
  $= e^{-4} + e^{-4} * 4 + \frac{e^{-4}4^2}{2!} = e^{-4}(1 + 4 + 8) = 0.238$

b) When we approximate the binomial distribution.

Let X be an r.v. such that $X \sim \mathrm{Bin}(n,\ p)$.Now X can be approximated by a Poisson distribution with parameter $\lambda = np$ if n is large (> 50) and p is small (< 0.1).

The approximation gets better as $n \to \infty \ \ and \ \ p \to 0$.

Example:

A factory packs bolts in boxes of 500. The probability that a bolt is defective is 0.002.

Find the probability that a box contains 2 defective bolts.

Solution

Let X be the random variable. The number of defective bolts in a box:

$X \sim Bin(n, p)$ since n=500, p=0.002, then $X \sim Bin(500, 0.002)$

Thus the pdf of X is given by:

$$p(X = x) = \binom{500}{x}(0.002)^x\ (0.998)^{500-x} \quad , x = 0,1,2,.....,500$$

$$p(X = 2) = \binom{500}{2}(0.002)^2\ (0.998)^{498} = 0.184$$

Since n=500, p=0.002, we use the Poisson approximation with $\lambda = n\,p = 500 * 0.002 = 1$

$X \sim P_0(1)$ With pdf $p(X = x) = \dfrac{e^{-1} * 1^x}{x!}$, $x = 0,1,2,.....500$

We require computing:

$$P(X = 2) = \frac{e^{-1} * 1^2}{2!} = 0.184$$

Mode

Let $X \sim P_0(\lambda)$. Assume that $\lambda = 1$,thus the pdf of X is given by:

$$P(X = x) = \frac{e^{-1} * 1^x}{x!}; x = 0,1,2,.....$$

$x = 0$, $P(X = 0) = e^{-1} = 0.3678$

$x = 1$, $P(X = 1) = e^{-1} = 0.3678$

$x = 2, P(X = 2) = \dfrac{e^{-1}}{2!} = 0.1839$

$x = 3, P(X = 3) = \dfrac{e^{-1}}{3!} = 0.0613$

……

……

We notice that the modes are 0 and 1.

In general, if λ is an integer, then there are two modes and these occur when $x = \lambda - 1$, $x = \lambda$

Let $\lambda = 1.6$, $X \sim P_0(1.6) = P(X = x) = \dfrac{e^{-1.6} * 1.6^x}{x!}$ $x = 0,1,2,.....$

$x = 0$, $P(X = 0) = e^{-1.6} = 0.20189$

$x = 1$, $P(X = 1) = e^{-1.6} * (1.6) = 0.3230$

$x = 2$, $P(X = 2) = \dfrac{e^{-1.6} * (1.6)^2}{2!} = 0.2584$

$x = 3$, $P(X = 3) = \dfrac{e^{-1.6} * (1.6)^3}{3!} = 0.1378$

The mode is 1.

In general if λ is nor an integer then the mode is an integer such that $\lambda - 1 < \mathrm{mod}e < \lambda$

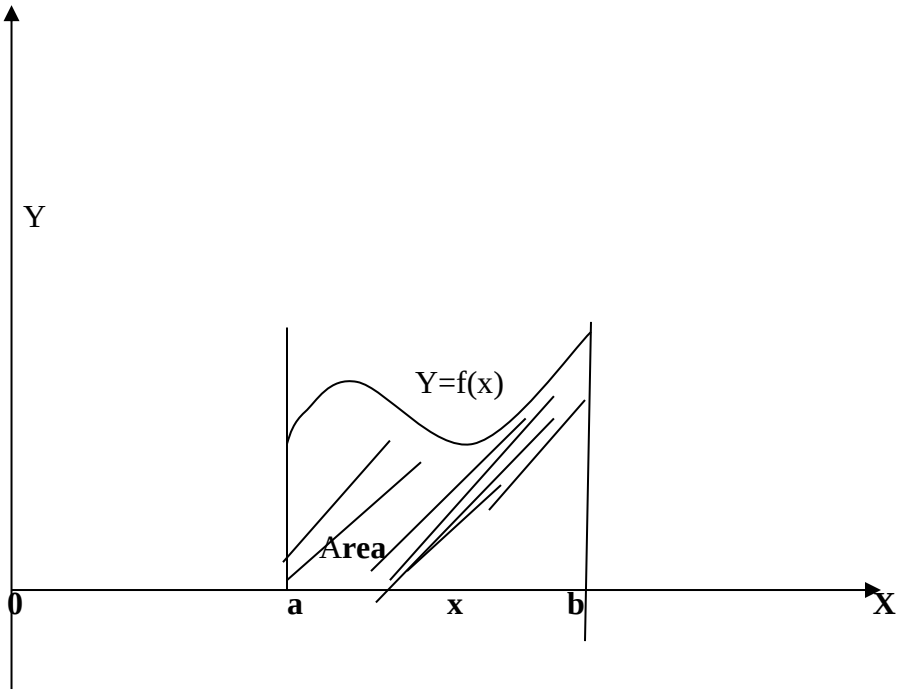
Continuous random variables

(i) Introduction

A continuous random variable is a theoretical representation of a continuous variable such as height, mass or time.

Let X a random variable such that

$$X : \Omega \to X(\Omega) = [a, b]$$
$$\underbrace{\omega}_{Event} \to \underbrace{X(\omega)}_{\mathrm{Re}al\ number}$$



One says that X is a continuous random variable and f(x) is the probability density function.

This pdf satisfies the following conditions:

$$1.\ f(x) \geq 0 \ \ and \ \int_a^b f(x)dx = 1$$

$$2.E(X) = \int_a^b x\, f(x)dx$$

$$3.Var(X) = \int_a^b x^2 f(x)dx - E^2(X)$$

$$4.\sigma_X = \sqrt{Var(X)}$$

If f(x) is the probability density function on $(-\infty, +\infty)$, then:

1. $\int_{-\infty}^{+\infty} f(x)dx = 1$

2. $E(X) = \int_{-\infty}^{+\infty} x\, f(x)dx$

3. $Var(X) = \int_{-\infty}^{+\infty} x^2 f(x)dx - E^2(X)$

4. $\sigma_X = \sqrt{Var(X)}$

Example The continuous random variable X has pdf $f(x) = \begin{cases} \frac{3}{4}(1+x^2), & 0 \le x \le 1 \\ 0, & otherwise \end{cases}$

Find the expectation, the variance and the standard deviation of X.

Solution

$$E(X) = \int_{all\ x} f(x)dx = \frac{3}{4}\int_0^1 x\,(1+x^2)dx = \frac{3}{4}\int_0^1 (x+x^3)dx = \frac{3}{4}\left[\frac{x^2}{2} + \frac{x^4}{4}\right]_0^1 = \frac{3}{4}\left(\frac{1}{2} + \frac{1}{4}\right)$$

$$= \frac{3}{4}\left(\frac{3}{4}\right) = \frac{9}{16} = 0.5625$$

$$But\ E(X^2) = \int_{all\ x} x^2 f(x)dx = \frac{3}{4}\int_0^1 (x^2 + x^4)dx = \frac{3}{4}\left[\frac{x^3}{3} + \frac{x^5}{5}\right] = \frac{3}{4}\left[\frac{1}{3} + \frac{1}{5}\right] = \frac{3}{4} * \frac{8}{15} = \frac{2}{5} = 0.4$$

$$Var(X) = \int_{all\ x} x^2 f(x)dx - \mu^2$$

$$Then,\ Var\ X = 0.4 - (0.5625)^2 = 0.0835$$

$$\sigma_X = \sqrt{Var\ X} = 0.289$$

Mode

The mode is the value of X for which f(x) is greatest in the given range of X.

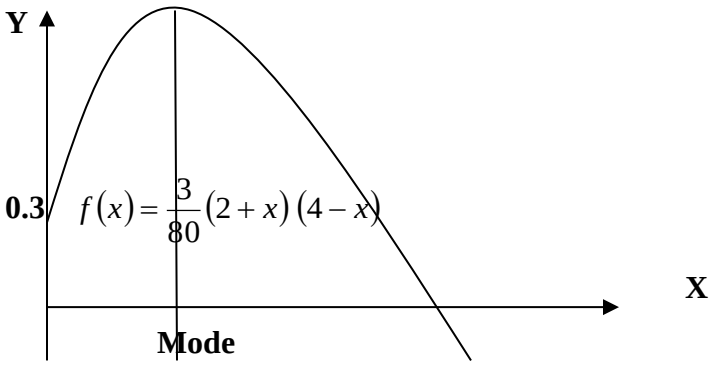
For some pdf, it is possible to determine the mode by finding the maximum point on the curve y=f(x).

Example

The continuous r.v. X has pdf $f(x) = \begin{cases} \frac{3}{80}(2+x)(4-x), & 0 \le x \le 4 \\ 0, & otherwise \end{cases}$

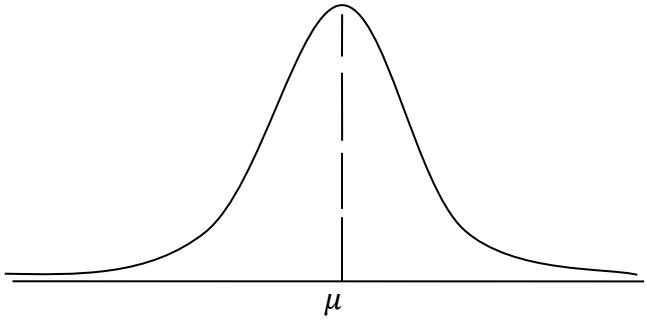
Find the mode.

Solution



Mode=1

ii) THE NORMAL DISTIBUTION



The normal distribution is important in statistics for four main reasons:

1. Numerous phenomena measured on continuous scales have been shown to follow (or can be approximated by) the normal distribution.
2. We can use the normal distribution to approximate various discrete probability distributions, such as the binomial and the Poisson.
3. It provides the basis for the statistical process-control
4. It provides the basis for classical statistical inference.

The Standard Normal distribution

In general we have:

$X \sim N(\mu, \sigma^2)$ Such that the density distribution is given by this mathematical expression:

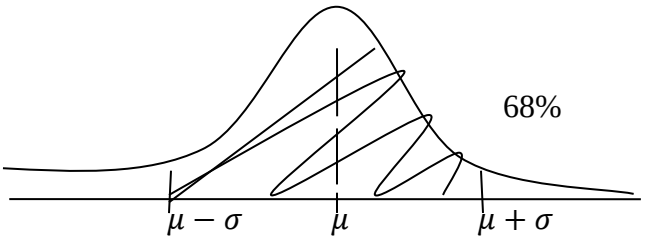
$$f(X) = \frac{1}{\sqrt{2\pi}\sigma} e^{\left\{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}}$$

By standardizing a normally a normally distributed random variable, we will need only one table.

By using the transformation formula for Z, the **standard normal score** $\boldsymbol{Z = \frac{X-\mu_x}{\sigma_x}}$

In this case we have**:** $\boldsymbol{Z \sim N(0,1)}$

The distribution of a random variable which follows the normal law is symmetric according to its expected.



$P(\mu-\sigma \leq X \leq \mu+\sigma)=0.68$

$P(\mu-2\sigma \leq X \leq \mu+2\sigma)=0.95$

$P(\mu-3\sigma \leq X \leq \mu+3\sigma)=0.997$

$\int_{-\infty}^{\infty} f(x)=1$

Properties

1. $P[a<X<b]=P[X<b]-P[X<a]$
2. $P[X>a]=1-P[X<a]$
3. $P[X<-a]=1-P[X<a]$
4. $P(a<X<b/X>c)=\frac{P[(a<X<b)\cap(X>c)]}{P[X>c]}$

Examples:

1. $X\sim N(4,2)$. Calculate$P[X<2]$?

Solution:

$$P[X<2]=P\left[\frac{X-4}{2}<\frac{2-4}{2}\right]=P[Z<-1]=1-\underbrace{P[Z<1]}_{=0.8413}=0.1587$$

2. $X\sim N(1,3)$. Calculate$P[0<X<2/X>-2]$?

$$P[0<X<2/X>-2]=\frac{P[(0<X<2)\cap(X>-2)]}{P[X>-2]}=\frac{P[(0<X<2)]}{P[X>-2]}=\frac{P[X<2]-P[X<0]}{1-P[X<-2]}=$$

$$\frac{P\left[\frac{X-1}{3}<\frac{2-1}{3}\right]-P\left[\frac{X-1}{3}<\frac{0-1}{3}\right]}{1-P\left[\frac{X-1}{3}<\frac{-2-1}{3}\right]}=\frac{P\left[Z<\frac{1}{3}\right]-P\left[Z<\frac{-1}{3}\right]}{1-P[Z<-1]}=0.31$$

EXERCISES

1. Consider Z a variable follows a standard normal distribution.

Calculate:

a. $P[Z<0]$ b. $P[0<Z<2]$ c. $P[Z>2.21]$ d. $P[Z>2.73/Z>2]$ e. $P[Z<2.04]$

f. $P[|Z| < 2]$

2. Consider Z a variable follows a standard normal distribution. Determine $z \in R$ in each follows cases:

a. $P[|Z| < z] = 0.95$ b. $P[|Z| < z] = 0.90$ c. $P[|Z| < z] = 0.8238$

d. $P[z < Z < 1] = 0.6826$ e. $P[0 < Z < z] = 0.4878$

3. Number of cars that pass through a car wash between 16:00 and 17:00 has the following probability distribution the table below:

X	4	5	6	7	8	9
$P(x)$	$1/12$	$1/12$	$1/4$	$1/4$	$1/4$	$1/4$

What is the expected number of cars washed?

4. Find the expected number of boys on a student committee of 3 selected at random from 4 boys and 3 girls?

5. For the standard normal distribution, find the area:

a. between -1.27 and 1.86

b. below 1.7

c. above 1.18

d. between -0.47 and -0.35

3.4. Simple linear regression analysis.

General View

Very often in practice a relationship is found to exist between two or more variables. For example, weights of adult males depend to some degree on their heights. It is frequently desirable to express this relationship in mathematical form by determining an equation that connects the variables. **Regression analysis** is a statistical technique that is used to solve these types of problems.

In simple linear regression, we attempt to model the relationship between two variables, for example, income and number of years of education, height and weight of people, temperature

and output of an industrial process, altitude and boiling point of water, or dose of a drug and response.

Simple Linear Regression Model

The simple linear regression model for n observations can be written as

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, i = 1, 2, \cdots, n$$

The designation **simple** indicates that there is only one explanatory variable x to predict the response y, and **linear** means that the model (3.1) is linear in β_0 and β_1. For example, a model such as $y_i = \beta_0 + \beta_1 x_i^2 + \varepsilon_i$ is linear in β_0 and β_1, whereas the model $y_i = \beta_0 + e^{\beta_1 x_i} + \varepsilon_i$ is not linear. The variable x is referred to as independent variable and y is a dependent variable.

In this course, we assume that:

- y_i and ε_i are random variables and
- $x_i, i = 1, 2, \cdots, n$ are known constants.

In addition, we make the following assumptions:

1) $E(\varepsilon_i) = 0 \; \forall i = 1, 2, \cdots, n$, or, equivalently $E(y_i) = \beta_0 + \beta_1 x_i$.

This states that y_i depends only on x_i, and that any other variation in y_i is random.

2) $Var(\varepsilon_i) = \sigma^2 \; \forall i = 1, 2, \cdots, n$, or, equivalently $Var(y_i) = \sigma^2$.

This assumption asserts that the variance of ε_i or y_i doesn´t depend on the value of x_i. This assumption is also known as the assumption of **homoscedasticity**, homogeneous variance or constant variance.

3) $\text{cov}(\varepsilon_i, \varepsilon_j) = 0 \; \forall i \neq j$, or, equivalently $\text{cov}(y_i, y_j) = 0$.

Under this assumption, the ε_i´s variables (as well as the y_i´s) are uncorrelated each other.

Estimation of β_0, β_1

The Least Squares Method

Using the random sample of n observations $y_i, i = 1, 2, \cdots, n$ and the fixed constants $x_i, i = 1, 2, \cdots, n$, we seek estimators β_0 and β_1 that minimize the sum of squares of the deviations $d_i = y_i - y_i$ of the n observed values $y_i, i = 1, 2, \cdots, n$ from their predicted values $y_i = \beta_0 + \beta_1 x_i$. In fact, for a given value $x_i, i = 1, 2, \cdots, n$, there will be a difference between the

observed value $y_i, i = 1, 2, \cdots, n$ and the corresponding value $y_i, i = 1, 2, \cdots, n$ as determined from the fitted or estimated regression line given by

$$y = \beta_0 + \beta_1 x$$

Figure 3.1 illustrates the situation.

The sum of the squares of the deviations of the observations from the true regression line is

$$L = \sum_{i=1}^{n} d_i^2 = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x_i \right)^2$$

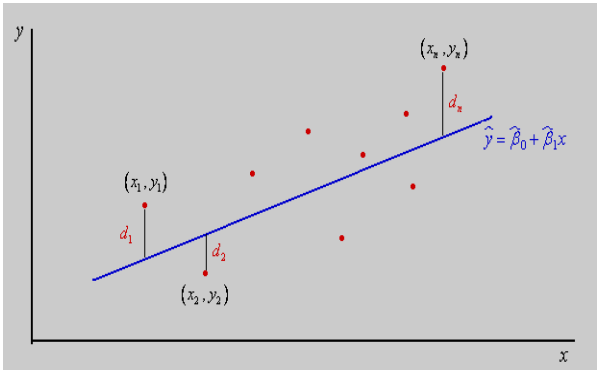


Figure 3.1: Deviations $d_i = y_i - y_i, i = 1, 2, \cdots, n$

The least squares estimators of β_0 and β_1, say β_0 and β_1, must satisfy

$$\frac{\partial L}{\partial \beta_0} = -2 \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x_i \right) = 0$$

$$\frac{\partial L}{\partial \beta_1} = -2 \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x_i \right) x_i = 0$$

Simplifying these equations leads to the following least squares normal equations:

$$\sum_{i=1}^{n} y_i = n\beta_0 + \beta_1 \sum_{i=1}^{n} x_i$$

$$\sum_{i=1}^{n} y_i x_i = \beta_0 \sum_{i=1}^{n} x_i + \beta_1 \sum_{i=1}^{n} x_i^2$$

The solution to the normal equations results in the following least squares estimators β_0 and β_1 :

$$\beta_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}$$

$$\beta_0 = \frac{\sum_{i=1}^{n} y_i}{n} - \beta_1 \frac{\sum_{i=1}^{n} x_i}{n}$$

or

$$\beta_1 = \frac{\sum_{i=1}^{n} y_i x_i - n\bar{x}\bar{y}}{\sum_{i=1}^{n} x_i^2 - n\left(\bar{x}\right)^2} = \frac{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)\left(y_i - \bar{y}\right)}{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

where $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$ and $\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n}$.

Example

Table 3.1 gives the annual consumption (y) of 10 households each selected from a group of households with a fixed personal income (x). Both income and consumption are measured in $ 10,000. Fit the least squares regression line of y on x.

Table: Annual Consumption of 10 Households

Observation i	Consumption y_i	Income x_i
1	4.6	5
2	3.6	4
3	4.6	6
4	6.6	8
5	7.6	8
6	5.6	7
7	5.6	6

8	8.6	9
9	8.6	10
10	9.6	12

Solution

Table 3.2 shows basic calculations to be done.

The regression model is $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i; i = 1, 2, \cdots, 10$.

Using the results of Table, we obtain $\beta_0 = 0.4286$ and $\beta_1 = 0.8095$.

The estimated regression line is $y = 0.4286 + 0.8095x$.

Table 3.2: Basic Calculations.

y_i	x_i	x_i^2	$x_i y_i$	y_i^2
4.6	5	25	23	21.16
3.6	4	16	14.4	12.96
4.6	6	36	27.6	21.16
6.6	8	64	52.8	43.56
7.6	8	64	60.8	57.76
5.6	7	49	39.2	31.36
5.6	6	36	33.6	31.36
8.6	9	81	77.4	73.96
8.6	10	100	86	73.96
9.6	12	144	115.2	92.16
$\sum_{i=1}^{10} y_i = 65$	$\sum_{i=1}^{10} x_i = 75$	$\sum_{i=1}^{10} x_i^2 = 615$	$\sum_{i=1}^{10} x_i y_i = 530$	$\sum_{i=1}^{10} y_i^2 = 459.4$

Figure 3.2 indicates the estimated regression line $y = 0.4286 + 0.8095x$.

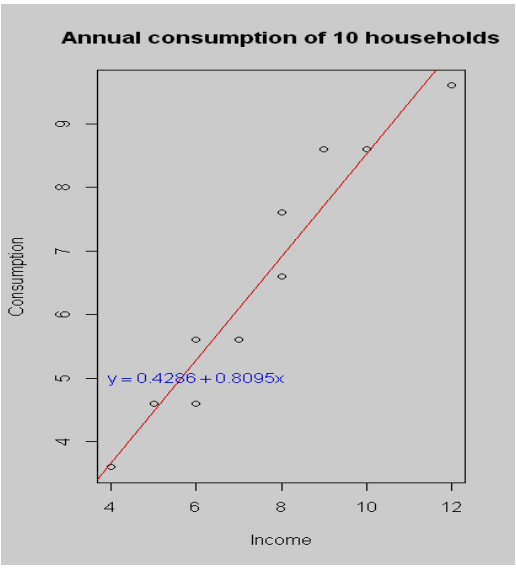


Figure 3.2: Estimated Regression Line $y = 0.4286 + 0.8095x$.